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Structural Bioinformatics Tools for Drug Design

Extraction of Biologically Relevant Information from Structural Databases



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Chapter 1

Introduction

**Jaroslav Koča, Radka Svobodová Vařeková, Lukáš Pravda,
Karel Berka, Stanislav Geidl, David Sehnal and Michal Otyepka**

The rise of computers has revolutionized every single field of human activity. Medicine and drug design is no exception. In fact, understanding the basis of different diseases together with the development of appropriate cures has taken a gigantic leap forward in the last few decades. In the past, drug design was mainly the domain of chemists and medical doctors carrying out wet lab experiments and subsequent testing on live subjects. Nowadays, it is a complicated synergy of a vast spectra of different interoperable life-science fields combined with available biological data. Every day we take advantage of available computational power and combine it with knowledge of the disease's biological nature in order to grasp its true molecular basis and try to suggest potential drug substances using up-to-date bioinformatics methods.

Structural bioinformatics is a well-defined part of bioinformatics. It is related to the analysis and prediction of the three-dimensional structure of biomacromolecules.

In this context, structural bioinformatics has become a very powerful tool, applicable in drug design. This branch strongly benefits from the fact that a great amount of data about various types of molecules is available. For example, we can obtain a complete human genome of a selected person in less than 14 days, nearly 90 million small molecules are described in freely accessible databases (e.g., Pubchem [1], ZINC [2], DrugBank [3], ChEMBL [4]), more than 120 thousand biomacromolecular structures have been determined and published (Protein Data Bank [5]). Thanks to these advances we can relatively routinely solve and analyze the structures of the causative agents of many diseases – proteins, nucleic acids and their complexes. Indeed, solving and examining the atomic structure of hemoglobin aided in understanding the molecular basis of sickle-cell disease [6]. This is caused by a single-nucleotide

polymorphism in DNA, which leads to a substitution of a glutamic acid with a valine residue. As a result a hydrophobic patch is exposed, enabling hemoglobin molecules to aggregate, hence causing sickle-cell disease. Such a detailed level of understanding of biological processes is enabled thanks to the availability of atomic resolution models and their bioinformatic analysis. The importance and usefulness of structural bioinformatics is also highlighted by the several Nobel Prizes related to this research field [7].

Nobel Prizes related to structural bioinformatics:

- Chemistry 2013: Martin Karplus (1/3), Michael Levitt (1/3) and Arieh Warshel (1/3). Development of multiscale models for complex chemical systems
- Chemistry 2009: Venkatraman Ramakrishnan (1/3), Thomas A. Steitz (1/3) and Ada E. Yonath (1/3). Studies of the structure and function of the ribosome.
- Chemistry 2006: Roger D. Kornberg. Studies of the molecular basis of eukaryotic transcription.
- Chemistry 2003: Roderick MacKinnon (1/2). Structural and mechanistic studies of ion channels.
- Chemistry 2002: Kurt Wüthrich (1/2). Development of nuclear magnetic resonance spectroscopy for determining the three-dimensional structure of biological macromolecules in solution.
- Chemistry 1997: John E. Walker (1/4) Elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP).
- Chemistry 1991: Richard R. Ernst. Contributions to the development of the methodology of high resolution nuclear magnetic resonance (NMR) spectroscopy.
- Chemistry 1988: Johann Deisenhofer (1/3), Robert Huber (1/3), Hartmut Michel (1/3). Determination of the three-dimensional structure of a photosynthetic reaction centre.
- Chemistry 1982: Aaron Klug. Development of crystallographic electron microscopy and his structural elucidation of biologically important nucleic acid-protein complexes
- Chemistry 1972: Christian B. Anfinsen (1/2). Work on ribonuclease, especially concerning the connection between the amino acid sequence and the biologically active conformation
- Chemistry 1964: Dorothy Crowfoot Hodgkin. Determinations by X-ray techniques of the structures of important biochemical substances.
- Medicine 1962: Francis Harry Compton Crick (1/3), James Dewey Watson (1/3), Maurice Hugh Frederick Wilkins (1/3). Discoveries concerning the molecular structure of nucleic acids and its significance for information transfer in living material

- Chemistry 1962: Max Ferdinand Perutz (1/2), John Cowdery Kendrew (1/2). Studies of the structures of globular proteins.
- Chemistry 1946: James Batcheller Sumner (1/2). Discovery that enzymes can be crystallized.

Note: The fractions (1/2), (1/3) or (1/4) express a share of each researcher in the Nobel Prize.

Structural bioinformatics supports our understanding of key cell processes, and the execution of the required analyses via a plethora of databases, algorithms and tools, which store, categorize and analyze the biological message. On the other hand, the richness and variability of the services, constant development of their novel functionality and continuous updating of the information in the databases make structural bioinformatics extremely dynamic and difficult to orient in for a novice. For example, a non-exhaustive list of software tools held by Nucleic Acids Research for 3D structure comparison contains almost 80 different online tools and services. These facts motivated us to write this book, which introduces the main structural bioinformatics databases and the key steps of data analysis that are applicable in drug design.

As the computational world has been moving towards online and cloud services, we have paid special attention to the selection of tools and services available online for everyone, free of charge. First we focus on the examples of biomacromolecular fragments, which are also denoted as biomacromolecular patterns (Chap. 2). Fragments often have a biologically relevant function for many biological processes and phenomena. These fragments, however, have to be identified in the available structures, which are deposited in structural databases. Popular databases that often serve as the primary source of biologically relevant information are therefore overviewed in Chap. 3. Trust, but verify is the motto of Chap. 4, as it has been shown that not all of the structures available in public databases are structurally sound. This chapter describes the methods and tools for validating biomacromolecular structures and therefore deciding whether the structures are reliable. Pattern detection and extraction is the key feature for understanding and modulating many vital processes as well as diseases. Example tools tailored for such a purpose are introduced in Chap. 5. Chapter 6 focuses on the detection of channels and pores, biomacromolecular fragments of high biological importance, which allow the passage of a drug into the active site or through a membrane. Chapters 7 and 8 deal with the characterization of biomacromolecular patterns, a task of great importance for inferring biological function. Specifically, we discuss the employment of partial atomic charges and the analysis of channels leading to or through the buried volumes of biomacromolecules. Each chapter contains practical examples and is followed by exercises. Alternatively, these examples can be accessed on-line at <http://fch.upol.cz/en/teaching/structural-bioinformatics-tools-for-drug-design/>. Last but not least, in Chap. 9 we provide two examples, which puts all of the above-mentioned bits and pieces together in a complete and easily understandable bioinformatics project that provides meaningful

biological information. Finally, Chap. 10 summarizes the mission and goals of the book and Chap. 11 contains solutions to the exercises.

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Part I
Patterns, Fragments and Data Sources

Chapter 2

Biomacromolecular Fragments and Patterns

Lukáš Pravda

The function of biomacromolecules such as proteins is intimately connected with their three-dimensional (3D) structure, and as such it is a reasonable starting point for structure-based drug design. Since the tertiary structure is more evolutionarily conserved than the primary sequence, the analysis of 3D structure provides key insights, not only in terms of classification, but has many implications in biotechnologies and drug design. On one hand, we can search for novel binding partners of characterized and validated target proteins; on the other hand, we can infer the function of as-yet uncharacterized proteins responsible for various diseases.

The question is, which part of a biomacromolecule or biomacromolecular properties do we want to evaluate? In general, we are mainly interested in the parts exhibiting biological functions. These are usually small and well-conserved spatial arrangements of amino acids and/or interacting ligands, such as cofactors; substrates or products of enzymatic reactions, inhibitors, or messenger molecules. In this book we collectively refer to these protein substructures as biomacromolecular patterns or fragments. A pattern can, in principle, take a number of different forms. It may be amino acids constituting catalytic or binding sites, sequence patterns responsible for cell signaling [1], allosteric regions [2–4], protein pockets and cavities [5–7], channel lining residues [8, 9] etc.

One of the first steps in every in silico analysis for not only drug design, is the detection of these biologically important patterns. There can be many reasons behind them. We can identify similar binding sites in off-target proteins,¹ discover new inhibitors, facilitate the identification of protein-protein interactions, or evaluate ligand-accessible pathways to the enzyme reaction site to name a few.

¹Off-target protein binding implies an undesirable binding of a small molecule with a therapeutic effect to a protein target other than the primary target for which it was intended. Such binding often causes unintended side effects.

2.1 Pattern Examples

2.1.1 Active Site and Their Inhibition – Cyclooxygenase Inhibitors

The cyclooxygenase enzymes (COX-1 and COX-2) are responsible for the bis-oxygenation of arachidonic acid to prostaglandins. This process is critical during inflammation, cancer, but also in kidney development or maintaining gastrointestinal integrity and it is responsible for pain [10]. As such they are primary targets for a large body of nonsteroidal anti-inflammatory drugs such as aspirin or ibuprofen. While COX-1 is expressed constantly in the majority of cells and possesses a housekeeping function, COX-2 is only induced by inflammatory stimuli. Therefore, the development of selective inhibitors which can in turn be used for example as anti-inflammatory and anticancer agents with as few side-effects as possible, is of great interest. The first structures of COX enzymes were solved some 20 years ago, revealing the binding pocket and their inhibitors as highlighted in Fig. 2.1.

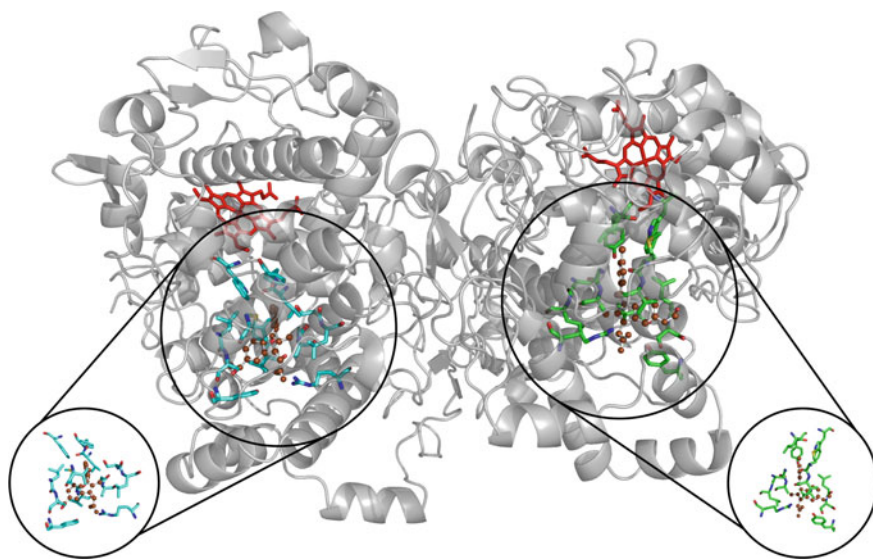


Fig. 2.1 Structure of COX-2 complexed with the indomethacin non-selective inhibitor (PDB ID 4cox). COX-2 is a homodimer, with each unit containing a cyclooxygenase active site and a peroxidase active site. The peroxidase active site is involved in activating the heme group (*red*), which is crucial for further cyclooxygenase reaction. The molecular patterns of the COX-2 inhibitor (in *brown*) together with its interacting partners (*green* or *cyan* with respect to a protein unit) in the enzyme active sites are highlighted. The inhibitor is stabilized both by polar and nonpolar interactions (color figure online)

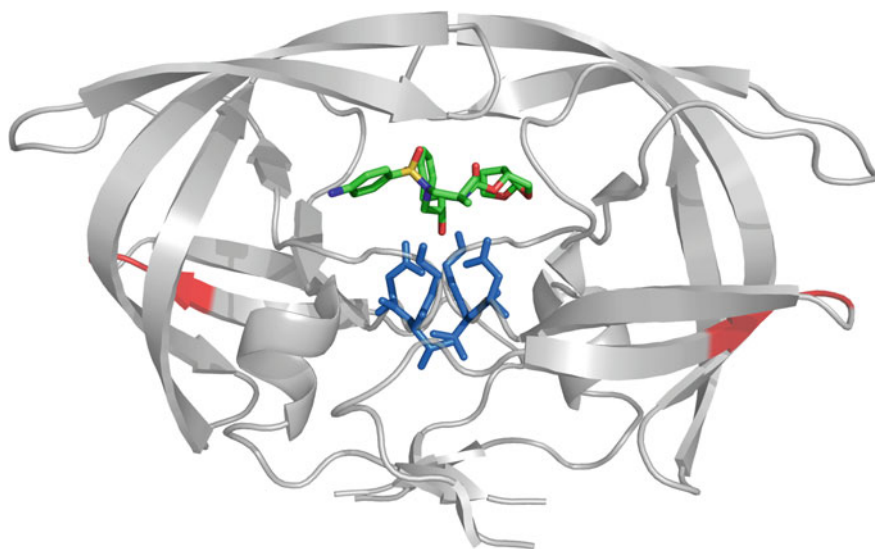


Fig. 2.2 HIV-1 protease complexed with the inhibitor darunavir (PDB ID 3lzv). The molecular pattern of a catalytic triad is highlighted in *blue*. Elbow allosteric regions presumably responsible for the protein flexibility are shown in *red* (color figure online)

2.1.2 Allosteric Site – Structural Flexibility of HIV Protease

Inhibition of the HIV-1 protease is considered to be one of the three key avenues for blocking HIV replication, and therefore prevention of the development of AIDS [11]. Inhibition of the HIV-1 protease active site with drugs like ritonavir, nelfinavir, and amprenavir was considered to be an efficient approach. As a consequence of the drug binding, HIV-1 protease loses its dynamic behavior, which is crucial for its proteolytic function. However, many drug-resistant variants emerged, so inhibitor development continues. A number of NMR and MD experiments revealed putative regions responsible for the enzyme flexibility. As such these allosteric regions can be rationally targeted by novel allosteric inhibitors, in order to inactivate the enzyme's function [12]. Figure 2.2 displays a catalytic triad of the enzyme active sites together with putative allosteric sites responsible for the enzyme's flexibility.

2.1.3 Transcription Factor – Zinc Finger Motif

The DNA-binding class of enzymes called zinc fingers (ZnF) is the most abundant across all biota. The first classical ZnFs denoted as C₂H₂ were extracted from the *Xenopus* transcription factor, where they specifically bind DNA and control transcription [13]. Besides this, ZnFs are responsible for DNA recognition, the regulation of

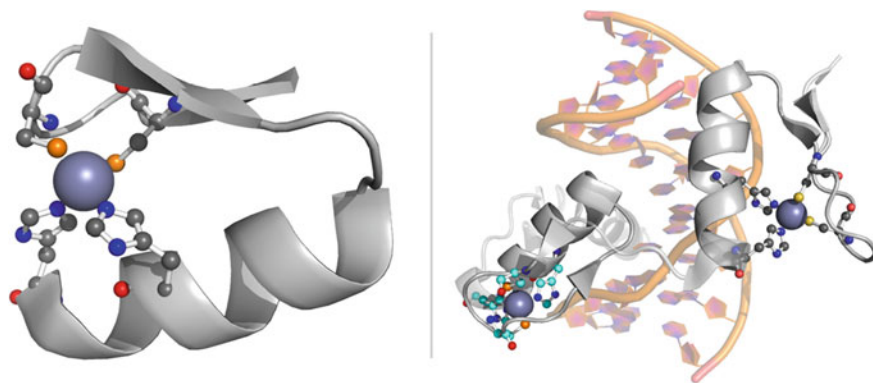


Fig. 2.3 C_2H_2 zinc finger motifs of the transcription factor early growth response protein 1 (Egr1) (PDB ID 4r2a). The figure on the *left* depicts the overall cartoon model of a zinc finger motif, with the residues (two cysteines and two histidines) responsible for zinc ion binding. The zinc ion is shown as a sphere, while cysteine and histidine are denoted in a ball-and-stick model. In the other figure, two zinc fingers are bound to the major groove of the DNA strand

apoptosis and lipid binding. This motif is usually defined by a simple primary structure pattern called a consensus profile. Nevertheless, atypical motifs exist deviating from the consensus profile $X_2-C-X_{2-4}-C-X_{12}-H-X_{3-5}-H$ (X stands for any amino acid, C is cysteine and H represents histidine in the consensus profile), that recognize specific genomic sites. The X_{12} region is usually further decomposed into the sequence $X_3-[FY]-X_5-\psi-X_2$, where [FY] represents either a phenylalanine or tyrosine residue, and ψ denotes a hydrophobic residue. At the 3D level, this sequence has a simple $\beta\beta\alpha$ fold, which is stabilized with a zinc ion coordinated with two histidine and two cysteine residues as shown in Fig. 2.3.

2.2 Pattern Prediction

Over the past few decades a plethora of software tools have been developed for the detection and extraction of biomacromolecular patterns from protein structures. The individual tools differ in the level of pattern description, the employed algorithms and of course their applicability. Drug design usually aims to identify potential binding sites in target and off-target proteins. These are often located in shallow protrusions in the protein surface referred to as pockets or clefts, as well as deeply buried in the protein structure. Therefore, the majority of the software is designed for predicting suitable pockets in apoproteins and holoproteins (e.g. CASTp [14], Pass [15], Q-SiteFinder [16], or FTSite [17]). Others may identify accessible pathways for the small ligands interacting with the proteins (e.g. MOLE 2.0 [18], Caver 3.0 [19] or MolAxis [20]). These are discussed in more detail in Chap. 6 – Detection of

Channels. Generally, pocket prediction for binding protein inhibitors can be classified into two groups: *geometry-based algorithms* and *energy-based algorithms*.

The geometry-based algorithms involve a couple of approaches. The most popular group of algorithms involves the projection of the protein structure onto a *3D grid* with a custom spacing. Next, grid points are evaluated, given their position on the protein and clustered in order to identify putative binding sites. The second approach covers the protein surface with dummy *spheres*, checks if they satisfy the given conditions and again, clusters the results. The final group of geometry-based algorithms utilizes *α -shape theory*. Here the protein structure is preprocessed using Delaunay triangulation/Voronoi diagrams and the pocket is identified based on a variety of filtering criteria.

In comparison to the geometry algorithms, energy-based algorithms instead of calculating favorable distances among sidechain atoms calculate the interaction energy between dummy spheres and sidechain atoms. These spheres are further clustered and ranked based on the energies. The top scoring clusters are in turn reported as favorable ligand binding pockets.

It is hard to define which of the highlighted approaches is the most suitable for binding site prediction, as they under or overestimate certain characteristics. Usually the best approach is to try a couple of them and select the most relevant result based on the consensus between different algorithms. This is the approach taken by the popular service MetaPocket [21], which is discussed in detail in Chap. 5 – Detection and Extraction of Fragments.

Below you can find an example of the successful application of this technique in the life-science domain.

2.2.1 Ubiquitin-Binding Domain Prediction

The family of small regulatory proteins – ubiquitin is responsible for a remarkable range of functions. Ubiquitin can be covalently attached to a specific substrate protein, the process is referred to as ubiquitination. Ubiquitination is responsible for the trafficking of endogenous and retroviral transmembrane proteins. Additionally, it was shown that the blocking of distinct ubiquitin binding domains (UBDs) in vivo can influence retroviral budding. Therefore, the successful identification of novel ubiquitin binding domains can contribute to the design of novel selective drugs. A database-wide study has been successfully conducted [22] in order to identify previously undiscovered UBDs. They found the apoptosis-linked gene 2 interacting protein X (ALIX) to contain a potential new UBD, specifically the central V domain. These in silico findings were later confirmed experimentally by biophysical affinity measurements.

2.2.2 Pattern Detection

In contrast to the prediction of protein structural patterns, there are software tools and approaches capable of their direct detection. The subtle difference between the two is rather simple. Prediction strives to make an educated guess as to whether or not an arrangement of amino acids will have the desired characteristics, while direct detection only identifies patterns with user-defined properties. For example you can specify a pattern composition at the atomic, residual or secondary structure level; restrict inter-atomic distances, or bond connections. This can be particularly useful for pharmacophore search and for the extraction of more general patterns of interest.

In the following section we review some of the tools used for pattern detection. RASMOT-3D PRO [23] is a web service performing systematic searches of 3D structures given a user-defined structural pattern. The pattern exploration is limited to up to 10 selected protein structures or a non-redundant set of PDB chains. An estimate of whether or not a found pattern corresponds to the query structure is made based on the comparison of C_α and C_β atoms altogether with the RMSD.² Another powerful service, which is directly incorporated into the Protein Data Bank in Europe [24] is PDBeMotif [25]. This web application allows a wide range of pre-defined search functions; however its customization is limited to the pre-defined parameters. Another drawback to this approach is the fact that the precomputed data in the database are stored for individual protein chains, therefore neglecting all patterns concerned with the interface of chains. In comparison, PatternQuery [26] is a language and a web-service covering the majority of the former search, taking into consideration the PDB entry as a whole. The advantage is that by using clear and highly customizable syntax, all the queries can be accurately tailored according to the user's needs, even covering complex patterns. More information on the functionality of PatternQuery is provided in Chap. 5. Finally, IMAAAGINE [27] is designed for the identification of patterns up to 8 amino acids (AA) in size with pre-defined distances, thus completely neglecting the bound ligands. Last but not least, ASSAM [28] identifies user-defined patterns of up to 12 AAs.

Below you can find an example of a pattern detection protocol successfully applied in the field of drug design.

2.2.3 Phosphorylation of Drug Binding Pockets

Roughly half of eukaryotic proteins are subject to a post-translational modification – phosphorylation. This addition of a phosphate group to certain amino acid residues can greatly influence the properties of a binding site which is subject to drug inhi-

²RMSD is a metric describing the structural difference between two molecules (patterns) in Ångströms, i.e. how well would two or more structures fit on top of each other. The higher the RMSD is, the more divergent the structures are. Two molecules with identical conformation (same atomic positions) have an RMSD equal to 0.

bition. A recent database-wide survey [29] examined mammalian proteins with the bound drug ligand. In particular, target-bound ligands together with residues within 12 Å of the binding site have been extracted and inspected for phosphorylation. Over 70 % (453) of the proteins exhibited phosphorylation. Almost one third of them (132) exhibited this phosphorylation in the vicinity of the binding site, and therefore can alter ligand binding. For 70 out of the 132 examples, it is known whether or not phosphorylation alters drug binding. 27 of them exhibited similar effects on activity even after phosphorylation, in contrast to the other 43, whose effects were the opposite.

For example, cyclin-dependent kinase 2 (CDK2) is an enzyme catalyzing the phosphoryl transfer of ATP phosphate group to serine or threonine hydroxyl in a protein substrate, a process important in cell cycle regulation. In particular, the enzyme exhibits phosphorylation both at a positive and negative regulatory site [30]. While the phosphorylation of threonine 160 in the vicinity of the active site activates the enzyme function [31], the phosphorylation of tyrosine 15 negatively affects substrate binding [32, 33].

This is just one example of how the database-wide identification, extraction and analysis of structural patterns can provide a fresh insight into the phosphorylation of an inhibitor's binding sites in the context of rational drug design. Using sophisticated tools like PatternQuery can tremendously simplify the complexities of obtaining input data for various types of analyses, and therefore enable analyses to be carried out that were not feasible before.

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Chapter 3

Structural Bioinformatics Databases of General Use

Karel Berka

3.1 How a Biomacromolecule Looks Codes What It Does

Biomacromolecules have complex structures that are difficult and expensive to obtain. These structures are however the key to understanding their function. In human genetics, one often wonders whether an observed disease causing a mutation is located in the structure related to the active site, channel leading into it, interface between individual proteins or another functionally important site. The structural motifs are also responsible for molecular recognition. Conserved residues can be an important clue to protein function and observed protein-ligand interactions are very helpful for *in silico* drug design. For these and other reasons, macromolecular structures are stored and analyzed by a bunch of essential and specialized databases and web services (Table 3.1).¹

The analysis and mainly annotation of individual structures is necessary, because merely the position of individual atoms – which is the information provided by experiment during structure elucidation – is not enough for our understanding of these complex 3D structures. The primary information added is therefore through fitting the sequence of a macromolecule to the atom positions and by an overall validation that the structure fits. Additional layers of information are also added from comparisons to similar structures and sequences or by the annotation of ligands and specification of their interactions with the macromolecule. However, the structures can also be studied by more specialized analysis of their physicochemical properties, e.g. membrane inclusion or disorder. In order to assist researchers, all major databases are nowadays interlinked from one source – The Protein Data Bank.

¹For more complete list of structural databases please refer to http://www.oxfordjournals.org/our_journals/nar/database/subcat/4/14.

Table 3.1 Overview of several (structural) bioinformatics databases for general use

Database	Description	Web address	Ref.
Worldwide Protein Data Bank (wwPDB)		http://wwpdb.org/	[1]
BMRB	Biological Magnetic Resonance Data Bank (NMR)	http://www.bmr.b.wisc.edu/	[2]
PDBe	Protein Data Bank in Europe	http://www.ebi.ac.uk/pdbe/	[3]
PDBj	Protein Data Bank Japan	http://pdbj.org/	[4]
RCSB PDB	Research Collaboratory for Structural Bioinformatics Protein Data Bank	http://www.rcsb.org/	[5]
Other views on PDB data			
PDBsum	Pictorial analysis of macromolecular structures	http://www.ebi.ac.uk/pdbsum/	[6]
PDB_REDO	Re-refined PDB files	https://xtal.nki.nl/PDB_REDO/	[7]
CCD	Chemical Component Dictionary	http://www.wwpdb.org/data/ccd/	[8]
Classification			
CATH	Domain classification of structures	http://www.cathdb.info/	[9]
Pfam	Classification of sequence families	http://pfam.xfam.org/	[10]
Flexibility and disorder			
PDB Flex	Intrinsic flexibility in proteins	http://pdfflex.org/	[11]
PED3	Protein Ensemble Database	http://pedb.vib.be/	[12]
Pocketome	Encyclopedia of ensembles of druggable binding sites	http://www.pocketome.org/	[13]
DisProt	Database of Protein Disorder	http://www.disprot.org/	[14]
Membrane proteins			
OPM	Orientations of proteins in membranes	http://opm.phar.umich.edu/	[15]
MemProtMD	Membrane proteins models	http://sbcb.bioch.ox.ac.uk/memprotmd/	[16]
Other biomacromolecules			
NDB	Nucleic Acids Database	http://ndbserver.rutgers.edu/	[17]
GFDB	Glycan Fragment Database	http://www.glycanstructure.org/	[18]
Other databases			
UniProt	All about Protein Sequences	http://www.uniprot.org/	[19]
ChEMBL	Small drug-like molecules and targets	https://www.ebi.ac.uk/chembl/	[20]
ChEBI	Chemical Entities of Biological Interest	https://www.ebi.ac.uk/chebi/	[21]

3.2 Worldwide Protein Data Bank (PDB) – Essential Structure Repository

PDB is the worldwide essential repository of macromolecular structure information coordinated by the Worldwide Protein Data Bank (wwPDB) consortium [1]. The wwPDB consortium is coordinated by four data centers which serve as the deposition, annotation, and distribution sites of the PDB archive. Each site offers tools for searching, visualizing, and analyzing PDB data:

- *Biological Magnetic Resonance Data Bank* (BMRB) collects NMR data and captures assigned chemical shifts, coupling constants, and peak lists for a variety of macromolecules; contains derived annotations such as hydrogen exchange rates, pK_a values, and relaxation parameters.
- *Protein Data Bank in Europe* (PDBe) provides rich information about all PDB entries, multiple search and browse facilities, advanced services including PDBePISA, PDBeFold and PDBeMotif, advanced visualisation and validation of NMR and EM structures, tools for bioinformaticians.
- *Protein Data Bank Japan* (PDBj) supports browsing in multiple languages such as Japanese, Chinese, and Korean; SeSAW identifies functionally or evolutionarily conserved motifs by locating and annotating their sequence and structural similarities, tools for bioinformaticians, and more.
- *Research Collaboratory for Structural Bioinformatics Protein Data Bank* (RCSB PDB) provides simple and advanced searches for macromolecules and ligands, tabular reports, specialized visualization tools, sequence-structure comparisons, RCSB PDB Mobile, Molecule of the Month and other educational resources at PDB-101, and more.

All partners share the responsibility for annotating macromolecular structure depositions to the PDB. More than 120,000 experimentally determined atomic structures (summer 2016) in the PDB archive are a treasure trove for scientists in fields such as structural biology, biochemistry, bioinformatics, protein engineering, drug design, human genetics, and molecular biology. As such it is an indispensable data source for today's life sciences.

Each structure in the PDB archive is assigned a four-character-long access code – PDB ID, e.g., 1tcn, which serves as a key identifier. Any PDB ID leads to a page in the PDB archive, which also covers additional layers of information – name, authors, citation, source organism, interacting compounds, sequence, known function (e.g. reactions for enzymes), gene ontology (GO) annotation, validation and description of the experiment, and links to other databases. PDB presents data either in a so-called asymmetric unit, which is a minimal irreproducible representation of the structure, or as a biological unit, which should represent an actual functional complex. All data are also accessible in a plaintext version for each PDB ID. The former format of PDB that was defined with the establishment of the PDB archive in the 70s underwent several updates, but its rigid structure started to be a problem with successes in macromolecular structure elucidation for large complexes, such as viral

capsids or ribosomes. For this reason, the wwPDB consortium agreed on the new format PDBML/mmCIF, which uses extendible XML/XSD schemas (<http://pdbml.rcsb.org/>).

3.2.1 Protein Data Bank in Europe (PDBe)

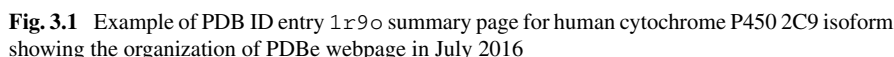
PDBe is actively involved in managing three core archives in structural biology. In addition to its role in the wwPDB consortium (<http://wwpdb.org>) in the annotation of all European and African depositions (>35 % of all depositions), it also established Electron Microscopy Data Bank (EMDB) in 2002 to archive macromolecular structure volumes determined using cryo-microscopy and tomography. *EMDataBank* (<http://emdatabank.org>, [22]), an international consortium of which PDBe is a founding member, now manages EMD. The final archive – the Electron Microscopy Pilot Image Archive (EMPIAR; <http://pdbe.org/empiar>, [23]) established in 2014 stores raw image data for a number of entries in EMD.

The PDBe database underwent a complete redesign in 2015 to improve the accessibility of macromolecular structure data.² Its integration with the UniProt database (<http://uniprot.org>) via the SIFTS resource provides necessary cross-referencing, which enables better searches than merely protein or gene names and provides a quick link to the vast UniProt archive with additional biological information.

The PDBe webpage for an individual PDB ID contains a summary of the structural analysis provided by numerous annotations and citations where available (see Fig. 3.1). Information about the structure itself is sorted into four main sections, which summarize various levels of knowledge about the structure.

- The first section contains analysis of the *function* and *biology* connected with the macromolecule – source organism, and Gene Ontology (GO) terms connected with the structure or its sequence – its biochemical function within the biological process together with cellular component localization. The sequence family from Pfam and the structure domain from CATH can be used to analyze an individual domain. If the structure is an enzyme, then EC classification is also provided.
- The second section provides detailed information about the *structure* itself and about possible quaternary structure – assembly of several chains. Each macromolecular chain is also described by its sequence and is annotated from external sources, e.g. UniProt, Pfam, CATH, and from a structural point of view, e.g. quality or secondary structure. PDBe is also capable of interactive visualization of all 3D, 2D and 1D structural information about individual annotations at the same time, which enables a better understanding of the interconnectedness of individual sequence positions. It is also possible to search for similar structures via PDBeFold or sequences via PDBeXplore.

²Latest version of PDB format documentation can be found at <http://www.wwpdb.org/documentation/file-format-content/format33/v3.3.html>.



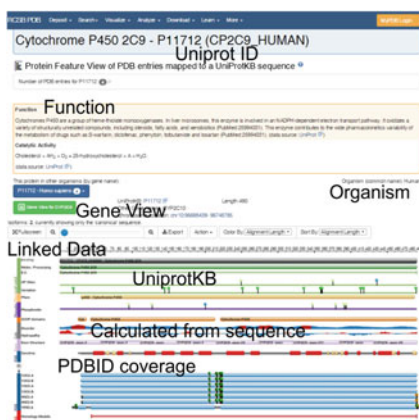
- The third section investigates *ligands* and their pose and interactions with macro-molecule. It contains a description of each ligand together with its surrounding environment in a 2D diagram from LigPlot and the same in 3D representation. Links are provided to other EBI chemical databases, such as similarity search from ChEBI, identification of the molecule within the Chemical Component dictionary in PDBeChem, or bioassay data from ChEMBL.
- The last section covers all information about *experimental results* and *validation* of the model quality. The detailed description of the experiment is given with links to depositories with raw data or refined structures. The major value used in the quality assessment is model resolution R, which describes which atomistic details are visible within the structure. A value below 2 Å usually means that there are enough details to also localize individual atoms, and a value below 3 Å is usually usable for structural analysis. However the quality of the structure is more complex, and therefore additional validation about the structure is available to discern which parts of the structure are trustable compared to others, as we will discuss in the Chap. 4 about Validation.

PDBe also uses advanced search within the PDB database, which not only searches for the names of macromolecules given by depositors but also within protein families, enzymes, GO terms, genes, authors, or even journals. A search can be interactively focused on specific species, interacting compounds, resolutions, citations, and more. As such, PDBe is a nice starting point for the analysis of biological questions from the structural point of view.

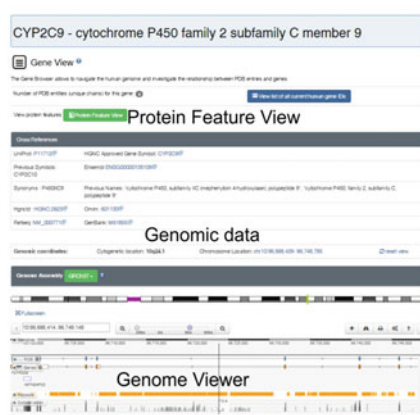
3.2.2 RCSB PDB

RCSB PDB is a US partner of the wwPDB consortium, which also delivers annotated data from the PDB database. Interesting features of RCSB PDB in comparison with PDBe are Protein Feature and Gene Views, which combine information about the sequence from the UniProtKB database and related structures (Fig. 3.2). Protein Feature View lists all available PDB ID with their coverage and observed secondary structure together with linked information from various sources – UniProt, Pfam, phosphorylation sites, domains, predicted disordered parts, calculated hydropathy profile for the analysis of possible transmembrane regions, exon structure and available homology models. As such it allows the position of a sequence within a structure to be connected with its function. Gene View enables navigating the human genome and investigating the relationship between PDB entries and genes. It shows the position of the gene on the chromosome, exon structure, the presence of nearby repeats and conservation.

In addition to the information shared via all wwPDB partners, RCSB PDB also contains PDB-101 education resources, which show possibilities of structural bio-



(a) Protein Feature View



(b) Gene View

Fig. 3.2 Protein Feature View (left) and Gene View (right) for human cytochrome P450 2C9 in July 2016

ogy for our understanding of health and diseases, the molecules of life and bio and nanotechnologies. It enables us to understand how individual macromolecules contribute to whole body functions and how we can use this knowledge e.g. for disease treatment and synthetic biology engineering. Upon entering a new field or just out of pure curiosity, PDB-101 resources can guide a user's first steps in structural biology.

3.3 Other Notable Databases

3.3.1 *PDBsum – Pictorial View on PDB Database*

PDBsum provides an at-a-glance overview of the contents of each 3D structure deposited in the wwPDB. It shows the molecule(s) that make up the structure (i.e., protein chains, DNA, ligands and metal ions) and schematic diagrams of their interactions; it also adds information about structural motifs such as clefts, channels and pores and ligand validations (see Chap. 4 on Validation). *PDBsum* also contains visualizations in 2D structures not only for ligand-protein interactions, but also for protein secondary structure and interactions between individual chains. For enzymes, catalytic residues are listed from the *Catalytic Site Atlas* database (<http://www.ebi.ac.uk/thornton-srv/databases/CSA/>, [24]). Additional information about the conservation of amino acid sequences are obtained from multiple alignments, and known sequence variants from the 1000 Genomes Project are also mapped on the corresponding protein sequences in the Protein Data Bank, cross-referenced with UniProt via SIFTS.

3.3.2 *PDB_REDO and WHY_NOT Databases for Curated Structures*

As the methodology of macromolecular structure advanced over the years, errors in structures were unavoidable. Many PDB entries are old and can suffer from issues that can be fixed with current programs. *PDB_REDO* is built from automatically re-refined PDB entries. In cases where the PDB entry is not suitable for publishing in *PDB_REDO*, the reasons for its exclusion are explained.

3.3.3 *CATH and Pfam Databases for Classification of Protein Folds and Sequences*

The classification of proteins within a certain hierarchy can be helpful for the analysis of an unknown protein or of evolutionary relationships. These databases use different

classification methods, which can be sometimes nicely complementary. The CATH database organizes proteins according to their structures, whereas the Pfam database uses multiple sequence alignments to known protein sequences.

Protein structures in *CATH* are classified by four major levels of hierarchy: (i) Class according to secondary structure composition; (ii) Architecture according to overall shape as determined by the orientations of the secondary structures in 3D space; (iii) Topology by fold groups in terms of both the overall shape and connectivity of the secondary structures; and finally (iv) Homologous superfamily which are thought to share a common ancestor.

Protein sequences in the *Pfam* database are classified into large collections of protein families, each represented by multiple sequence alignments and hidden Markov models and sorted into “clans” of related entries containing one or more functional regions – domains.

3.3.4 *PDB Flex, Pocketome and PED3 Databases to Analyze Protein Flexibility and Disorder*

PDB entries are useful frozen snapshots of the macromolecular structures. Even such static 3D images of a macromolecule can help to ascertain their function and to explain a number of scientific questions. However, in reality, all molecules undergo constant unstoppable stochastic movements organized by the intrinsic flexibility of the macromolecular chain, which is necessary for macromolecular function. Therefore it is helpful to study the flexibility of the macromolecular chain.

The *PDB Flex* database explores this intrinsic flexibility by the analysis of structural variations between different depositions of the same protein in PDB. The structures of protein chains with identical sequences (sequence identity >95 %) were aligned, superimposed and clustered. Then global and local structural differences were calculated within these clusters and visualized not only in terms of identifying the most flexible parts, but also in idealized molecular movies obtained from the interpolation of individual structures.

The plasticity of the binding site in its interaction with small molecules is captured by the *Pocketome* database. The automatic Pocketome generation procedure includes only proteins that have an entry in the reviewed part of the UniProt knowledge base, are represented by at least two PDB ID codes, and have been co-crystallized in complex with at least one drug-like small molecule – in a pocket. Such binding pockets can be further analyzed for conformational clusters, important residues, binding compatibility matrices and interactive visualization of the ensembles using the ActiveICM web browser plugin.

On the other hand, the PDB database only contains those proteins which have to some extent a static structure – otherwise the image would be unobtainable by the most common X-ray crystallography. However some parts of the proteins or even whole protein classes can be unstructured – disordered – and therefore uncrystaliz-

able. In such cases, one single structure is not enough to describe such protein parts, and a whole ensemble of structures is necessary for a proper description of such a disordered and yet not random protein part. Ensembles can be in general obtained by NMR and these structures can be found in the *BMRB* and therefore in the PDB database. However, there are also other techniques which can be used for structure ensemble generation, e.g., Small Angle X-ray Scattering (SAXS) or molecular dynamics simulations (MD). These ensembles can be found in the Protein Ensemble Database (*PED3*), but unfortunately only for a handful of proteins so far.

3.3.5 *OPM and MemProtMD Databases for Membrane Protein*

Around 20–30 % of protein-encoding genes and more than 50 % of drug targets are membrane proteins [25]. It can therefore be seen that membrane proteins are vital for many biological processes, such as cellular metabolism, molecular sensing, intracellular communication, and others. However there are only about 3,000 membrane protein structures in the PDB database, which translates to a mere 3 % of all available structures, as these proteins are difficult to obtain. As PDB does not contain the position of lipids around membrane proteins, while this is an important feature for their activity, there are several structural databases focused on membrane immersion.

The Orientation of Proteins in Membranes (*OPM*) database solves membrane immersion via the minimization of protein transfer energies from water to membrane with an implicit solvent model. *OPM* provides a reasonable orientation of the protein structure and its localization in individual cell membranes (e.g. cell membrane, endoplasmic reticulum or mitochondrial membranes, etc.). The server also calculates the orientation of user-uploaded structures such as homology models and its results can be used for the start of MD simulations.

MD simulations of membrane proteins are an especially important tool for studying this protein class, as this technique enables experimental difficulties with the establishment of their structure to be overcome. However the equilibration of a protein in membrane is a computationally intensive task. The *MemProtMD* database provides membrane proteins immersed and equilibrated in explicit coarse-grained lipid bilayers. As such, this approach is usable not only for integral membrane proteins, but also for peripheral ones.

3.3.6 *NDB and GFDB Databases for Other Macromolecules*

Most of the above databases are focused mainly on proteins. However other macromolecules are important as well – the importance of nucleic acids is growing every

day and other molecules such as sugars can have also some impact on biological processes.

The Nucleic Acids Database (*NDB*) is the main reference in the field of the structures of nucleic acids. It uses the structures of RNA and DNA extracted from PDB database, but its annotation focuses on nucleic-acid-specific information, which is hardly accessible from the original PDB entry. Structures stored in NDB are searchable by sequence, secondary structure and structural patterns, with special interest in hydrogen bonding motives. The database allows visualization not only in 3D, but also in 2D, as well as links to other databases, tools and educational resources related to nucleic acid structures. The NDB database is therefore a good start for the analysis of these macromolecules.

The Glycan Fragment Database (*GFDB*) is focused on known structures with glycosylation, since especially secreted proteins are expected to be glycosylated. Glycan Reader is used to build the Glycan Fragment DB and Glycan-Protein DB. Glycan Builder generates glycan structures through fragment-based threading approaches. This portal is tightly integrated with the *CHARMM-GUI* server (<http://www.charmm-gui.org/>, [26]) for MD simulation input generation and online electrostatic potential visualization.

3.3.7 *UniProt and ChEMBL Databases – Power of Connection*

While 3D structures are important information about given macromolecule, more data about the function, localization, mutations, interactions with other (macro) molecules, etc. are commonly available from scientific literature. Unified view over the corpus of literature is however hard to establish by simple reading as the amount of data is stockpiled in an ever increasing rate. In such case, the database focused on unification and curation of available data starts to be important starting point for any focused study.

UniProt database is centered on protein sequences – for each protein sequence, UniProt adds all known annotations of Gene Ontology on molecular function, connected biological processes, subcellular location, but also about processing of the sequence, its pathology, expression in individual tissues, interactions with small molecules or other macromolecules, available structures or structural models, classification of family and individual domains and cross references to other databases and scientific literature. As such it is a valuable resource hub of protein-related data.

ChEMBL database is built around small drug-like molecules and their targets. It stores not only structures of ligands and link to their targets, but also essays over binding, functions or pharmacokinetics connected with individual compounds and targets. In addition it provides calculated molecular properties of ligand such as molecular weight, octanol-water partition coefficient ($\log P$), surface area, acid dissociation constants (pK_a) and number of rotatable bonds, hydrogen bonds acceptors

and donors. ChEMBL database contains about 2 millions of compounds and around 14 million of bioactivity data collected from about 62 thousand publications making it large data trove for any data mining activities over drug-like molecule functions.

3.4 Conclusion

Four decades of intense development of macromolecular databases have compiled enormous amount of data, which is a treasure trove for today's life sciences. Thanks to interconnectedness among individual databases and their openness, it is easier than ever to delve into these immense data sources to quickly gain insight into a research task in question from almost anywhere in the world.

3.5 Exercises

3.5.1 Use of PDBe

1. Na^+/K^+ ATPase

Sodium-potassium pumps take almost $1/5 - 2/3$ of energy produced within cells. In this exercise, we will try to find, how can be such important protein analyzed from the structural point of view. Search PDBe for Na^+/K^+ ATPase structures for the one with best resolution and obtain:

- (a) PDB ID with resolution and source organism,
- (b) present ligands,
- (c) number of individual chains,
- (d) secondary structure of gamma subunit,
- (e) functions and other GO annotations.

2. Larger structures

Macromolecules are usually present in PDB database as an asymmetric unit, whereas biological function can be provided by macromolecular assembly. Prime examples of such behavior are viral capsid proteins.

- (a) Find how many viral capsid proteins are needed for building of empty canine parvovirus viral capsid.
- (b) PDBe contains also data from electron microscopy (EM) therefore it is possible to compare experimental EM data directly with built atomistic model. One of canine parvovirus capsid proteins was resolved using EM. Try to compare built in model with EM volume map from EMDB. Does the EM map support preferred theoretical assembly from previous question?

3.5.2 Use of RCSB and ChEMBL

3. Kinase inhibitor example – roscovitine

Human protein kinases are regulation proteins important not only for cell cycle, however their involvement in the cell cycle regulation is their key property for certain class of cancerostatic therapeutics in current development and clinical trials. In this exercise, we will look into a prime example of kinase inhibitor – roscovitine. Look for roscovitine among ligands in RCSB.

- (a) Find its 2D structure.
- (b) With what proteins it was crystalized? List their PDB ID as well as protein name.
- (c) The typical target for roscovitine is cyclin-dependent kinase 2 (CDK2), look into the complex of ligand ID RRC with CDK2 and list all aminoacids, with which it interacts.
- (d) How active inhibitor is roscovitine? Find any values indicating how well will roscovitine bind to CDK2.
- (e) Finally – look into ChEMBL for CDK2 inhibitors, which undergo clinical trials.

3.5.3 Use of PDBsum

4. Na⁺/K⁺ATPase

Let's return to sodium-potassium pump – PDBsum can be also used for further analysis, which is not present in PDBe, nor in RCSB, so let's have once again have a look on PDB ID 2zxe:

- (a) Identify ligand clusters present in sodium-potassium pump α subunit.
- (b) Is the structure of cholesterol present in the structure correct?
- (c) Identify catalytic residues.
- (d) Which parts of β subunit are the least conserved?
- (e) Does FXYD protein have more protein-protein contacts to α or β subunit? What type of amino acids form majority on the interfaces? Are there present inter-protein disulphide bonds?

3.5.4 Use of CATH

5. Cytochrome P450 proteins

CATH database sorts structures into homologous CATH superfamilies, where it collects common properties of proteins within given superfamily.

- (a) Find CATH superfamily for cytochrome P450 proteins.
- (b) Decode CATH code for this family with structural description.
- (c) What are the most typical GO terms annotated with this superfamily?
- (d) What is the typical reaction catalyzed by this enzyme?
- (e) Use Gene3D to find how frequent this protein domain is between species.
Which kingdom uses this domain the most?
- (f) Identify the smallest and largest representatives of this domain family?

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Chapter 4

Validation

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4.1 Introduction and Motivation

The three-dimensional structure of biomacromolecules can be determined via *X-ray crystallography*, *Nuclear magnetic resonance (NMR) spectroscopy* and *Cryo-electron microscopy (Cryo-EM)* [1]. X-ray crystallography is currently the most favored technique for the structure determination of proteins and biological macromolecules. As a first step, it requires crystallization of the biomacromolecule. Afterwards, the crystal is exposed to an X-ray beam, which diffracts through the crystal. By measuring the angles and intensities of these diffracted beams, a crystallographer can produce a three-dimensional picture of the electron density within the crystal. From this electron density, the mean positions of the atoms in the crystal can be determined, as well as their chemical bonds, their disorder and various other information. The better the quality of the crystal, the better resolution of the structure can be obtained. Resolution is the distance corresponding to the smallest feature that can be observed. High numeric values of resolution, such as 4 Å, mean poor resolution, while low numeric values, such as 1.5 Å, mean good resolution. NMR spectroscopy is the second-most used technique for biomacromolecular structure determination. It is based on utilizing the phenomenon of nuclear magnetic resonance, therefore the molecule in a solute must be first placed into a strong magnetic field. This field changes the resonance frequency of the atoms, the NMR spectrum of the molecule is measured and its evaluation provides detailed information on the electronic structure of a molecule. The applicability of NMR spectroscopy is limited to small or medium-sized biomacromolecules. To date cryo-EM has only produced a small part of the known biomacromolecular structures. However its importance is growing, because its resolution is improving steadily and it can handle really large structures such as viruses, mitochondria and ribosomes. Cryo-EM performs an imaging of radiation-sensitive specimens via a transmission electron microscope at cryogenic temperatures (usually the temperature of liquid nitrogen).

The validation of biomacromolecular structures obtained by all these approaches has become a very important topic, because some published structures have been

found to contain serious errors [2]. Major validation issues even caused a retraction of articles from prestigious journals such as Science and Nature [3]. The reason was that researchers based their discoveries on incorrect structures and deduced results, which cannot be seen from the real (correct) structures. Unfortunately, a problem with PDB structure quality still remains for two reasons: The structures deposited in the past, were not validated at all or were validated only partially. Even recently, authors can deposit the structure despite it containing quality issues (see the Nipah G attachment glycoprotein validation example below). Moreover, close to 20 % of the ligand structures from Protein Data Bank were reported to have some quality problems [4].

4.2 Nipah G Attachment Glycoprotein Validation Example

Nipah virus infection may lead to severe respiratory disease and fatal encephalitis in humans. The Nipah virus relies on the Nipah G attachment glycoprotein for host cell recognition. The crystal structure of the glycoprotein complexed with its receptor ephrin-B3 (PDB ID 3d12, [5]) was deposited into PDB in 2008. It contains the glycoprotein and 30 instances of 11 different carbohydrates, which all contain one ring and five chiral atoms. Validation of all carbohydrates using MotiveValidator [6] showed that 13 of these ligands had incorrect chirality. An example of a ligand with an incorrect chirality is shown in Fig. 4.1a. This particular ligand (i.e., NGA 1362 A) has a wrong chirality in all 5 of its chiral atoms. This means its real nature strongly differs from NGA. Therefore when we expect this ligand to have the properties and chemical behavior of NGA, we are misled. Summary information about chirality

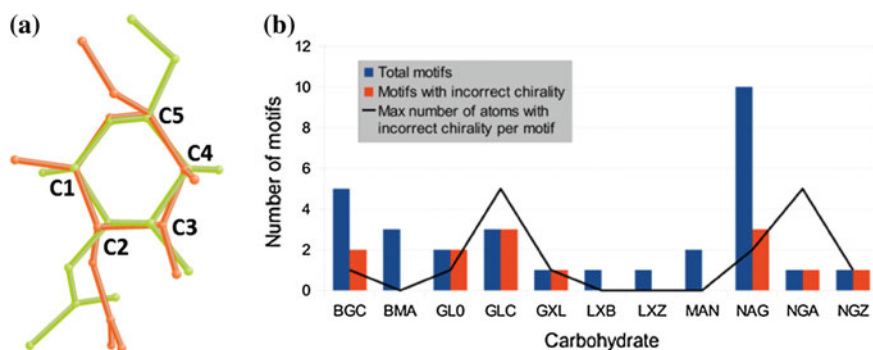


Fig. 4.1 Validation of carbohydrate ligands in Nipah G attachment glycoprotein and its receptor ephrin-B3 (PDB ID 3d12). **a** Ligand NGA 1362 A from 3d12 (in orange) superimposed with model NGA (in green). It shows that the chirality differs at all 5 chiral carbons (C1–C5), therefore the ligand has 5 chiral errors. **b** Summary information about validation of all carbohydrate ligands. 13 out of 30 ligands displayed incorrect chirality at 1–5 atoms (color figure online)

errors in all the ligands is in Fig. 4.1b. More details about the validation of 3d12 can be found in [6].

4.3 Objects of Validation

Which parts of structures should we validate? The first step in the validation of biomacromolecules and their complexes is checking the standard building blocks (residues), namely standard amino acids and nucleotides. The next key step is the validation of ligands (e.g., sugars, drugs, heme) and non-standard residues. The

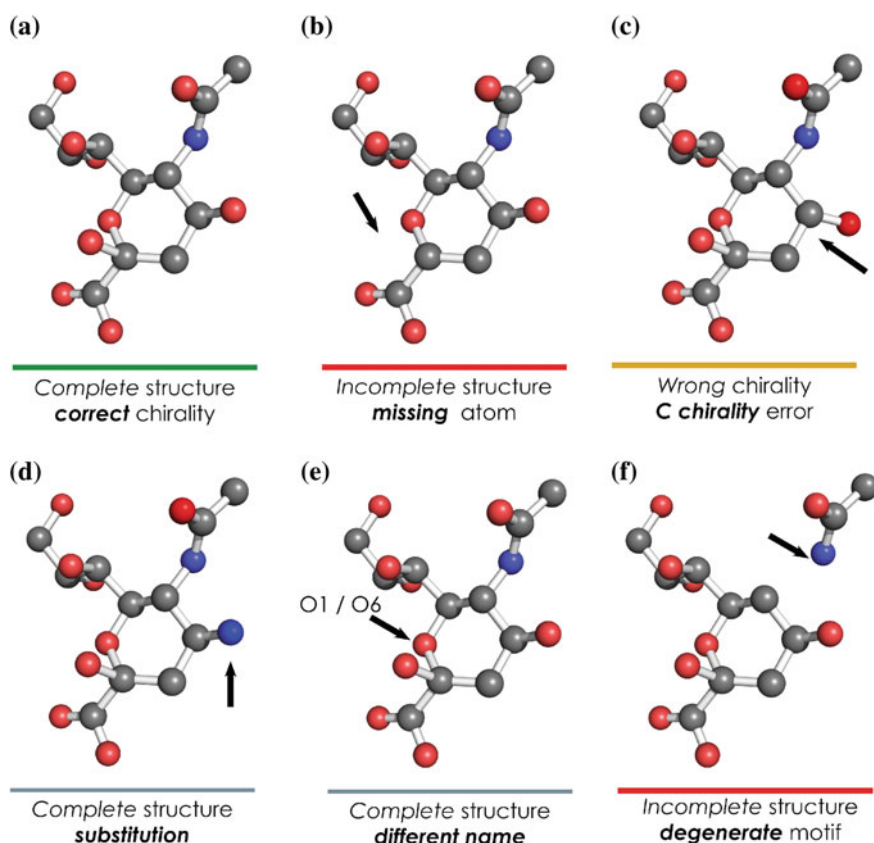


Fig. 4.2 Examples of validation issues detected via validating annotation. The color coding of the figures is the same as in MotiveValidator and ValidatorDB (i.e., a correct structure is marked *green*, a structure with missing atoms or a degenerated structure is marked *red*, a structure with substitution or a different atom name is marked *gray*) (color figure online)

ligands are the main sources of errors in structures [7], therefore their validation deserves increased attention.

What structural aspects should be validated? We have to focus on the topology (2D structure) and geometry (3D structure) of the residues. Within the topology, the main validation issues are missing atoms (Fig. 4.2b) and wrong chirality (Fig. 4.2c). When a missing atom is part of a ring, the validation issue is more serious and some software tools report these missing atoms as *missing ring atoms* or *missing rings*. Additionally, important validation issues are also atom *substitutions* (Fig. 4.2d) and *different (incorrect) atom naming* (Fig. 4.2e). In the field of geometry, the main errors are: *atom clashes* (i.e., two atoms are too close), *bond length problems*, *wrong bond angles* and *wrong torsion angles*.

4.4 Source Data for Validation

The validation process is independent of data source, can be performed on molecules obtained from X-ray, NMR, Cryo-EM (when the resolution is sufficient) and also on structures generated by structure generators such as Corina [8], Omega [9] etc.

4.5 Validation Approaches

We differentiate two validation approaches – validation against a template molecule (validation of annotation) and validation against tabular values of properties (structure validation).

The validation of annotation evaluates whether the molecule is annotated correctly (e.g., when it is named MAN, it is really α -D-mannose). Specifically, it focuses on the topology of individual residues and inspects, if its key aspects are the same as in the template molecule. Namely, if all the atoms are present or some of them are missing or replaced by another chemical element, if the chirality of atoms is identical to the template, etc. A particular implementation of validation of annotation can be done via a comparison of molecular graphs (template vs. validated molecule) or via a superimposition of molecular 3D structures (again template vs. validated molecule). In both of these approaches, we have to work with the fact that some atoms are shared by two interconnected residues. Therefore the validated molecule cannot be purely the residue itself, we must also include these shared atoms, located in neighboring residues (they are designated *foreign atoms*). When we are unable to map the validated molecule to the template (e.g., some bonds are missing), the validation of annotation cannot be performed and the validated molecule is denoted as a *degenerate motif* (Fig. 4.2f).

Structure validation evaluates whether the structure (geometry) is correct. Specifically, it compares the geometrical properties of a residue (e.g., atom distance, bond length, bond angle, torsion angle) with a correct (tabular) value of this property. If the

difference is higher than a defined tolerance, an error is reported. The implementation of this method is relatively straightforward, but this approach is very sensitive to the input tabular values. Also the interpretation is more demanding – it is nontrivial to differentiate which of the reported errors is a real problem and which is just a slide discrepancy.

4.6 Evolution of Validation Tools

Early validation software tools, e.g. WHAT_CHECK [2], PROCHECK [10], PROCHECK-NMR [11], AQUA [11], MolProbity [12] and OOPS [13], were focused on the structure validation of standard residues.

The next key step was the validation of ligands and non-standard residues, which was performed in a similar manner as the standard residues – therefore via the structure validation approach. Such a methodology is implemented in several validation software tools: ValLigURL [14], Mogul [15], Coot [16] and PHENIX [17].

Later, the validation of annotation methodology was developed, because structure validation was not able to cover errors such as missing atoms, atom substitutions and wrong chirality. The first proposal of this approach was published by Lütke et al. and implemented in pdb-care [18] – a software tool for validation of carbohydrates. A demand for a more universal ligand validation tool led to the development of MotiveValidator [6], which enables the validation of any ligand or non-standard residue. Then the ValidatorDB [4] database was published, containing ligand validation results precalculated using MotiveValidator for all the PDB ligands.

A big step forward was the release of PDB validation reports, which summarize all the validation results (both validation of structure and validation of annotation) for each PDB entry.

4.7 How to Handle Structures with Errors

Therefore what should we do, when we detect errors in a structure of interest? It depends on the location of the error and also on its type. If the error is located in part of the structure far from our location of interest (e.g., we are studying a binding site and on the opposite side of the structure is an atom clash), we can neglect it. On the other hand, if we detect missing atoms or several chirality problems directly in the location of interest, we have to take them into account. For example, remove the problematic structure from the analyzed set or at least expect these sorts of problems when drawing conclusions. An example of how to handle structures with errors is provided in the Sect. 9.1 Lectin example (Sect. 9.1.2 – Step 2: Validation of the obtained PDB entries).

4.8 Exercises

Overview: The practical exercises will be performed using two software tools – ValidatorDB and PDB validation reports. These two tools cover all of the validation analyses mentioned above (Fig. 4.1) and they provide precalculated and weekly updated results for the whole of PDB. A usage of ValidatorDB is shown in the solution of demo exercises 1, 4 and 7. A work with validation reports is introduced in the demo exercises 8 and 10. When you want to validate your structures, use MotiveValidator and Mogul.

1. Demo exercise: Detecting of problematic ligands in nipah G attachment glycoprotein

Validate all the ligands in the nipah G attachment glycoprotein (PDB ID 3d12). Detect which of them have missing atoms or wrong chirality and describe where the validation issues are (which ligand and which atom).

2. Validation of sucrose (SUC) in Plant Photosystem I – missing atoms

Validate a molecule of sucrose (SUC, residue ID 1226) in Plant Photosystem I (PDB ID 2WSC). Detect which atom is missing.

3. Validation of β -carotene (BCR) in Plant Photosystem I – missing rings

Validate a molecule of β -carotene (BCR) in Photosystem I (PDB ID 4rku). Detect which atoms within the ring of this molecule are missing.

4. Demo exercise: Validation of maltose (MAL) ligands in Protein Data Bank – missing atoms and rings

Detect all maltose (MAL) ligands in Protein Data Bank which have missing atoms and rings.

5. Detection of atom substitution in biotin (BTN) from 50S Complex (PDB ID 1kqs)

Validate biotin (BTN) from 50S Complex (PDB ID 1kqs) and detect which atom is substituted.

6. Detection of chirality problems in all sialic acids (SIA) from Protein Data Bank

Validate all sialic acids from PDB and identify the atoms of this molecule at which chirality errors occur.

7. Demo exercise: Detection of chirality problems in testosterone derivatives

Validate all samples of these testosterone derivatives: 5- β -dihydrotestosterone (BDT), 5- α -dihydrotestosterone (DHT), epi-testosterone (FFA), testosterone (TES) and testosterone hemisuccinate (TH2). Detect if there are any samples with chirality errors.

8. Demo exercise: Obtaining of structure validation data for a rhodostomin

Obtain information about structure quality for a rhodostomin (PDB ID 4rqq), specifically:

- (a) How many clashes are there in the structure?
- (b) Which atom clashes are the closest?
- (c) How many bond length outliers are in the structure?
- (d) Which bond length outlier is the highest?
- (e) How many bond angle outliers are in the structure?
- (f) Which bond angle outlier is the highest?

9. Obtaining of structure validation data for a mutant of cytochrome P450cam

Obtain information about structure quality for a mutant of cytochrome P450cam (PDB ID 2h7s), specifically:

- (a) How many clashes are in the structure?
- (b) Are there any atoms closer than 1 Å?
- (c) How many bond length outliers are there in the structure?
- (d) Are there any bond length outliers > 0.3 Å?
- (e) How many bond angle outliers are there in the structure?
- (f) Which bond angle outlier $> 20^\circ$?

10. Demo exercise: Understanding of PDB validation reports summary criteria for oxy-hemoglobine in methanol

Protein Data Bank in Europe uses summary quality criteria to summarize validation information about the structure. The summary criteria are: Clashscore, Ramachandran outliers, sidechain outliers, RSRZ outliers.

Obtain summary criteria for oxy-hemoglobine in methanol (PDB ID 11fz) and then learn more about their meaning.

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Part II

Detection and Extraction

Chapter 5

Detection and Extraction of Fragments

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The detection and extraction of biomacromolecular patterns can take a number of different forms, as was explained in Chap. 2. There are approaches which predict the existence of patterns with a certain biological function based on a variety of search criteria (geometry, physicochemical properties, sequence similarity, etc.). On the other hand there are tools capable of their *direct* detection i.e. ‘I know what I am looking for’. Internally, they usually implement a means of describing molecular patterns and there are a lot of them. Some directly encode the pattern of interest into a structure which is compared with other molecules stored in the database; other methods even contain different variations of query languages. Even popular molecular visualization tools such as PyMOL [1] or VMD [2] contain an internal language for pattern selection. However, their applicability is often limited to a narrow domain. Nevertheless, there is a paradigm for the easy description and extraction of complex biomacromolecular patterns of user interest from protein databases – PatternQuery [3].

5.1 PatternQuery

PatternQuery (PQ) is implemented in a simple web-based application designed for the detection and fast extraction of molecular patterns (<http://ncbr.muni.cz/PatternQuery>). The application uses a simple, yet intuitive query language based on the Python programming language syntax, which will be discussed in the following section in detail. Any pattern of interest can be described using query expressions based on the nature and relationship between atoms, residues and other structural elements. These expressions define the molecules composition, topology, connectivity and their molecular substructure.

This is particularly useful for many applications which often require using a combination of dedicated tools or writing single-purpose in-house programs for manipulating protein structures. On top of that, the web service allows both

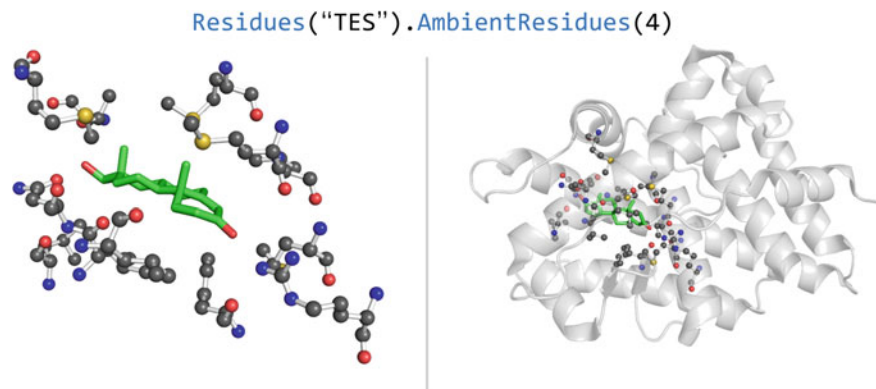


Fig. 5.1 Example of PatternQuery query identifying a testosterone ligand together with its binding cavity in the human androgen receptor (PDB ID 3zqt). Specifically, it detects a testosterone ligand (TES) according to the PDB residue annotation and any residue which is at most 4 Å away from it

querying single protein structures and the whole of the current Protein Data Bank. The library of available query expressions is quite rich (https://webchem.ncbr.muni.cz/Wiki/PatternQuery:Language_Reference) and is continuously being extended. An example query together with the result can be found in Fig. 5.1. For detailed instructions on how to use the service, please consult one of the many tutorials, which are available online at the software documentation page.

Besides querying smaller sets of input biomacromolecules in Explorer mode, PatternQuery offers database-wide search for querying the whole of Protein Data Bank. This search can be further restricted using a variety of criteria, such as PDB ID, resolution, organism of origin and more. Furthermore, all the results can be subjected to validation of annotation using ValidatorDB [4].

5.1.1 PatternQuery Explained

Nevertheless, the question remains. How do we construct queries? Is it difficult? Defining queries is fairly simple and the text below will guide you from constructing initial simple queries to the more robust expressions. There are three types of queries called *Generator* queries, *Modifier* queries and *Combinatory* queries. Generator queries, as the name suggests, generate patterns from the original input. They are the tool that transforms the input molecule into patterns that can be later modified or combined. Examples of these queries include `Atoms()`, `Residues()`, or `RegularMotifs()`.

Modifier queries operate on individual patterns and modify them or throw them away. Examples include `AmbientAtoms()`, `ConnectedResidues()`, and `Filter()`.

Combinatory queries take as input two or more sequence of patterns and combine them into a single new sequence that satisfies given criteria. Examples include `Or()`, `Near()`, and `Path()`.

5.1.2 Thinking in PatternQuery

Let us take a closer look at the following example. Although being somewhat artificial, it nicely covers all the steps for pattern identification using PQ. Let's assume we have loaded a protein stored in a PDB with correctly annotated HET groups and we want to identify all the HET residues in it for further processing.

The characteristics of an imperative approach is explicitly stating the steps that need to be performed in order to achieve a particular goal. In contrast the declarative approach used in PatternQuery states the goal we would like to achieve, leaving the individual steps as an *implementation detail*.

Using the imperative approach, we would do something like this:

```
1 result = List()
2 for residue in molecule.Residues:
3     if residue.IsHet():
4         result.Add(residue)
```

In the PQ, our code would look like this:

```
1 HetResidues()
```

Now, let us extend our example to *all HET residues and atoms within 4 Å around them* to include the binding partners of these HET residues.

In the imperative approach we would need to do something along the following lines:

```
1 temp = List()
2
3 for residue in molecule.Residues():
4     if residue.IsHet():
5         temp.Add(residue)
6
7 neighborhoodLookup = NeighborhoodLookup(molecule.Atoms())
8 result = List()
9
10 for residue in temp:
11     surroundings = neighborhoodLookup.Find(residue.Atoms, <= 4.0)
12     result.Add(union(residue, surroundings))
13
14 return result
```

Declaratively, the PatternQuery code is just:

```
1 HetResidues().AmbientAtoms(4.0)
```

Internally, the function `AmbientAtoms()` might run code similar to the imperative version. However, what is important is that this complexity is hidden from the user when the declarative approach is used.

Now, let us extend our example even further: *all HET residues and atoms within 4 Å around them*, where the entire structure contains at least one calcium atom.

We will not bother the reader with writing down the imperative version – implementing the condition *at least one calcium atom* is rather boring. However, using the declarative approach, the description of the pattern becomes simply:

```
1 HetResidues()  
2 .AmbientAtoms(4.0)  
3 .Filter(lambda m: m.Count(Atoms('Ca')) > 0)
```

Lambda is a concept of functional programming used for creating anonymous functions. We will not go into unnecessary details, instead let us explain its function in following example:

```
Residues().Filter(lambda m: m.Count(Atoms()) > 6)
```

The query `Residues()` returns a sequence of patterns. Each pattern in the filtering function is represented as 'm', afterwards it is a subject to a count function. Here all the atoms in the given pattern are counted. In case there is at least 7 of them, the function is evaluated as true and, therefore, the pattern is kept for further processing, in the other case it is discarded.

5.1.3 Basic Principles of the Language

5.1.3.1 Intuitive Description

As we've seen in the example above, it is very easy to compose our ideas about the final shape of the pattern we are interested in. The way this works is that the input molecule is decomposed into a sequence of patterns. These sequences can then be modified and combined into new sequences, which can be modified and combined again.

As an example, take the query `Atoms('Ca')`. What the PatternQuery language does is to extract all calcium atoms from the input molecule and represent them as a sequence of sets containing one atom each, as illustrated in Fig. 5.2.

Now, each element of this sequence can be modified, for example to include all atoms within 4 Å of the original calcium atom. Now we have a sequence of sets of atoms, where each set contains the original Ca atom and the atoms within the given radius. This would be represented by the query `Atoms('Ca').AmbientAtoms(4)` and is illustrated in Fig. 5.3.

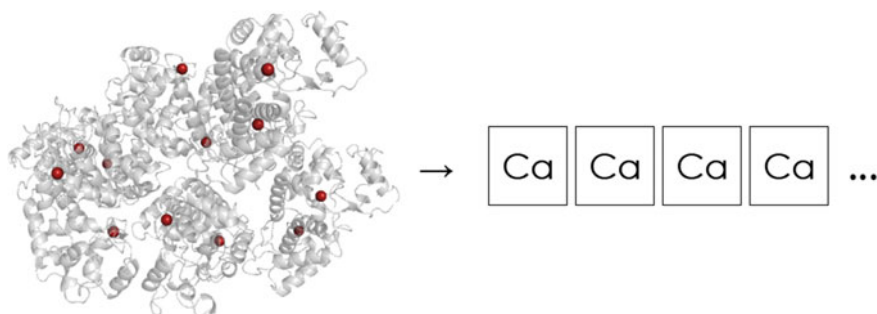


Fig. 5.2 First step in a pattern identification process. In this example only calcium atoms are identified

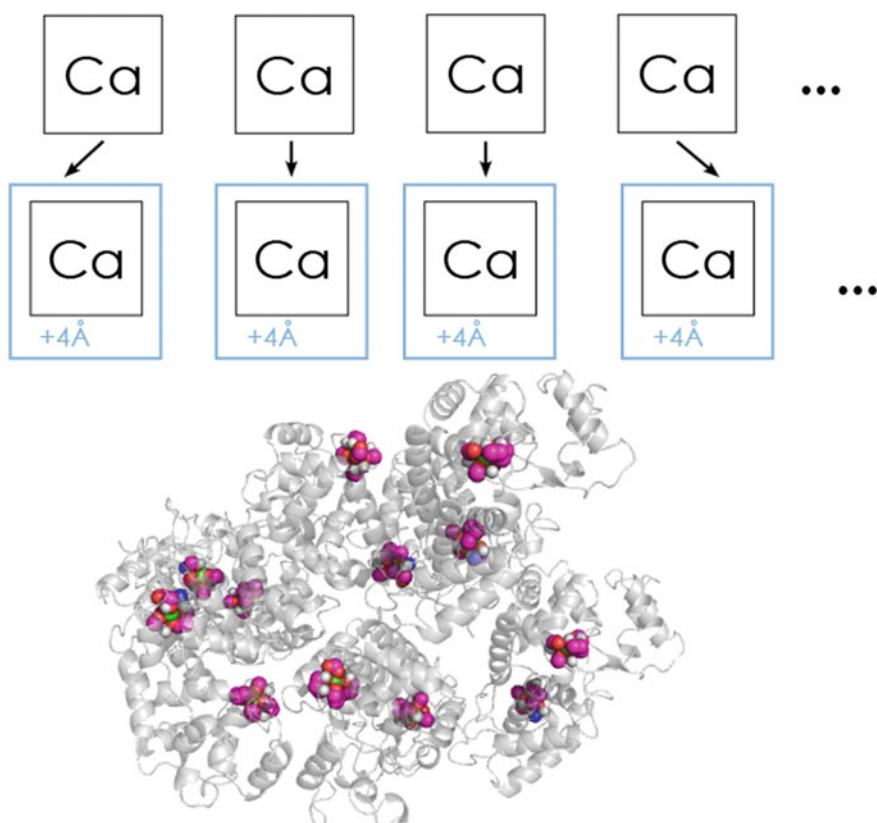


Fig. 5.3 Second step in the example. Patterns are extended to the 4Å surrounding the calcium atoms

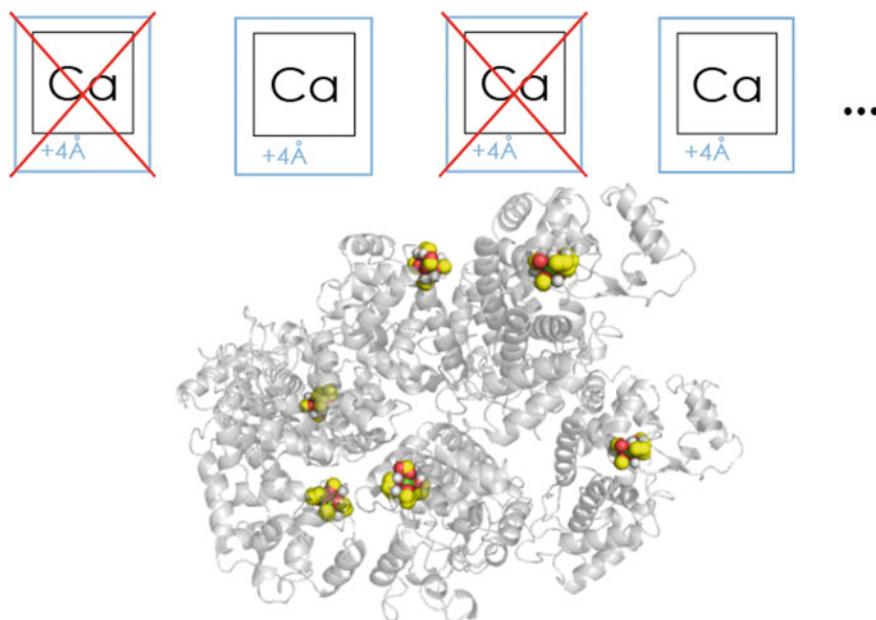


Fig. 5.4 Third step in the example. Patterns not satisfying the filtering criteria are filtered out

In the next step, we might wish to keep only these patterns that contain at least 6 atoms. This is achieved by looking at each pattern, counting the number of atoms and discarding those patterns that do not meet the criteria. Written as a query, this could be represented as:

```

1 Atoms('Ca')
2 .AmbientAtoms(4)
3 .Filter(lambda m: m.Count(Atoms()) >= 6)

```

This is shown in graphical form in Fig. 5.4.

The previous filter query also demonstrates another interesting concept of the language: the ability to identify patterns within patterns, which is what the expression `m.Count(Atoms())` does – the `Atoms()` query is executed for each pattern from the original input sequence provided by the expression `Atoms('Ca').AmbientAtoms(4)`, and creates a new sequence of patterns that each contain a single atom. Then the `Count` function takes over and returns the number of patterns produced by its argument. In this way, the query `Atoms()` inside the `Count` function can be replaced by any function that also produces a sequence of patterns, for example `Rings()`.

Finally, streams of patterns can be combined. For example, let's say we want to find all pairs of calcium atoms that are no further than 4 Å apart. This can be achieved using the query `Near(4, Atoms('Ca'), Atoms('Ca'))`. So this query takes as its input two identical streams of calcium atoms and for each pair of

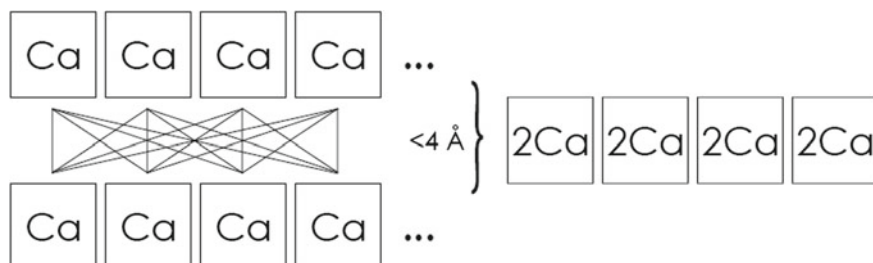


Fig. 5.5 Schematic illustration of the `Near()` query. The pattern is identified when 2 calcium ions are at most 4 Å apart

them determines if the atoms are closer than 4 Å to each other. For each pair that satisfies this condition, a new pattern from the 2 atoms is created. Therefore, the result of the above `Near()` query is a sequence of sets of atoms (patterns) that each contain two calcium atoms that are no further than 4 Å from each other (Fig. 5.5).

With these basic types of queries outlined in the previous paragraphs, the sky is the limit. Due to the composable nature of the language, if a new type of pattern emerges, only a single function needs to be added to the language for it to work with all of its other parts. As an example, assume we did not know that proteins had a secondary structure called a *sheet* and we just discovered it and a fancy algorithm to identify these *sheets*. Now we would be interested in how this new type of protein substructure interacts with other parts of the molecule. All that would be needed is to add a function called `Sheets()` to the language and immediately we would be able to analyze and filter its neighborhood using the functions `AmbientAtoms()` and `Filter()`.

5.1.3.2 Example, Revised

Now that we know the basic building blocks of the language, let's go back to our original example and analyze it:

```

1 HetResidues()
2   .AmbientAtoms(3.0)
3   .Filter(lambda m: m.Count(Atoms('Ca')) > 0)
```

This corresponds to the following process:

A generator query `HetResidues()` is executed that produces a sequence of patterns that are composed of atoms corresponding to HET residues.

Next, the original sequence is modified by adding atoms within 3 Å of any of the original atoms to each pattern.

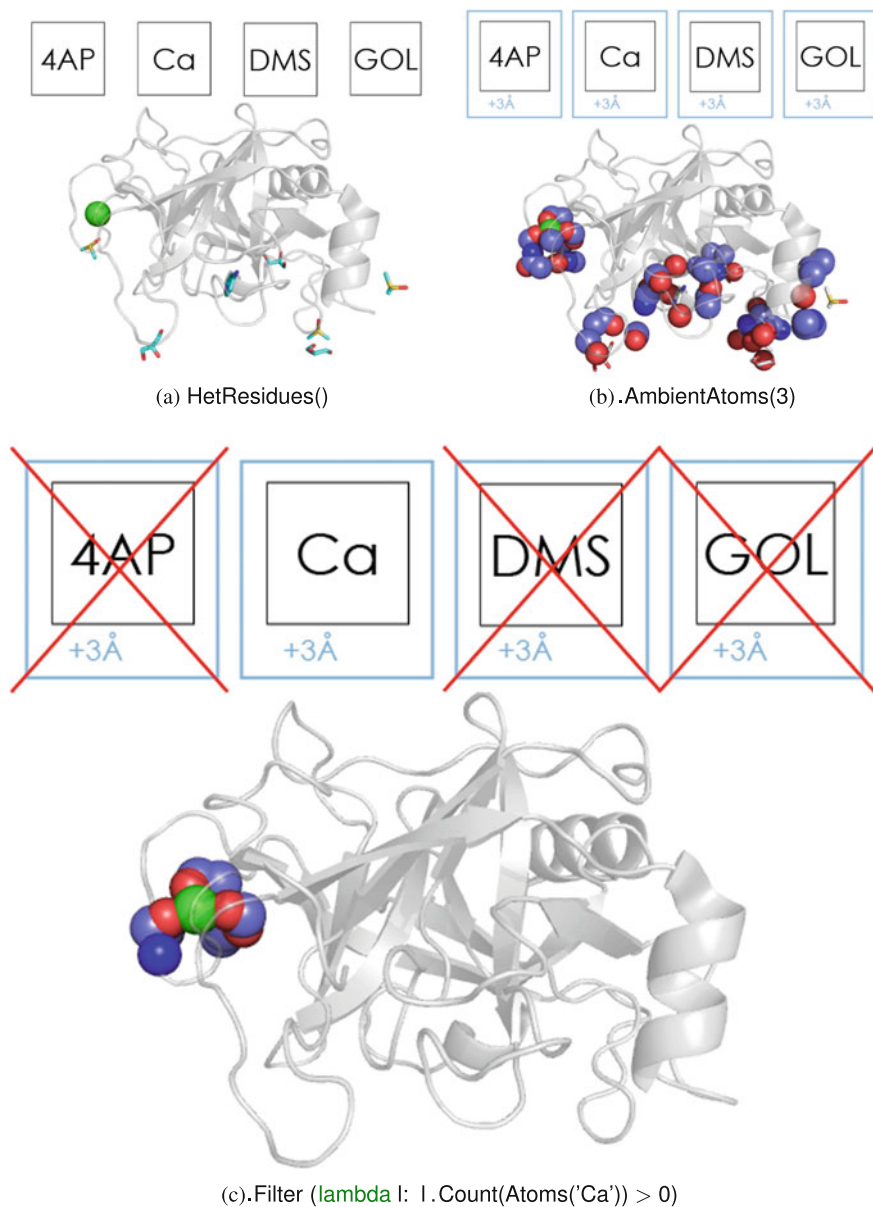


Fig. 5.6 Scheme of a pattern identification process. Ligands together with atoms at most 3 Å away are identified, where at least one calcium ion is present in the pattern

Finally, each pattern in the modified sequence is examined: all *calcium atom patterns* are identified and counted. Only those patterns that contain at least 1 Ca atom are kept.

This is shown graphically in Fig. 5.6c.

5.2 MetaPocket 2.0

MetaPocket [5] is a web-based consensus classifier for predicting putative ligand binding-sites in apo and holoproteins. In order to provide the best results, MetaPocket combines a variety of already established tools and methods in order to predict ligand-binding sites that are as relevant as possible. LIGSITE^{CSC} [6], PASS [7], Q-SiteFinder [8], SURFNET [9], GHECOM [10], Fpocket [11], ConCavity [12] and POCASA [13] are considered in the current method. The whole workflow combines energy-based, geometry-based or rolling-probe based methods and as such represents convenient combination of all popular approaches.

First, the structure under study is processed by all of the aforementioned methods. Next, a z-score [14] is calculated for each predicted binding site and just the 3 top scoring are kept for further consideration per tool. As a majority of the binding sites will inevitably overlap, a clustering algorithm is applied in order to identify such overlaps. The resulting clusters are then subjected to another z-score re-ranking in order to make the final decision. Finally, ligand binding residues per pocket are identified in the top scoring pockets and returned as the output of the analysis.

The web service is really simple to use, as only the PDB ID, or uploaded structure with the optional selection of chains is needed in order to successfully run the service.

It can be particularly useful for defining possible binding pockets of interest in proteins lacking ligands in their structure. Nevertheless, when the ligands are present in the structure, PatternQuery is a more straightforward approach for identifying the residues responsible for binding. The drawback of this approach could be the fact that the database-wide screening is rather computationally demanding and since some of the tools still heavily rely on the obsolete PDB file format, their applicability for large structures available only in the mmCIF file format might be limited.

5.2.1 Serotonin Receptor Example

Neurotransmission is a vital intercellular signaling process enabling neural signals to be distributed among other neurons. One of the ways to achieve neural signal promotion is the passage of ions and other messengers thorough the cell membrane via ion channels. However, trafficking ion channels is often thoroughly controlled by the binding of small ligands to the receptor domain, which in turn enforces a conformation change. There is a large body of such ion channels categorized into different families, which enable the selective permeation of messengers.

One example is the cation-selective serotonin receptor (5-HT₃ receptor), a member of the Cys-loop family. This receptor is responsible for the modulation of other neurotransmitters as well as hormones. As such it directly influences a lot of behavioral processes. Indeed, it has been shown that this receptor is also involved in various disorders (e.g. psychotic disorders, cognitive disturbance, pain disorders, or drug addiction) [15]. As such it represents an interesting target for alleviating of chemotherapy-induced and post-operative syndromes.

5.2.1.1 Detection of Serotonin Binding Site

We will try to identify a serotonin binding site in the 5-HT₃ receptor using the MetaPocket service. For this purpose we will extract the structure of this protein from the Protein Data Bank (PDB ID 4pir). Next, this receptor is a homo-pentamer containing over 20,000 atoms. Feeding the service with such a large structure might not return a result. Therefore, we have to restrict our search. It has been known, that the binding site is located at the interface of two subunits. Thus we can restrict our search to chains A and B.

After the search is completed we can retrieve the results in a number of different forms. First, we can see the overview of the analysis stating which methods were capable of predicting pockets and, on the other hand, which failed to process the structure. Finally the results are clustered, and based on the z-score, the best n deemed pockets is retrieved. The results can be downloaded for further processing in the PDB format or in three other formats.

When evaluating the results, we can start with the elimination of the top scoring pocket, as it is located in the transmembrane region and therefore unlikely to be accessible for the serotonin. Next, we can proceed with the evaluation of the second-best scoring pocket. The residues composing the pocket are highly conserved among the family of these receptors. Moreover, the binding site is close to the local narrowing of the channel, suggesting that the conformational change enforced by the binding can change the open/close conformation of the channel (Fig. 5.7). Indeed, the second pocket has been identified as a putative serotonin binding site [16].

5.3 Note on Pattern Comparison

In order to further polish the dataset created using pattern detection/prediction services, structural patterns can be compared from different perspectives. On the one hand, we can evaluate their atomic, or residual composition, on the other hand, we can infer their physicochemical properties etc. Additionally, we can directly compare their conformations and determine the root mean square deviation (RMSD). All of these metrics can help us to distinguish which patterns should be discarded before proceeding in our analyses, and which should be kept.

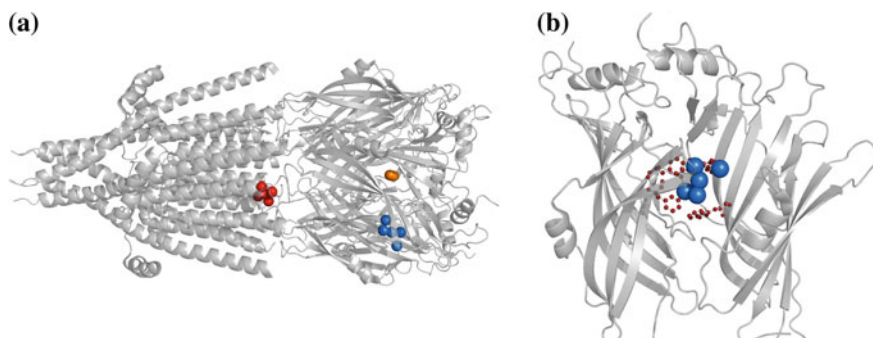


Fig. 5.7 Putative 5-HT₃ serotonin receptor binding sites identified by MetaPocket. **a** Overall view of all the center of mass binding sites predicted by different services. Ordered by MetaPocket according to their z-score red-blue-orange. **b** Closer look at the serotonin binding site, which is conserved throughout the family with the highlighted residues Trp 156, Trp 63, Tyr 126, Tyr 207 and Phe 199 [16].

RMSD is a metric describing the structural difference between two molecules (patterns) in Ångströms, i.e. how well would two or more structures fit on top of each other. First, the two molecules are superimposed on top of each other and then the atomic distances between the corresponding atoms are measured. The higher the RMSD is, the more divergent the structures are. Two molecules with identical conformation (same atomic positions) have an RMSD equal to 0. For protein differences, usually only C_α carbons are taken into consideration for inferring RMSD, whereas for smaller patterns or fragments all the atoms in question are taken into consideration.

Again a number of tools have been implemented in the past for the comparison of two structures. These are even integrated with popular visualization packages such as PyMOL (commands: align, cealign super, pair_fit) for protein comparison. However, an all-atom comparison of smaller patterns is a somewhat difficult problem to tackle, and therefore the number of available tools is restricted. An example is the web-service SiteBinder [17] (<http://ncbr.muni.cz/SiteBinder>), which is capable of a rapid pairwise or multiple-pattern comparison.

SiteBinder is an online application designed for the superimposition and comparison of large sets of structural patterns. Moreover, it provides a clean, user-friendly and very intuitive graphical user interface, which allows the user to visualize the motifs, select specific atoms or residues for superimposition, export the coordinates of the superimposed structures, as well as the RMSD values, etc.

5.4 Exercises

5.4.1 PatternQuery

Identify the following structural patterns in the specified entries, or in the entire Protein Data Bank. Try to pre-filter the Protein Data Bank, in order to restrict the

search space and speed up the overall calculation. For instance, if you are looking for a pattern containing platinum ions, only PDB structures with the Pt ion in the structure are worth querying for. Alternatively, you can use a predefined dataset of certain PDB IDs https://webchem.ncbr.muni.cz/PQ_set.txt, in order to get the results interactively. For smaller datasets (dozens of PDB entries) it is advised to use the interactive version of PatternQuery – PatternQuery Explorer.

1. Glycine residues in tubulin protein (PDB ID 1tub)

How many glycine residues are present in the tubulin protein (PDB ID 1tub)?
How many of them are not part of any type of secondary structure?

2. All experimental binding sites of the doxycycline antibiotic

Doxycycline (doxycycline, PDB residue ID DXT) is an antibiotic used for the treatment of a variety of bacterial infections. Search Protein Data Bank for all instances of the doxycycline ligand and extract the ligand together with its binding pocket. How can you speed up the process of identification?

3. Rubredoxin active site

Rubredoxins are a class of low-molecular-weight iron-containing proteins found in sulfur-metabolizing bacteria and archaea. The active site is composed of an iron ion coordinated by 4 cysteine residues. Find all patterns in the PDB that satisfy this description. How strong is the structural conservation of these patterns? Use SiteBinder (<http://ncbr.muni.cz/SiteBinder>) for this task and evaluate the RMSD metric.

4. Query constructions

Identify the following patterns:

- Residues within 2.5 Å of the iron ion located in the HET residue
- All HET residues not covalently bound to the protein structure
- Modified residues with the phosphorylation modification
- All the platinum-containing residues together with the residues covalently bounded to them.

5. HET residues in the vicinity of modified residues

What is the most abundant HET residue found within 5 Å of any of the modified residues? How many of them contain any structural discrepancies?

6. Manganese ion in a single residue

List the residues which contain a manganese ion in their structure at most 2 Å from an oxygen atom in the same residue.

7. RNA k-turn motif and its AAs surrounding

The kink-turn (k-turn) motif is commonly found in RNA structures. It performs an important architectural function, as it forms a binding site for a number of proteins. Identify k-turn motifs (specifically search for the Kt-7 subtype) in the PDB together with their 3 Å molecular environment.

8. Classical C₂H₂ zinc finger binding motif

A brief overview of the zinc finger transcription factors is given in Chap. 2 or in the literature [18]. Identify C₂H₂ zinc finger motifs in the entire Protein Data Bank.

9. C₂H₂ zinc finger binding motif with the zinc substitution

Identify a zinc finger binding motif where a zinc ion has been replaced with a different element. Do not specify the middle part of the primary sequence explicitly. Instead, use just a general regular expression `{12}`.

10. *Pseudomonas aeruginosa* Lec-B sugar binding site

Pseudomonas aeruginosa is an opportunistic pathogen, which mimics host cells using lectins sugar-binding membrane proteins, thus preventing itself from being discovered by an immune system.

The binding site is composed of two calcium ions approximately 4 Å away from each other, which binds a carbohydrate residue. What is the most common composition of the interacting partners?

5.4.2 MetaPocket

1. HIV-protease

Identify three ligand binding pockets for HIV-1 protease (PDB ID 2wkz) using MetaPocket server. Which one of them is the most promising? What are the other possible binding pockets?

2. COX-2 protein binding site

Try to use MetaPocket to identify a putative binding site in the COX-2 protein (PDB ID 3nt1), highlight the residues responsible for binding.

3. Apicoplast DNA polymerase

Identify the most prominent binding site in the exonuclease domain of the apicoplast DNA polymerase from *Plasmodium falciparum* (PDB ID 5dkt).

4. Detection of caffeine binding site

Identify the caffeine binding pocket in the adenosine A_{2A} receptor (PDB ID 3rfm) using the MetaPocket service and PatternQuery. Discuss the differences.

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Chapter 6

Detection of Channels

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6.1 Introduction and Motivation

The function of proteins usually depends on their interaction with other biomacromolecules as well as ions and small molecules, such as water, messenger and endogenous compounds, pollutants and drugs, which can occupy otherwise empty spaces in protein structures [1]. The empty volumes inside protein structures can be categorized into several independent groups.

While *Cavities* correspond to the void regions deep inside the protein structure, *Pockets* are shallow clefts and protrusions in the protein surface. Both of these empty spaces provide an increased surface area and, hence, increased opportunity for the protein to form interactions with other molecules, particularly small ligands [2]. Additionally, the selectivity of the enzymatic reaction can be further influenced by certain residues, which limit access to the active site. This is why an active site is usually either located in a shallow pocket or deeply buried in the protein structure. *Channels*, as access pathways to cavities, have been a particular focus of interest in recent years, although there is no unified nomenclature for them. Indeed, terms like channels, tunnels or pores are often used interchangeably. In literature, channels are commonly referred to as all ligand-accessible pathways in the protein body [3]. Some authors further specify channels to be tunnels leading to the buried active or binding sites inside the protein body, while pores represent a pathway passing through the whole structure and are usually found in transmembrane proteins. This is the nomenclature which is followed in this text. An overview of protein empty space is shown in Fig. 6.1.

In fact, channels play many roles in biologically relevant systems. In enzymes, they allow the passage of substrates and products to/from the active site. Specifically, they are present in a large fraction of enzymes [4]. In addition, photosystem II tunnels are involved in photosynthesis, ribosomal polypeptide exit tunnels enable nascent peptides to leave the ribosomes during translation, and pores in transmembrane channel proteins allow the permeation of various compounds through the cell membrane. Moreover the malfunction of ion channels, mostly due to genetic defects, has been

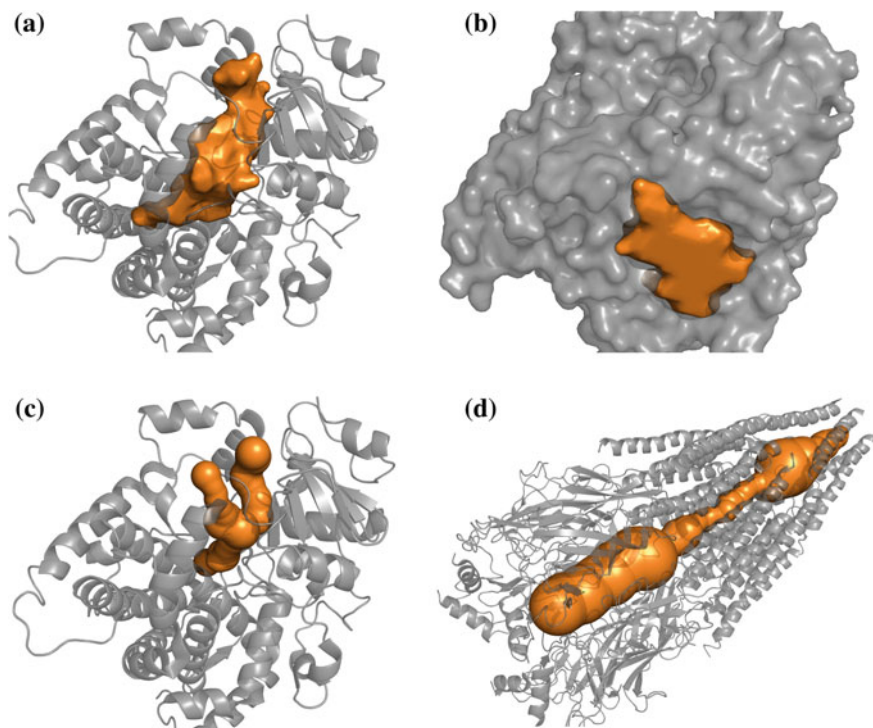


Fig. 6.1 Examples of empty volumes in protein structure: **a** cavity, **b** pocket, **c** tunnels, **d** pore

linked with a number of diseases in the past. Usually a neuromuscular system is affected, causing diseases such as epilepsy, ataxia, myotonia or cardiac arrhythmia [5]. As a result, the analysis of a channel's structure and its properties can significantly contribute to a better understanding of a number of vital processes.

The channels can be viewed as part of the empty space in the protein structure, however, there is another point of view. The empty space is fully encapsulated by the protein body – amino acid residues. These create a specific physicochemical environment. As such, the channel lining residues are a co-determining factor for the channel's width and flexibility. Some residues of the channel lining are of even greater importance. These are typically located in local or global narrowings (bottlenecks), where they can contribute to the specificity or selectivity of enzymatic reaction, prevent a solvent from accessing the active site or synchronize processes in different parts of biomacromolecules.

Indeed, there are studies e.g. [6] suggesting that the modification of certain residues in the channel's walls can greatly influence the function of the protein. As a consequence, this knowledge is being extensively used in biotechnologies.

Nevertheless, the question remains: how can we locate channels in the protein structures and how can their analysis contribute to understanding key biological

questions, which we can in turn use, for example, in drug development? Over time a number of different approaches have been postulated. Four main streams have been established since.

Grid-based methods project the molecular structure over a finite 3D grid of vertices with a given resolution, which is usually 1 Å. A custom scoring function is used for evaluating each of the vertices. Clusters of vertices satisfying the given criteria are then reported as channels. The disadvantage of this method is its computational complexity, which scales up cubically with the number of vertices in the grid. While this approach has been implemented in a variety of tools (e.g. Caver 1.0 [7], dxTuber [8], Hollow [9], it is currently no longer used for channel analysis. The main idea of sphere-filling methods lies in an iterative carpeting of the protein structure with layers of dummy atoms with a given radius. In the subsequent step, the protein structure is carpeted with dummy atoms of much larger radius. If these two intersect, the smaller is removed. As a result, the clustering of the smaller spheres represents the protein's empty volumes. All in all, these methods are generally used in pocket detection, and thus are not used much for channel analysis. This approach has been implemented for example in Surfnet [10] or PASS [11]. The third class involves slice and optimization methods. To begin with, these methods split the biomacromolecular structure into slices along a start vector defined by the user. Optimization methods are then used to determine the largest sphere in each slice along that vector. These approaches are implemented in the software tools HOLE [12] and PoreWalker [13]. Last but not least, the class of alpha-shape theory methods involves the utilization of Voronoi diagrams/Delaunay triangulation or their generalization, power diagrams. This approach is used by MOLE 1.0 [14], MOLE 2.0 [15], CAVER 3.0 [16], MolAxis [17] and the relatively new tools BetaVoid [18] and CHEXVIS [19]. Briefly, MolAxis and Caver cover the surface of all atoms with a certain amount of vertices in order to optimally approximate the channel centerline. In contrast CHEXVIS implements power diagrams. After the calculation of the molecular graph, paths from a defined starting point, usually an active site, are detected and given software-specific edge weights using Dijkstra's shortest path algorithm [20]. Further, each of these algorithms employs its own way of treating redundant channels.

To sum up, there are a number of tools available for channel analysis. However, they are mostly command-line based or provide a stand-alone graphical user interface (MOLE, CAVER). The only available online web-services are MOLEonline 2.0 [21], MolAxis and CHEXVIS.

In order to assist in answering key biological questions, we have to carry out a subsequent analysis of channels. The output of most of these tools is a channel centerline, together with the residues encapsulating the channel pathway. By analyzing the overall residual composition of the channel lining, its key regions and their properties, we can infer interesting conclusions in terms of a putative channel function. Below you can find examples of two systems, where channel analysis can contribute to a better understanding of vital processes.

6.1.1 *Bunyavirus Polymerase Example*

Negative-strand RNA viruses are serious human pathogens, causing diseases such as Crimean-Congo hemorrhagic fever, Lassa fever, or influenza. While influenza's symptoms are often mild, Crimean-Congo hemorrhagic fever is associated with a high level of mortality and morbidity. A single-chain L protein is the RNA-dependent RNA polymerase responsible for the virus' replication. Knowledge of its structure and the mechanism of replication can shed light onto the virus' life cycle and can be in turn used for the development of an antiviral treatment. This can be achieved for example by designing inhibitors selectively blocking a polypeptide exit tunnel [22]. *Bunyavirus* polymerase binds complementary 3' and 5' genomic RNA ends in distinct sites. The binding of the 5' influences the conformation of the polymerase active site, where the nucleotide addition occurs. Channels analysis is one way to approach the rationalization of the polymerase function. The research conducted [23] reveals the polymerase structure and suggests an elegant strategy for RNA synthesis with the channel system visualized in Fig. 6.2.

Fig. 6.2 Channel system of the *Bunyavirus* RNA polymerase. Nucleotide entry channel in *red*; RNA-template entry and exit channels in *blue* and *orange* respectively; and finally, product exit channel in *green*. All the tunnels are conserved among *Bunyavirus* polymerases and are similar for influenza polymerase as well (color figure online)



6.1.2 Aquaporin Example

Aquaporins (AQPs) are a large family of proteins permitting the permeation of various molecules. Classical aquaporin mainly allows for water permeation, while aquaglyceroporin transports glycerol, urea and small neutral molecules. These processes are crucial for pertaining homeostasis and their detailed understanding can be crucial for the development of novel therapeutics.

Careful structure analysis of aquaporin (AQP2) reveals a homotetrametric structure, however the functional unit is monomeric, with a single aqueous channel passing down through the structure. The channel represents a tight fit for the water, with its narrowest diameter close to the diameter of water, which is 2.8 Å acting as one of the selectivity filters. Further analysis of the channel lining sheds light on the additional filtering mechanism. Specifically, how can water permeate through the structure, while protons don't? One of the answers lies in the fixed positive charge location of arginine and histidine residues, which repels protons and prevents them from passing along the channel. Another explanation lies in the roughly 20 Å long trim, which allows for water to pass as single molecules forming hydrogen bonds with the channel lining rather than with each other. The visualization of the aquaporin (AQP2) channel together with the residues important for channel selectivity are highlighted in Fig. 6.3.

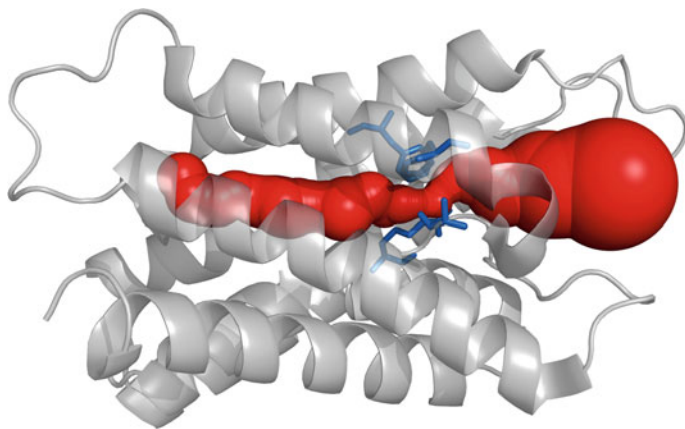


Fig. 6.3 One of the subunits of *Homo sapiens* aquaporin 2 (PDB ID 4nef) together with the channel enabling passage of water molecules throughout the plasmatic membrane. Arginine and histidine residues are highlighted in *blue*. This constriction region (ar/R - aromatic/arginine) plays an important role, due to its function as a selectivity filter of aquaporin 2, preventing the access of protonated water and other molecules (color figure online)

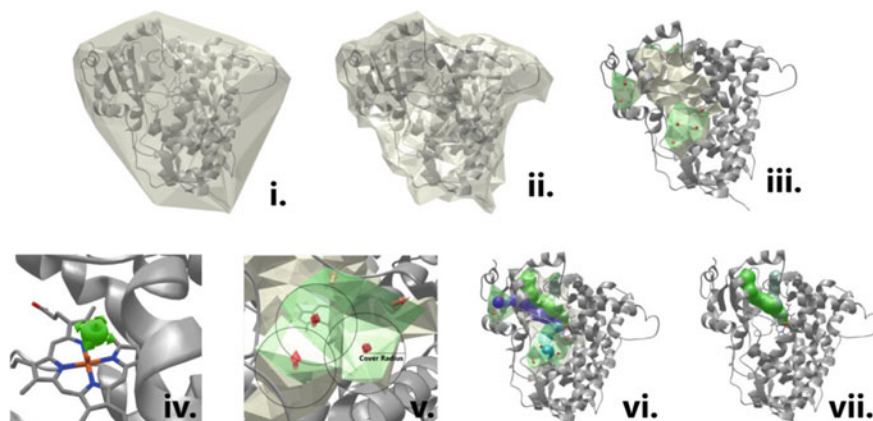


Fig. 6.4 Scheme showing the steps involved in the channel calculation algorithm. *i* Delaunay triangulation/Voronoi diagram calculation; *ii, iii* approximation of molecular surface; *iv, v* channel start and end points identification; *vi* channel computation; *vii* filtering of channels

6.2 MOLE - Channel Analysis Tool

MOLE 2.0 and MOLEonline 2.0 is a channel analysis software tool, which is capable of evaluating the physicochemical properties of the identified channels. First, the MOLE algorithm builds the Voronoi diagram from the atomic centers neglecting their van der Waals radii. It should be noted that this approach can result in the underestimation of channel radius over a short distance. In special cases, this error can be up to 2.0 Å [17], however, due the nature of this error, the problem only arises at a tiny portion of the channel and, therefore, can be handled in post-processing. The whole algorithm is described in detail in [15] and graphically illustrated in Fig. 6.4.

MOLE channels are reported as the union of spheres of a given radius (similar to the radius of expected ligand), together with the residues forming the channel walls, referred to as lining residues. It should be noted that the representation of the identified channels as a union of inscribed spheres is a simplification, as the real channel structure may be more complex, involving some protrusions etc. MOLE 2.0 was the first software tool providing some information about the physicochemical properties of channels (see Chap. 8).

6.3 Identification of Channels Using MOLEonline

6.3.1 Setup

Before one can start analyzing channels, it is vital to provide a protein structure. This can be either a user defined structure in the PDB format, or a PDB ID. By default

Fig. 6.5 Introductory setup of the MOLEonline service

Quick start

Enter PDB ID code: ⓘ

and optional Unit: ⓘ

and optional chain(s): ⓘ

or upload your own file: Browse... ⓘ

MOLE works with asymmetric units, if one wants to work with a biological unit instead, its ID is required (e.g. 1, 2, ...). When a search needs to be restricted to only certain chains, these can be optionally entered in as well (Fig. 6.5).

6.3.2 Geometry Properties

After the initial setup, several parameters need to be provided. Those of particular interest are *ProbeRadius*, *InteriorThreshold*, *OriginRadius*, *BottleneckRadius*, *MaxTunnelSimilarity*.

ProbeRadius is used for approximating the molecular surface. Increasing this value causes parts of the metric space further from the protein body to be also searched for channels. You can imagine this value to be the upper threshold for channel radius. Best practice is to use the value 3 Å for small and tightly packed proteins. For less packed proteins, values of 5–6 Å are recommended. When a protein contains large empty voids, values of around 10 Å or more can be suitable. This can be particularly useful when locating wide transmembrane pores.

While *ProbeRadius* represents the upper bound of a channel radius, *InteriorThreshold* represents a lower bound. As this value increases, channels possessing local narrowing around the value of *InteriorThreshold* or even smaller are unlikely to be detected. Both of these parameters are used to speed up the whole process and for customization according to one's needs. Usually 1.1 Å is considered to be well suited for the majority of calculations.

Additionally, channel starting point can be defined either as a residue in the format *chain name ID* (e.g. A HEM 508), or a group of residues separated by a comma. In this case a center of mass will be used for the particular start of the channel search. Optionally a point in a Cartesian coordinate 3D system can be provided as well. Nevertheless, sometimes the actual spot can be placed inappropriately, making channel detection impossible. This is why the *OriginRadius* parameter is introduced in order to look for a more appropriate channel starting points within a sphere of a given radius from the user defined starting spot. In most cases the default value of 5 Å is enough.

Next, we have to identify which parts of the input structure we consider to be integral. Besides amino acid residues or nucleotides, PDB entries generally contain a

Fig. 6.6 MOLE online form for providing parameters for the calculation

Settings

click to expand / contract

Structure:

1tqn ([structure summary on pdb.org](#))

☐ Automatic starting points

Starting point:

A GLU 308, A THR 309

Active sites from [CSA](#):

A GLU 308, A THR 309

A THR 309, A PHE 435, A CYS 442

☐ Ignore all HETATM

Advanced Settings

click to expand / contract

Ignored residues:

Starting point [X,Y,Z]:

Probe Radius:

3

Interior Threshold:

1.25

Surface Cover Radius:

10

Origin Radius:

5

Bottleneck Radius:

1.2

Max Tunnel Similarity Radius:

0.9

[reset](#) to default values

plethora of non-protein molecules. These are usually cofactors, substrates or products of enzymatic reactions, lipids, sugars or remainders of crystallization buffers. All of the above, can represent obstacles to the channel analysis, as they can occupy channel regions, therefore, effectively preventing the channel from being identified. We have to decide whether we want to get rid of all HET atoms prior to the calculation, or just selectively get rid of just certain residues and list them in the underlying box.

Last but not least, MOLEonline provides an advanced level of filtering, as both *ProbeRadius* and *InteriorThreshold* are rather approximate parameters. For this purpose *MaxTunnelSimilarity* and *BottleneckRadius* are introduced. When a channel possesses a local narrowing lower than *BottleneckRadius*, the channel is discarded. Similarly, when two channels share more than *MaxTunnelSimilarity* (scale 0–1) centerline, the longer channel is discarded (Fig. 6.6).

6.4 Exercises

1. Gramicidin D channel

Identify a pore penetrating the structure of the small gramicidin ion channel (PDB ID 1grm). How long is the pore and how narrow is the bottleneck?

2. *ProbeRadius* and *InteriorThreshold* settings for channel identification

What is the suitable value for the *ProbeRadius* and *InteriorThreshold* parameter for channel identification? Use the following systems:

- Acetylcholine receptor channel (PDB ID 4boi)
- Carbonic anhydrase tunnel from the active site Cys 57, His 112, Cys 115 (PDB ID 3eyx)
- Haloalkane dehalogenase from the buried iodine ion coordinates (PDB ID 1cqw)

3. Channel starting point

What is the best way to identify a starting point prior to channel identification?

- For channels leading to a buried volume in the protein structure
- For pores spanning the whole structure

4. Polyamine oxidase binding channel

Identify the U-shaped polyamine oxidase (PDB ID 1b37) binding channel, as described in the literature.

5. Polypeptide channel in ribosome

Identify a polypeptide channel in a large ribosomal subunit (PDB ID 1jj2). Use U 2620 and A 2486 located on chain 0 as the starting point for channel identification.

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Part III

Characterization

Chapter 7

Characterization via Charges

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7.1 Introduction and Motivation

Partial atomic charges are real numbers which provide information about the distribution of electron density within a molecule [1]. Namely, they reflect the amount of electron density attributed to individual atoms. Information about partial atomic charges is therefore very useful, because it provides us with a clue to the chemical behavior and reactivity of the molecule. Specifically, the strongly positively or negatively charged parts of the molecule tend to become reaction centers. See the Dinitrotoluene example Sect. 7.2, which demonstrates the application of charges for the determination of toluene and 2, 4-dinitrotoluene reactivity. On the other hand, the atomic partial charges cannot reflect some peculiarities, e.g., inhomogeneous electrostatic potential on heavy halogen atoms, which enables the formation of halogen bonds [2].

The concept of partial atomic charges began to be used in physical chemistry and organic chemistry. Afterwards, partial atomic charges were adopted by computational chemistry (e.g., applications in molecular dynamics, docking, conformational searches, binding site predictions) [3, 4]. Recently, partial atomic charges also became popular in chemoinformatics (descriptors for QSAR and QSPR modelling, virtual screening) [5, 6] and bioinformatics (e.g. similarity searches, the study of mechanisms and effects connected with certain chemical actions) [7, 8].

7.2 Dinitrotoluene Example

Toluene is known to have more negative charges in the positions ortho- and para- (comparable to meta- position) and therefore it directs electrophilic substitutions to these positions. This information agrees with the results of charge calculations (Fig. 7.1a). 2,4-dinitrotoluene has the most negative C atom in the ring at position 6- and therefore directs the electrophilic substitution to this position. A well-known

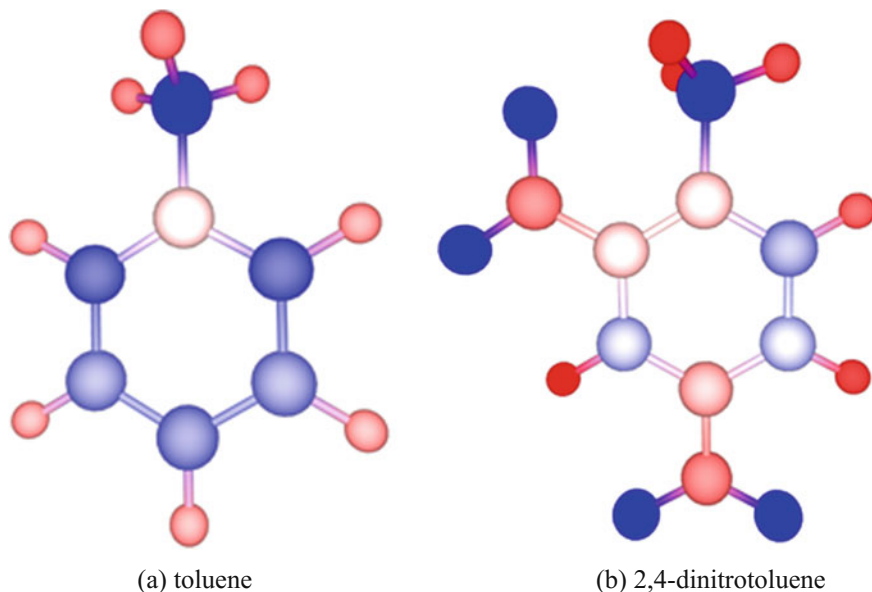


Fig. 7.1 Charge distribution in toluene and 2,4-dinitrotoluene. Calculated and visualized via AtomicChargeCalculator [9] with default settings, color scale from -0.4 (blue) to 0.4 (red) (color figure online)

example is the synthesis of 2,4,6-trinitrotoluene, which benefits from this fact. This information agrees with the results of charge calculations (Fig. 7.1b).

7.3 Charge Calculation Approaches

Partial atomic charges cannot be determined experimentally or derived in an unambiguous way from the results of quantum mechanics (QM), therefore many different methods have been developed for their calculation. We cannot determine which approach (and which charges) is the best, each is more suitable for some applications and less suitable for other applications.

The most common method for charge calculation is an application of the QM method and afterwards the utilization of a charge calculation scheme (i.e., an approach for the division of electron density into atoms). The most widely known charge calculation scheme is Mulliken population analysis [10]. This approach calculates the partial atomic charge of a certain atom as the sum of the electron population from atomic orbitals belonging to this atom plus half of the electron population from bond orbitals belonging to the atom. Other approaches divide the orbital electron population in a more sophisticated way, often-used charge calculation schemes are natural population analysis [11], atoms-in-molecules approach [12], and the Merz-Singh-Kollman method [13], just to name a few.

Unfortunately, QM charge calculation approaches are very time-consuming. A markedly faster alternative is to employ empirical charge calculation approaches, which can also provide reasonable estimates of partial atomic charges. These approaches can be divided into conformationally-independent, which are based on 2D structure (e.g., Gasteiger's and Marsili's PEOE [14], GDAC [15], KCM [16], DENR [17]) and conformationally-dependent, calculated from 3D structure (e.g., EEM [18], QEq [3] or SQE [19]). The conformationally-dependent charges are considered to be more suitable for the characterization of biomacromolecules and their fragments. The reason for this is that these charges contain extensive information not only about the chemical surroundings of atoms, i.e., its topology (2D structure based charges) but also the geometry (chemical structure) of the surroundings. Such information is missing, for example, in force field charges which use averaged atomic charges from large sets of structures.

EEM (electronegativity equalization method) is the most frequently used conformationally-dependent empirical charge calculation approach. It calculates charges using the following system of linear equations:

$$\begin{pmatrix} B_1 & \frac{\kappa}{r_{1,2}} & \cdots & \frac{\kappa}{r_{1,N}} & -1 \\ \frac{\kappa}{r_{2,3}} & B_2 & \cdots & \frac{\kappa}{r_{2,N}} & -1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\kappa}{r_{N,1}} & \frac{\kappa}{r_{N,2}} & \cdots & B_N & -1 \\ 1 & 1 & \cdots & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_N \\ \bar{\chi} \end{pmatrix} = \begin{pmatrix} -A_1 \\ -A_2 \\ \vdots \\ -A_N \\ Q \end{pmatrix} \quad (7.1)$$

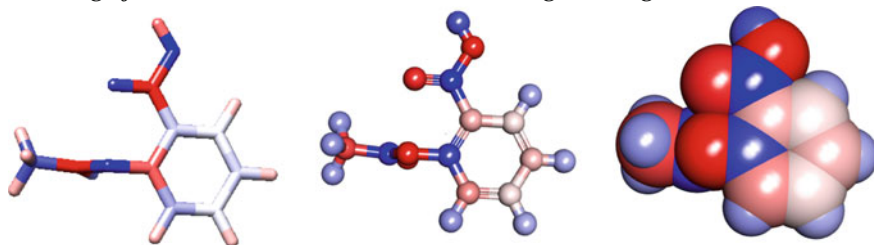
where q_i is the charge of an atom i ; $R_{i,j}$ is the distance between atoms i and j ; Q is the total charge of the molecule; N is the number of atoms in the molecule; $\bar{\chi}$ is the molecular electronegativity, and A_i , B_i and κ are empirical parameters. EEM is not only a fast charge calculation approach, but it can also provide highly accurate charges, i.e., they can mimic the QM charges for which EEM has been parameterized. Therefore, many EEM parameter sets for various QM charge calculation approaches have been published in the past and more recently [20].

7.4 Charge Visualization

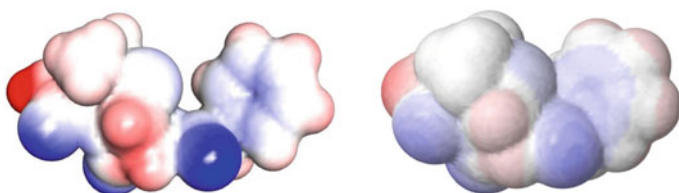
The visualization of partial atomic charges provides us with a way to quickly and easily obtain the information stored in the charges. Based on the visual representation of the charges, we can immediately recognize the most charged regions, compare the charge distribution of molecules or understand charge transfer.

The models for the visualization of partial charges are based on common models for molecular visualization. These common models can be divided into three classes – standard models (sticks, balls-and-sticks, CPK), surface models (van der Waals surface, solvent accessible surface) and schematic models (cartoon, ribbon). The charges can be later visualized on top of these common models, and it can be done

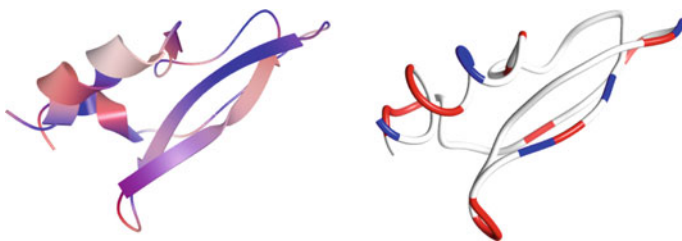
Coloring of molecular structure models according to charges



Standard models (sticks, balls-and-sticks, CPK)

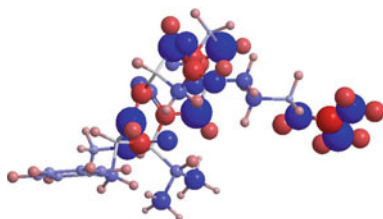


Surface models (van der Waals surface, solvent accessible surface)



Schematic models (cartoon, ribbon)

Size of atoms based on charges



Balls-and-sticks model only

Fig. 7.2 Examples of individual charge visualization models. The coloring can be defined by the user, the most common coloring is to use *blue* for negative charge and *red* for positive charge (color figure online)

in two ways – via coloring according to the charges or via setting the atom size according to the charges (applicable only in balls-and-sticks model). Examples of charge visualization models are depicted in Fig. 7.2.

7.5 Formats for Saving of Charges

Two main formats for storing of partial atomic charges are available – MOL2 and PQR.

The *MOL2 format* was developed by Tripos Inc. and is frequently used, especially for storing the structures of organic molecules. This file format contains information about partial atomic charges directly in lines, which describe individual atoms. The general syntax of this line is:

```
atom_id atom_name x y z atom_type [subst_id [subst_name [charge [status_bit]]]]
```

where: *atom_id* is the index of the atom; *atom_name* is the name of the atom; x, y and z are the coordinates of the atom; *atom_type* is the SYBYL atom type; *subst_id* is the index of the substructure containing the atom; *subst_name* is the name of this substructure; charge is the charge of atom and *status_bit* is a string determining an internal SYBYL status bit associated with the atom (e.g., BACKBONE, WATER etc.). The brackets '[' and ']' denote the optional parts of the syntax. An example is:

```
1 CA -0.149 0.299 0.566 C.3 1 ALA1 -0.216 BACKBONE
```

PQR format is a modification of the PDB format, which was introduced by Protein Data Bank [21] and is the most popular format for storing protein 3D structures. Again, the information about partial atomic charges is part of the lines describing individual atoms. The general syntax of this type of line is:

```
field_name atom_id atom_name residue_name chain_id residue_id x y z charge radius
```

where: *field_name* specifies the type of PQR entry (i.e., ATOM or HETATM); *residue_name* is the name of the residue; *chain_id* contains the identifier of the chain; *residue_id* is the index of the residue; and *radius* is the atomic radius. An example of such a line is:

```
ATOM 1 N ALA 1 46.457 12.189 21.556 0.1414 1.8240
```

7.6 Exercises

Overview: The practical exercises will be performed using the software tool AtomicChargeCalculator (ACC), because it provides both charge calculation and visualization, and is currently the only available web application for charge calculation. A usage of the ACC is shown in the solution of demo exercise 1 and also in the ACC manual.

1. Demo exercise: Detection of the first dissociating hydrogen in 3-hydroxybenzoic acid

3-hydroxybenzoic acid contains two hydrogens, which can potentially dissociate. The hydrogen from the COOH group and the hydrogen from the OH group. The more positively charged hydrogen will dissociate first. Calculate atomic charges for the 3-hydroxybenzoic acid using ACC (with `Bult2002_mpa` parameters) and detect the first dissociating hydrogen.

2. Comparison of charges in phenol molecules and detection of correlation between charges and pK_a

Download from PubChem the 3D structures of the molecules mentioned in the table below. For all the molecules calculate partial atomic charges using ACC (with `Bult2002_mpa` parameters). Analyze, if there is any dependency between the pK_a of the molecules and the atomic charges of certain atoms. Specifically, focus on charges on the H and O from the phenolic OH group, the C connected to the OH group and the charges on the C atoms in the phenolic ring.

Molecule	Pubchem ID	pK_a
2,4,6-trinitrophenol	6954	0.38
2,3-dinitrophenol	6191	5.96
3-hydroxybenzaldehyde	101	8.5
2,4,6-trimethylphenol	10698	10.9

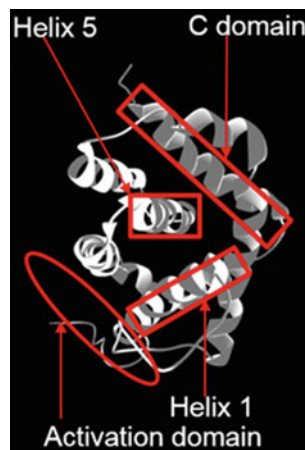
3. Comparison of charge distributions in cocaine binding sites

Calculate partial atomic charges for two cocaine (PDB residue ID COC) antibodies, specifically Anti-Cocaine Antibody M82G2 (PDB ID 1q72) and Cocaine catalytic Antibody 7A1 Fab' (PDB ID 2ajv). Afterwards, compare charge distribution in cocaine binding sites. Note: In both cases, delete water and cocaine molecules before starting the charge calculation. Use ACC with default settings for the calculation. Visualize the charge distribution in ACC using the Display Mode *Surface* and Probe Radius 1.4.

4. Comparison of a charge distribution in activated and inhibited apoptotic proteins

BAX is a protein participating in regulation of apoptosis – programmed cell death. When the cell is healthy, BAX is inactive and its C domain is tightly bound to its structure (see the figure below). During apoptosis, BAX is activated (i.e., an activator is bound to the activation site), its C domain becomes free, can insert into the mitochondrial membrane and penetrate it. It was found [7] that the activation is realized via a charge transfer and that helices 1 and 5 (see Fig. 7.3) play a role in this process. In this exercise, you will compare the charge distribution in active and inhibited BAX.

Fig. 7.3 Key helices and domains in BAX protein



Calculate charges in inactive BAX (PDB ID 1f16), inactive BAX in complex with its inhibitor (PDB ID 2l1r1) and active BAX (PDB ID 2k7w) via ACC. Describe how the charges in helix 1, helix 5 and the C domain changed.

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Chapter 8

Channel Characteristics

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The identification of channels is just the first step of a study. The more important part is analysis of the information we retrieved from the identification. Specifically, we have to evaluate channel characteristics and carefully examine whether the obtained characteristics and features match those which were expected. In the following section we will use the channel analysis software MOLE. As a feature for aiding this process, MOLE provides an estimation of basic channel characteristics using physicochemical properties. Specifically, MOLE can infer polarity [1], hydropathy [2], hydrophobicity [3], mutability [4] and the formal charge computed as the average of all lining residues.

In brief, the method sums up the length-weighted physicochemical properties of the amino acids lining the channel. The values for individual amino acids can be found in Table 8.1. As the algorithm is rather approximate and relies on a lot of averaging, it should be noted that the estimated physicochemical properties should be interpreted with care.

8.1 Physicochemical Properties

8.1.1 Hydropathy

Hydrophobicity and hydrophilicity are two extremes of a spectrum, commonly referred to as Hydropathy, and relate to the tendency of a molecule to interact with water. Several hydropathy scales have been developed in order to grasp the overall character of proteins or their parts. The Kyte–Doolittle scale is a widely used measure for expressing the hydrophobicity of amino acid residues. Regions with values above 0 are considered hydrophobic in character.

Table 8.1 Physicochemical properties of single amino acid residues as interpreted by MOLE 2.0

Residue	Charge	Hydropathy	Hydrophobicity	Polarity	Mutability
ALA	0	1.8	0.02	0	100
ARG	1	−4.5	−0.42	52	83
ASN	0	−3.5	−0.77	3.38	104
ASP	−1	−3.5	−1.04	49.7	86
CYS	0	2.5	0.77	1.48	44
GLU	−1	−3.5	−1.14	49.9	77
GLN	0	−3.5	−1.1	3.53	84
GLY	0	−0.4	−0.8	0	50
HIS	0	−3.2	0.26	51.6	91
ILE	0	4.5	1.81	0.13	103
LEU	0	3.8	1.14	0.13	54
LYS	1	−3.9	−0.41	49.5	72
MET	0	1.9	1	1.43	93
PHE	0	2.8	1.35	0.35	51
PRO	0	−1.6	−0.09	1.58	58
SER	0	−0.8	−0.97	1.67	117
THR	0	−0.7	−0.77	1.66	107
TRP	0	−0.9	1.71	2.1	25
TYR	0	−1.3	1.11	1.61	50
VAL	0	4.2	1.13	0.13	98

8.1.2 Polarity

Polarity is the property of a molecule given by the separation of electric charge, leading to the molecule having electric poles. Generally speaking, polar molecules are hydrophilic, while non-polar molecules are usually hydrophobic, but there can be exceptions. In terms of amino acids residues, hydrophilic/polar and hydrophobic/non-polar can be considered synonyms.

8.1.3 Mutability

Relative mutability quantifies the tendency of an amino acid to be substituted (mutated) in a protein structure. Substitution with similar amino acids generally retains protein function, while substitution with amino acids with different properties may affect the protein structure or function. Relative mutability is high for easily substitutable amino acids, such as small polar residues and low for amino

acids which play a significant role in the protein structure, i.e. substrate binding or catalytic activity. Alanine has a normalized value of 100.

8.1.4 Charge

Some amino acids can be (de)protonated based on the pH and, therefore, charged. At physiological pH, lysine and arginine are positively charged, whereas aspartic and glutamic acids are negatively charged. On the other hand, the protonation state of histidine is dependent on its micro-environment. MOLE treats all histidines as positively charged.

From Chap. 6, we know that a channel can be represented in two ways. First, as an empty space in a protein. MOLE uses a spherical probe for this kind of representation. These probes are placed alongside the channel centerline in the most appropriate way in which they do not penetrate the vdW sphere of any of the protein's atoms. Second, as the residues encapsulating this empty void. The situation for three channels of Cytochrome P450 is highlighted in Fig. 8.1.

To sum up, MOLE represents channels as a union of spheres with a given length and radius. Moreover all the residues lining the centerline are reported. The centerline is uniformly divided into layers, and each layer is defined by the residues lining it. A new layer starts whenever there is a change in the list of residues lining the channel along its length. The lining of the channel is then described as a sequence of layer lining residues. For each layer, the length (distance between the first and last atom of the layer projected to the channel centerline) and radius (bottleneck) are computed. Additionally, the orientation of each residue is determined to check whether the residue faces the channel with its backbone or side-chain moiety.

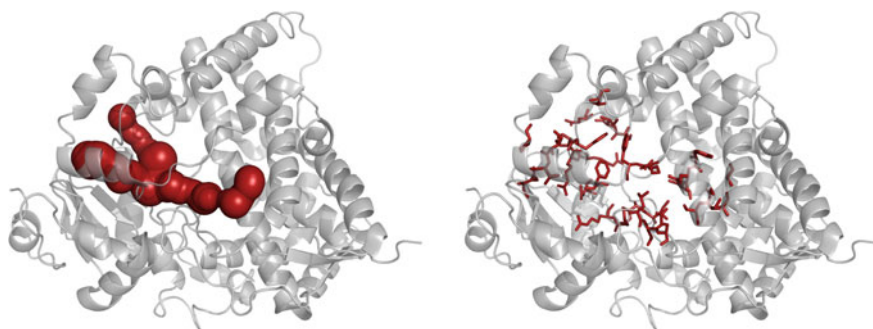


Fig. 8.1 Channels of cytochrome P450 (PDB ID 1tcqn). On the *left* channels represented using void spheres in the protein body and *right* residues lining channels walls

8.2 Characterization of Channels Using MOLEonline

Channel calculation together with the tool setup is described in detail in Chap. 6.

8.2.1 Results Analysis

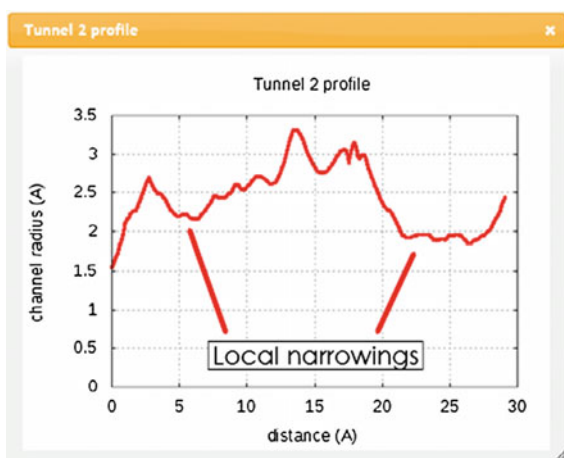
When the setup is completed and the calculation executed, the result is retrieved within a couple of seconds. Now comes the interesting part, we have to analyze the results and infer putative channel function.

First, we can base our guess on the channel's centerline and its width. As highlighted in Fig. 8.2, we have to take into consideration that proteins are not rigid structures, instead they are rather flexible. Therefore, even if the radius does not seem to be sufficient, the ligand can still pass through the channel. What is more important are the residues forming the channel lining walls and the corresponding physicochemical properties. These are highlighted in Fig. 8.3.

Best practice for inferring channel polarity is to use the Hydropathy index, as the scale is symmetrical and not subject to rapid changes like the Polarity index. Indeed, values over the Hydropathy scale are symmetrical and uniformly distributed. The problem with the Polarity scale is that just the presence of a few polar residues in the channel lining can totally change the overall perception of the channel, as it shifts the average polarity index towards polar values.

When interpreting the Hydropathy index, negative values are considered to be hydrophilic, while positive values are considered hydrophobic. Nevertheless, there is a grey zone ($-1, 1$) where it is hard to exactly define whether or not a channel is hydrophilic/hydrophobic in its behavior. Overall, comparing the particular

Fig. 8.2 Channel profile as a function of length (from the beginning of the channel towards the bulk solvent) and channel radius



Lining residues for tunnel 2

1 Unique lining residues set - all
 212 ARG A, 308 GLU A, 312 SER A, **308 GLU A**, 369 ILE A, 309 THR A, 482 LEU A, **483 LEU A**, 369 ILE A, 371 MET A, 213 PHE A, 215 PHE A, 370 ALA A, 372 ARG A, 370 ALA A, 105 ARG A, 108 PHE A, 119 SER A, 106 ARG A, 120 ILE A, 120 ILE A, 107 PRO A, 122 GLU A, 107 PRO A, 111 VAL A

Unique lining residues set - sidechains
 212 ARG A, 308 GLU A, 312 SER A, 369 ILE A, 482 LEU A, 371 MET A, 215 PHE A, 370 ALA A, 105 ARG A, 108 PHE A, 120 ILE A, 122 GLU A, 107 PRO A, 111 VAL A

2 Physicochemical properties of lining side-chains
 Charge: 0 (2-2)
 Hydropathy: -0.1
 Hydrophobicity: 0.15
 Polarity: 16.52
 Mutability: 82

3 Lining residues
[show all](#) | [hide all](#)

#	Res	Bln	Dist	Hpa	Hpb	Pol	Mut
1	☐ 212 ARG A, 308 GLU A	1.56	0.84	-4	-0.78	50.95	80
2	☐ 212 ARG A, 308 GLU A, 312 SER A	1.99	1.36	-2.93	-0.84	34.52	92
3	☐ 212 ARG A, 312 SER A, 308 GLU A , 369 ILE A	2.21	1.51	-0.3	-0.1	14.3	101
4	☐ 212 ARG A, 312 SER A, 308 GLU A , 369 ILE A, 309 THR A	2.26	1.97	-0.32	-0.24	12.11	101
5	☐ 212 ARG A, 312 SER A, 308 GLU A , 369 ILE A	2.34	2.26	-0.3	-0.1	14.3	101
6	☐ 212 ARG A, 312 SER A, 369 ILE A	2.49	2.49	-0.27	0.14	17.93	101
7	☐ 212 ARG A, 312 SER A, 369 ILE A, 482 LEU A	2.61	2.69	0.75	0.39	13.48	89
8	☐ 212 ARG A, 312 SER A, 369 ILE A, 482 LEU A, 483 LEU A	2.7	2.81	0.52	0.15	11.46	89
9	☐ 212 ARG A, 312 SER A, 369 ILE A, 482 LEU A	2.66	3.01	0.75	0.39	13.48	89

Fig. 8.3 1) Unique list of lining residues. **Bold** residues correspond to the lining formed by a backbone of the protein. 2) Overall physicochemical properties for the channel. 3) The list of layers is given in Chap. 6. This decomposition of a channel aids the interpretation of parts of the channel. Specifically, local and global narrowing can be easily identified together with their composition and their hydrophilic/hydrophobic properties

values with the values for individual residues (Table 8.1) is often useful for inferring characteristics of the channel or parts of the channel.

8.2.1.1 Aldolase-Dehydrogenase Example

The aldolase-dehydrogenase (HsaF-HsaG) complex [5] of *Mycobacterium tuberculosis* is part of the pathway responsible for cholesterol uptake and degradation. The process is vital for this lung-infecting bacteria's survival. The structure of the HsaF-HsaG complex (PDB ID 4jn6) is organized as a heterotetramer with two distinct aldolase-dehydrogenase units. Both the enzymes are interconnected via a molecular channel connecting their active sites. In this process, often referred to as substrate

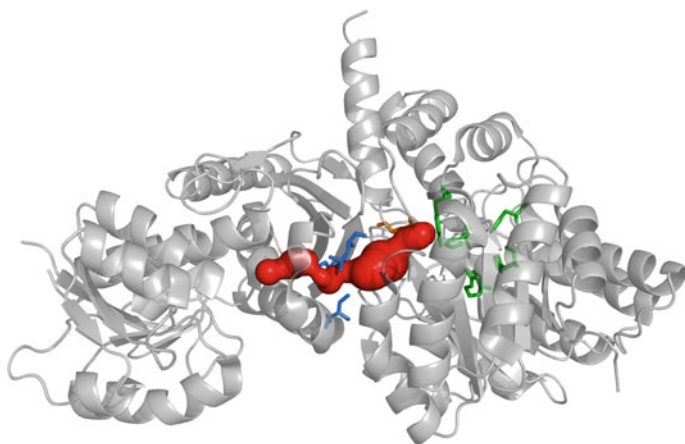


Fig. 8.4 Molecular channel connecting active sites of Aldose-Dehydrogenase complex (PDB ID 4jnb). The active site of Aldose and Dehydrogenase is visualized using a *green* and *blue* stick model respectively. The bottleneck referred to in the literature [5] formed with glycine residues is highlighted in *orange* (color figure online)

channeling, a toxic aldehyde can reach the other active site, while minimizing interference with other competitive metabolites. Let us identify this molecular channel!

From the literature [5] we can extract the information about the composition of the active sites in Aldolase. This active site contains a manganese ion, so we can use this information for finding a rational start point for the channel. Next we have to take into consideration that the active site also contains an oxalate residue, these should be removed prior to the start of the analysis. Either we can list all the unwanted residues or we can directly filter out all the HET residues. Another approach taken by the authors is to move the starting point of the channels further in the direction of the channel. Tyr 289 A has been used as a rational channel starting point, so we will use it as well. Since this protein complex is rather small and tightly packed, we can keep the default settings for *ProbeRadius* and *InteriorThreshold*. However there is a bottleneck present, so it is vital to decrease the *BottleneckRadius* from the default value to 0.9. The identified channel is visualized in Fig. 8.4.

The resulting channel is ~ 30 Å in length and both the active sites are separated by an ~ 20 Å length of the channel. The Hydropathy index of this channel is 1.19, suggesting hydrophobic behavior, which is in agreement with its function. This is not surprising, as the channel is mostly lined with leucine, isoleucine or methionine.

8.3 Common Errors in Channel Calculation and Characterization

8.3.1 *No Channels Have Been Identified*

When you have set up the starting point of your calculation, provided all the parameters, but no channels have been identified. One of the following scenarios is likely to have occurred:

- (a) Wrong set up of *ProbeRadius*, *InteriorThreshold* radius or filtering criteria.
- (b) Substrate is blocking channels
- (c) Active site is located at the protein surface
- (d) There is no channel at all.

8.3.1.1 Wrong Set up of *ProbeRadius*, *InteriorThreshold* Radius or Filtering Criteria

ProbeRadius and *InteriorThreshold* are key parameters for channel identification, as they figuratively speaking represent the lower and upper radius of the identified channel. For example, if the protein under study contains a bottleneck with a radius of 1 Å, than setting *InteriorThreshold* to 2 Å will inevitably decrease the chance of identifying the channel. The best values for consideration are (1.1, 1.25 and 1.4).

On the other hand, *ProbeRadius* represents an upper bound for channel radius. If the radius of the channel is thought to be wide, then setting up a small *ProbeRadius* will also lead to a decreased chance of channel identification. Setting *ProbeRadius* to be smaller than *InteriorThreshold* is also clearly nonsense.

Finally, channel filtering criteria can be a restrictive factor as well. After channel identification, a last filtering step is carried out by MOLE which strives to report channels that are as relevant as possible. Channels are filtered out based on the length and size of their bottleneck or by the similarity of their centerline. Try adjusting the values of *MaxTunnelSimilarity*, *BottleneckRadius* or *BottleneckLength*. It is always good practice to set *BottleneckRadius* a little lower than *InteriorThreshold*.

8.3.1.2 Substrate is Blocking Channels

For channel identification, by default MOLE takes into consideration all atoms in the input structure. The only residues which are discarded prior to the calculation are HOH residues, which correspond to water. The majority of protein structures are stored in the Protein Data Bank with non-protein residues, these are usually either protein cofactors, substrates, products of protein activity, or various chemical compounds present from the crystallization experiment. Since substrates and products are most likely to occupy channels, it is good practice to either remove all HET atoms prior

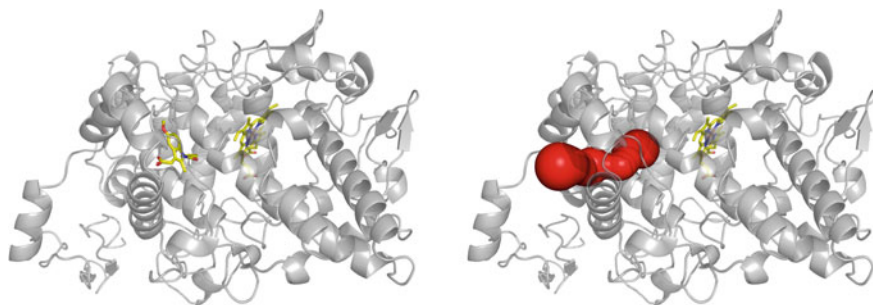


Fig. 8.5 Single subunit of cyclooxygenase-2 (PDB ID 4cox) complexed with non-selective inhibitor indomethacin (IMN). *Left* part is with the indomethacin, preventing the channel leading to the hydrophobic cavity from being identified. *Right*, after discarding the IMN residue prior to the calculation, the channel can be properly identified

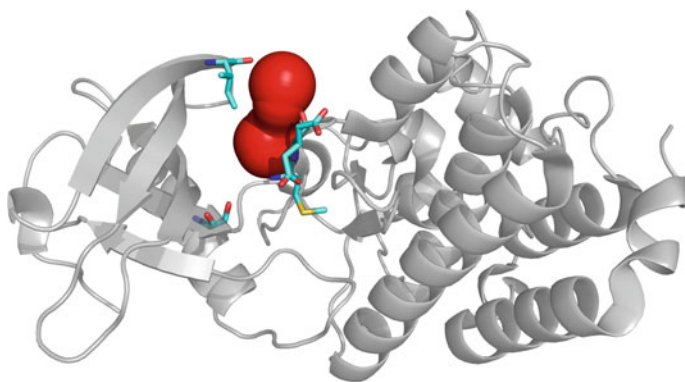


Fig. 8.6 Visualization of 8 Å-long pocket-like channel for tyrosine kinase EphB4 (PDB ID 2vwx). The catalytic site is represented by a stick model (*ProbeRadius* 10)

to the calculation, or selectively remove only those you want to get rid of (i.e. a substrate and not a cofactor) – Fig. 8.5.

8.3.1.3 Active Site is Located on the Protein Surface

MOLE is meant to only identify channels leading to the buried sites of protein structures. Therefore, the identification of shallow pockets is off the table. Strictly speaking you could artificially ‘bury’ the pocket by increasing the *ProbeRadius* parameter, however a channel identified in this way would be of little biological significance. So it is worth considering a tool meant for pocket identification instead. The situation is highlighted for the binding site of the tyrosine kinase EphB4 (Fig. 8.6).

8.3.1.4 There is No Channel at all

This is the final possibility. If none of the above helped and still no channels could be identified, it is likely that no reasonably sized channel leads to the site you have provided. In that case double check your parameters. If you are still persuaded that a channel should be identified, consider contacting the authors of the software.

8.3.2 A Lot of Different Channels Are Identified, However None of Them Seems to be Relevant to My Expectations

Most likely this is caused by a wrong setting of the *ProbeRadius* parameter. Try increasing it.

All or some of the channels contain an artificial large extension towards the end of the channel, which does not seem to be natural. It is likely that you have overestimated the *ProbeRadius* parameter. The other function of this parameter is approximation of the molecular surface. When the parameter is way too high, the approximate surface is far from the actual protein surface, which leads to the unnatural channel ending towards the protein surface (Figs. 8.7 and 8.8).

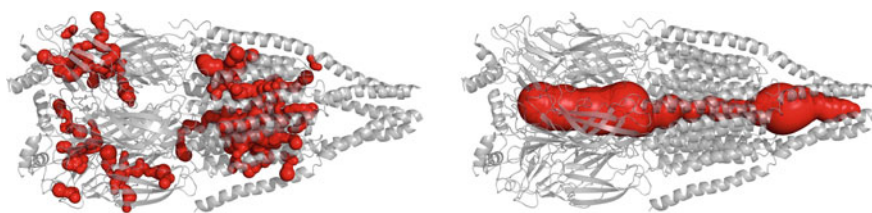


Fig. 8.7 On the *left* only small and irrelevant channels are identified for the acetylcholine receptor (PDB ID 2bg9) (*ProbeRadius* 3) due to the *ProbeRadius* being too low, which is not the case for the *right* figure (*ProbeRadius* 9), where the pore spanning the whole structure can be identified

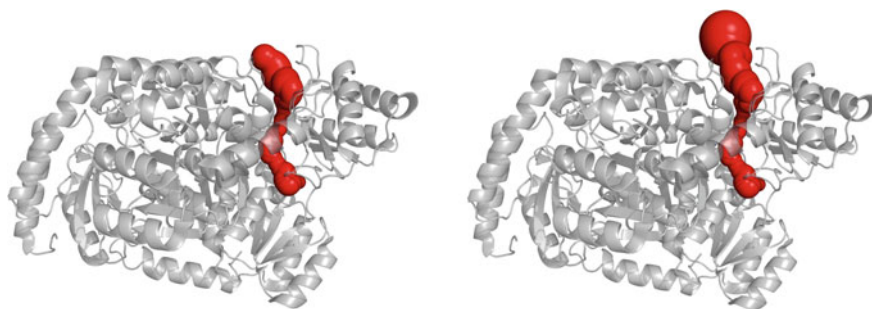


Fig. 8.8 Channel leading towards the active site of PLP-Dependent Acyl-CoA transferase CqsA (PDB ID 3kki). *Left* *ProbeRadius* set to a reasonable value of 5. *Right* overestimation of this parameter – *ProbeRadius* set to 20

8.4 Exercises

1. Gramicidin D pore

Identify a pore spanning through the structure of gramicidin D (PDB ID 1grn). What is its hydropathy? Which compounds are likely to pass through the channel?

2. Cytochrome P450 3A4

Using the default settings of MOLEonline, identify channels for Cytochrome P450 (PDB ID 1tqn). How many channels can you identify? Which of them is most likely the water channel based on the Hydropathy index?

3. Substrate tunnel

How do you have to preprocess the structure of sterol 14- α demethylase (PDB ID 3k1o), in order to identify the access channel for the posaconazole inhibitor. What is the nature of this channel?

4. Bottleneck

Identify a bottleneck in aquaporine (PDB ID 3gd8). What is the composition and Hydropathy of the bottleneck region?

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Part IV
Complete Process of Data Extraction
and Analysis

Chapter 9

Complete Process of Data Extraction and Analysis

Radka Svobodová Vařeková and Karel Berka

9.1 Lectin Example (Validation, Extraction, Comparison, Charge Calculation)

Pseudomonas aeruginosa is an opportunistic pathogen associated with a number of chronic infections. This pathogen forms a biofilm, enabling it to survive both the response of the host immune system and antibiotic treatment. One of the cornerstones of biofilm formation is the presence of the sugar-binding protein LecB (PA-IIL). Its inhibition is considered to be a promising approach for anti-pseudomonadal treatment.

For this reason, we will examine a sugar-binding site of LecB. Specifically, we will find all occurrences of this binding site in Protein Data Bank, validate them (and remove potential wrong structures) and ask the following questions:

1. Does this binding site also occur in organisms other than *Pseudomonas aeruginosa* and in proteins other than LecB?
2. Do the binding sites have a common amino acid composition? Specifically, are there amino acids which are present in all occurrences of the binding site (or most of them)? And are there binding sites which have a different amino acid composition, and thus seem to be outliers?
3. Is the 3D structure of the common amino acid part (or parts) similar? Thus does the binding site have some conserved structural pattern(s)?
4. Is there any common charge distribution within the binding site?

9.1.1 Step 1: Detection of All Occurrences of the Binding Site

We first define the pattern we will search for in the biomacromolecule. This pattern must be characteristic for the LecB sugar-binding site. Because the binding site

contains two close Ca atoms and it binds sugars (i.e. pyranoses), we can define the pattern in the following way.

A pair of Ca atoms (less than 4 Å apart), which are near to (less than 2 Å from) the pyran ring (i.e., a ring containing 5 carbons and one oxygen). Note: These distances were obtained from known structures of LecB. Because we are interested in the whole binding site, we will also add residues surrounding the Ca pair and the pyran ring to the pattern, specifically all residues 4 Å or less from them. Note: We selected such a relatively small surrounding region because we would like to provide a transparent and easy-to-follow example. Of course, in praxis you can also examine a larger surrounding region.

This pattern is depicted in Fig. 9.1 and described by the following PatternQuery expression:

```

1 Near(2,
2   Rings(5 * ["C"] + ["O"]),
3   Near(4, Atoms("Ca"), Atoms("Ca")))
4   .AmbientResidues(4)

```

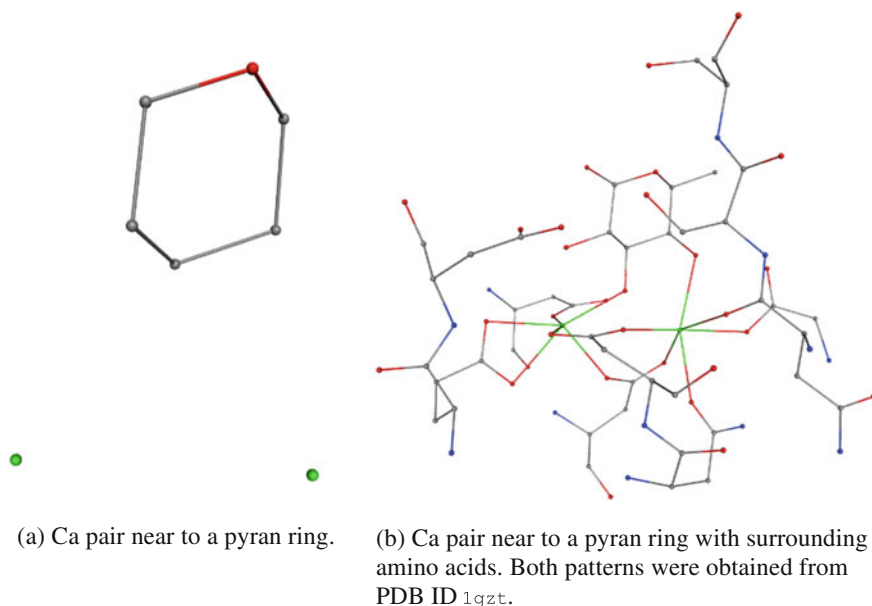


Fig. 9.1 LecB binding site pattern identified

Then we use `PatternQuery` and search for all the PDB entries containing this pattern. Because the pattern includes Ca, we can markedly speed up the search via adding Metadata information. The specific procedure for performing this task is described in the Sect. 9.1.7 - *Methodology of data analysis*.

The query provided us with a dataset containing 39 PDB entries and 127 relevant patterns (August 2016).

9.1.2 Step 2: Validation of the Obtained PDB Entries

In the validation step, we focused purely on ligands, because we are examining the ligand binding site and because the ligands are the main sources of errors. If we see some outlier structures in the subsequent steps of the analysis, we can study the validation reports of the outliers.

We used `ValidatorDB` to validate ligands in the obtained 39 PDB entries. The validation procedure is described in the Sect. 9.1.7 - *Methodology of data analysis*.

The validation found problems in 3 PDB entries: 2 PDB entries had missing atoms (3dcq, 5a6z) and one had a chirality error at a carbon (1oxc). We inspected the validation results for these PDB entries and we found the following:

- 3dcq: The missing atoms are in residues 2G0 201 A (3 missing atoms) and 2G0 201 B (7 missing atoms). These residues belong at the LecB sugar binding sites (i.e., they are the bound sugars). Therefore, these errors can influence the results of our analyses. For this reason, we excluded this PDB entry from our analysis.
- 5a6z: The missing atoms are in residuum GLA 204, which is outside the LecB binding site. Therefore, this PDB entry can remain in the dataset.
- 1oxc: The chirality problem is at the C1 atom of residues FUC 115 B and FUC 115 C. It seems that the residues are correct, just that they should be annotated FUL (α -L-fucose), like the two remaining sugars in the entry, instead of FUC (α -L-fucose). Therefore, this PDB entry can remain in the dataset.

Therefore, based on the validation results, we removed the PDB entry 3dcq from the dataset and we continue with a dataset of 38 PDB entries and 123 relevant patterns.

9.1.3 Step 3: Analysis of Organisms and Proteins, from Which the Obtained Binding Sites Originate

We used Protein Data Bank to find information about the organisms and proteins that the obtained binding sites originate from. The procedure is described in the Sect. 9.1.7 - *Methodology of data analysis*.

We found, that most of the PDB entries originate from the bacteria:

- *Pseudomonas aeruginosa* (27 entries) or its synthetic constructs (2)
- *Burkholderia cenocepacia* (4 entries)

- *Ralstonia solanacearum* (2 entries)
- *Chromobacterium violaceum* (2 entries)
- *Bacillus subtilis* (1 entry)

We also discovered that most of the PDB entries are (according to their UniProt molecule name) the following lectins:

- hypothetical protein LecB (24 entries)
- LECTIN (5 entries)
- FUCOSE-BINDING LECTIN PA-IIL (3 entries)
- CV-IIL LECTIN (2 entries)
- BC2L-A LECTIN (1 entries)
- LECB (1 entry)
- LECB LECTIN (1 entry)

The only exception is one entry (PDB ID 2o04), which is pectate lyase.

9.1.4 Step 4: Analysis of Common Amino Acid Composition

We used the output files from PatternQuery to analyze the amino acid composition of the binding sites. The procedure is described in the Sect. 9.1.7 - *Methodology of data analysis*.

We found the following information:

- All the binding sites contain 4 ASP, 2 ASN and 1 GLU. The only exception is the binding site from the PDB ID 2o04, which contains only 3 ASP, no ASN and no GLU. This entry seems to be an outlier in our data set, because it is not a lectin, but pectate lyase, and it is the only protein originating from *Bacillus subtilis*. Therefore, we will remove it from our data set of sugar-binding sites.
- Most of the binding sites (i.e., 112 from 123) contain GLY.
- All the binding sites contain 2 SER or 2 ALA or 1 ALA and 1 SER. Specifically:
 - 22 entries with 2 SER. They all are from *Pseudomonas aeruginosa* or synthetic constructs and they all are the hypothetical protein LecB or LECB.
 - 6 entries contain 2 ALA. They all are from *Burkholderia cenocepacia* and *Ralstonia solanacearum* and they are annotated as LECTIN or BC2L-A LECTIN.
 - 9 entries contain 1 ALA and 1 SER. They are from *Pseudomonas aeruginosa* (mainly mutants of its LecB) and *Chromobacterium violaceum*. Their annotations are highly heterogeneous: hypothetical protein LecB, FUCOSE-BINDING LECTIN PA-IIL, CV-IIL LECTIN, LECTIN or LECB.
- Most of the binding sites with 2 ALA (i.e., 12 from 15) also contain HIS.

Summary: The binding sites contain several common amino acids (4 ASP, 2 ASN, 1 GLU and 1 GLY). It seems that the binding site has two main variants – the first (from *Pseudomonas aeruginosa*) contains 2 SER, the second (from *Burkholderia cenocepacia* and *Ralstonia solanacearum*) contains 2 ALA and HIS. Furthermore, there are combined binding sites, containing one ALA and one SER.

9.1.5 Step 5: Analysis of Common 3D Structure Parts

We used SiteBinder and superimposed the output files from PatternQuery to analyze the amino acid composition of the binding sites. The procedure is described in the Sect. 9.1.7 - *Methodology of data analysis*.

We found the following information.

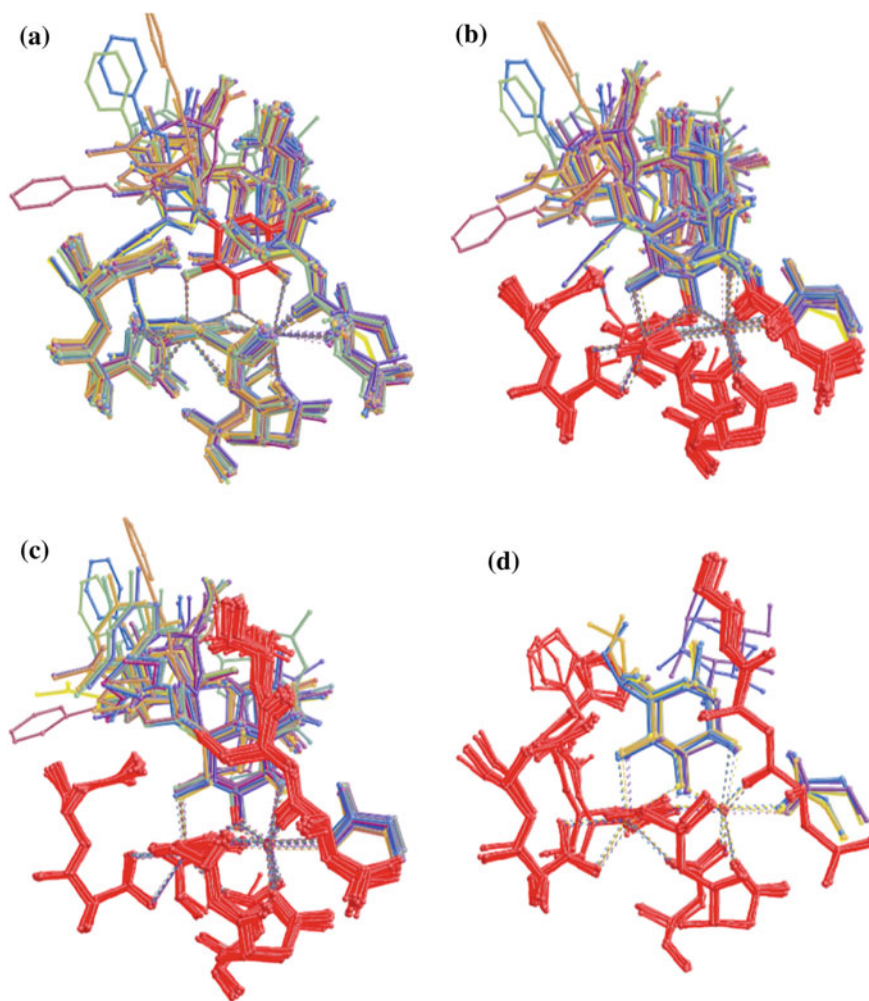


Fig. 9.2 **a** Superimposition of pyran rings in all sugar-binding sites. **b** Superimposition of 4 ASP, 2 ASN and GLU in all sugar-binding sites. **c** Superimposition of 4 ASP, 2 ASN, GLU and 2 SER in sugar-binding sites containing 2 SER. **d** Superimposition of 4 ASP, 2 ASN, GLU, 2 ALA in sugar-binding sites containing 2 ALA

The common part of the binding site is highly conserved. This can be seen even when we only superimpose the pyran ring and Ca atoms (Fig. 9.2a). After superimposition of the common amino acids (4 ASP, 2 ASN and GLU), we see the conservation of the binding site very clearly (Fig. 9.2b) and the RMSD of this superimposition is also very low (i.e., 0.3 Å).

Also, the two main structural variants are highly conserved. Specifically, when we superimpose all the binding sites containing 2 SER according to their common amino acids, the RMSD will be 0.18 Å (see Fig. 9.2c). When we performed the same procedure for binding sites with 2 ALA and HIS, we also saw their conserved structure (Fig. 9.2d) and the RMSD was 0.46 Å.

9.1.6 Step 6: Analysis of Charge Distribution

During the previous analyses, we recognized two main variants of the examined sugar-binding site. Despite their common parts, the two variants also have differences. We would like to analyze whether their charge distribution also differs or is similar.

We selected one representative of the 2 SER variant (PDB entry 1gzt) and one representative of the 2 ALA and HIS variant (PDB entry 2vnnv). Then we removed the ligands and water molecules from the structure – to have the binding site uncovered. Afterwards, we use ACC to calculate atomic charges. The procedure is described in the Sect. 9.1.7 - *Methodology of data analysis*. We found that the charge distribution in both the binding sites is very similar (see Fig. 9.3), which is in agreement with their similar chemical behavior.

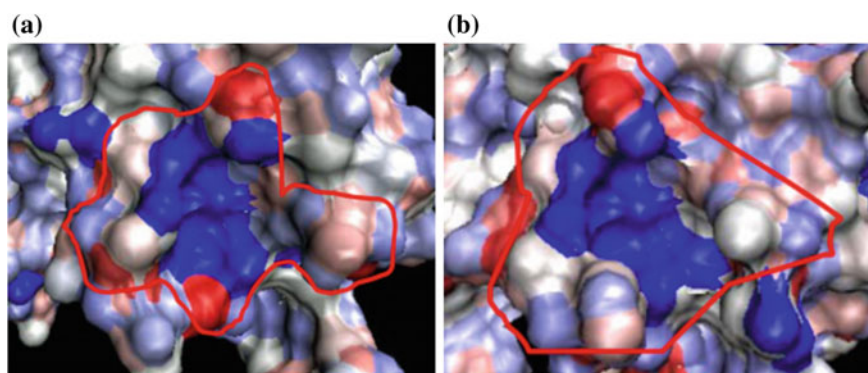


Fig. 9.3 **a** Charge distribution in sugar binding site of 1gzt. **b** Charge distribution in sugar binding site of 2vnnv

9.1.7 Methodology of Data Analysis

9.1.7.1 Step 1: Binding Site Extraction

Open PatternQuery, go to *Query Protein Data Bank*, enter the above query into the input box, fill in the name of the query (i.e. *LecB_binding_sites*) and click on *Add*. Then click on *Filtered by Metadata*, select *Category Atoms*, enter the atom ID *Ca* and click on *Add*. Then submit the query.

9.1.7.2 Step 2: PDB Entries Validation

Download the results of PatternQuery and unzip them. Open the file *patterns.xls* and copy from it the PDB IDs of the found PDB entries (they are in the second column, named *ParentID*). Open ValidatorDB, go to *Search*, click on *PDB Entry*, put the PDB IDs there and click on *Slow search*. Then click on *Details by the PDB Entry* and it will provide information about problematic structures. You can also sort the data according to the validation type (e.g. by clicking on “x Atoms”) and you can see details about validation problems by clicking on individual PDB IDs.

9.1.7.3 Step 3: Analysis of Organisms and Proteins

Open RCSB Protein Data Bank, click on *Advanced Search*, select query type *PDB ID(s)* and put the PDB IDs of the PDB entries from the dataset into the input box.

9.1.7.4 Step 4: Analysis of Common Amino Acid Composition

Open the file *patterns.xls*, which is part of the PatternQuery results. Look at the *Signature* column and you will see the amino acid composition of the binding site. When you see an amino acid repeating, you can test its occurrence in all the binding sites via an Excel function *SEARCH*.

9.1.7.5 Step 5: Analysis of Common 3D Structure Parts

Open SiteBinder and put all the inspected binding sites there (select them in the directory and move them to SiteBinder with the mouse). Click on *Atom Selection* and on *Clear*. Then enter the selection of atoms you would like to superimpose, for example:

- Ca pair and furan ring:

```
1 Near(2,
2   Rings(5 * ["C"] + ["O"]),
3   Near(4, Atoms("Ca"), Atoms("Ca")))
```

- 4 ASP, 2 ASN and GLU:

```
1 Residues("ASP", "ASN", "GLU")
```

- 4 ASP, 2 ASN, GLU, ALA and HIS:

```
1 Residues("ASP", "ASN", "GLU", "ALA", "HIS")
```

When you would like to only superimpose part of the binding sites (e.g., only binding sites with 2 SER) you must select them: Click on *Group by Selection*, go to the list of binding sites and only leave checked the relevant group of binding sites (i.e., the group with the highest count of selected atoms).

Then click on *Connect* and afterwards on *Superimpose*.

9.1.7.6 Step 6: Analysis of Charge Distribution

Download the PDB files of entries 1gzt and 1vny from Protein Data Bank. Using a text editor (e.g. WordPad) remove the ligands and water molecules from both PDB files (they are denoted HETATM). Note: Do not remove Ca atoms. Then calculate the charges of both molecules via ACC (default settings). Afterwards download the results of ACC, unzip them, get the PQR file and visualize it via VMD (<http://www.ks.uiuc.edu/Research/vmd/>), which has a more detailed visualization of charges on the surface. Use VMD in the following way: Load the molecule. In the tab Graphics, go to Representations, select Coloring Method *Charges* and Drawing Method *Surf*.

9.2 Cytochrome P450 Example (Database Search, Detection of Channels, Channel Characterization)

We have already encountered cytochromes P450 (CYPs) in previous examples in databases and channel detection, however here we focus on overall analysis of the given biomacromolecule from known sources and question the hypothesis of whether we can link the effect of known mutations to amino acids in the channels or rather to amino acids binding ligands.

First, for any new macromolecule for analysis, it is wise to look up known data in a somewhat concise form. For proteins, the UniProt database is such a place to start. So let's focus on data about the human CYPs presented in this database.

9.2.1 Database Search

1. Find human CYPs with the largest number of crystal structures. Note its UNIPROT ID.
2. What are the molecular functions and biological processes connected with this protein according to its GO annotation? Restrict yourself to major keywords.
3. State the most generic catalytic activity of this selected CYP. Write the equation of this chemical reaction.
4. What is the EC number of this protein?
5. Where is this protein located within the cell?
6. List the interactions of this protein with small molecules available in ChEMBL, DrugBank and BindingDB. Which database contains the most chemical interaction data?
7. Find known problematic mutations for this protein. List any variants with a known effect on protein function.
8. Find the closest protein partners via cross-link to the STRING database. List those which are known from experiment.

After collecting information about the protein in general, it is usually a good option to look at the structure in structural databases:

9. Select the structure of the protein with the best resolution and open it in the PDBe database to find whether this protein dimerizes or forms any other macromolecular assemblies.
10. Find similar 3D structures using PDBeFold – which other protein has the most similar structure? What is its sequence similarity?
11. Try to find the active site by using the ligands present within the structure.
12. Use the Protein Feature View in RCSB to compare the coverage of the sequence with the extracted information about the sources of disorder within the structure.
13. Compare the structure of the CYP protein with others from its family using PDB Flex. Which region exhibits the largest local flexibility?
14. Based on global flexibility analysis, find representative structures of individual clusters of conformations of the protein. Also select the two most distant structures.
15. Using PDBsum, find which ligands occupy the active site of the most distant structures from the previous task.
16. Analyze how different their surrounding residues are using LigPlot, and compare them to catalytic residues from the Catalytic Site Atlas.

9.2.2 Channels Detection

17. Analyze whether these two most distant structures share all channels from the catalytic site. Use MOLEonline 2.0 without HETATMs to even include channels blocked by ligands.

9.2.3 Channels Characterization

18. One of the structures contains a channel which is wider than the *ProbeRadius* (you can check the molecular surface). In order to analyze this channel as well, enlarge the *ProbeRadius* to twice the original value (i.e. to 6 Å) and redo the calculations. Are there visually similar channels in both structures now?
19. Compare the lining residues in the channels in both structures and list the channel-lining residues for channels with the largest overlap. How much do they differ in their composition, charge, hydropathy and polarity?
20. Compare residues lining the channels with mutated amino acids with a known effect. Are there any overlaps?

9.2.4 Solution

1. UniProt gives a large number of entries upon searching for *human cytochrome p450*. The first level of selection filtering is to only select the 71 reviewed entries. If filtered by protein family, only 61 entries remain. Under the Columns button at the top of the result table, almost any information on the proteins can be added to the result table as a new column (i.e. Names & Taxonomy, Sequences, Function, Structure, Gene Ontology, etc.). In the Structure section, tick the 3D option and the table with results will now show the number of available structures. Cytochrome P450 3A4 (CYP3A4) with UNIPROT ID P08684 has the most (24) 3D structures available, followed by CYP2A6 (11), CYP2D6 (10), and others.
2. The Function section shows a list of GO annotations, where it highlights keywords:

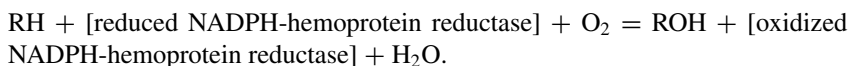
Keywords for Molecular function: Monooxygenase, Oxidoreductase

Keywords for Biological process: Lipid metabolism, Steroid metabolism, Sterol metabolism

3. The Function section also contain a description of the function itself, where it says that the enzyme *performs a variety of oxidation reactions (e.g. caffeine 8-oxidation, omeprazole sulfoxidation, midazolam 1'-hydroxylation and midazolam 4-hydroxylation) of structurally unrelated compounds, including steroids,*

fatty acids, and xenobiotics.

The most generic catalytic reaction is thus given by the equation:



4. The Enzyme and pathway database links give two individual EC numbers: 1 . 14 . 13 . 67 (quinine 3-monooxygenase) and 1 . 14 . 99 . 38 (cholesterol 25-hydroxylase), however a closer look in the databases (e.g. Metacyc) shows a large number of EC numbers annotated with CYP3A4.
5. The Subcellular localization section states that CYP3A4 is a single-pass membrane protein located in the endoplasmic reticulum membrane. The Topology section also lists that residues 2–22 form a helical transmembrane anchor.
6. The Pathology & Biotech and Interaction sections contain links to the following Chemistry databases: the ChEMBL link leads to the cytochrome P450 3A group, from which CYP3A4 has to be selected (ChEMBL340), which lists over 44,000 bioactivity types for about 27,000 ChEMBL compounds.

DrugBank entries are shown individually, but in the DrugBank database it is possible to list all drugs with CYP3A4 as a target (BE0002638), which leads to a list of 591 drug entries (mainly approved) of which 484 drugs are CYP3A4 substrates and 222 drugs are CYP3A4 inhibitors.

And finally, BindingDB lists 5,473 hits with affinity data for CYP3A4 with some known binding or inhibition.

7. There are several ways to find problematic mutations.

The first are links to the Polymorphism and mutational databases at the end of the Pathology & Biotech section. For example, BioMuta lists 420 known single nucleotide polymorphisms of human CYP3A4 and sorts them according to predicted benignity/possible damage and disease from several sources.

Another option is the Feature viewer section in the top left corner, which shows not only consequence-color coded Variants, but also Domains, Structural features, Topology and more over the sequence in graphical form. The final option is the list of Natural variants in Sequence section, which sometimes adds a short annotation of known experimental effects. It also provides links to the specialized genomic Ensembl and dbSNP databases. This option enables the user to gather a simply list of mutations with known effects (Table 9.1).

8. The STRING database provides known protein–protein interactions from several sources: experimental evidence about binding from primary sources, known metabolic pathways from curated databases, text mining for proteins that are

Table 9.1 Mutations with known effects for CYP3A4 gathered from UniProt database

Mutation	Variant	Effect
F189S	CYP3A4*17	Exhibits lower turnover numbers for testosterone and chlorpyrifos
S222P	CYP3A4*2	Exhibits a lower intrinsic clearance toward nifedipine
L293P	CYP3A4*18	Exhibits higher turnover numbers for testosterone and chlorpyrifos
T363M	CYP3A4*11	Unstable form
L373F	CYP3A4*12	Has an altered testosterone hydroxylase activity
P416L	CYP3A4*13	Lack of expression

often mentioned together, co-expression experiments, genomic neighborhood, gene fusions and co-occurrences across genomes. Based on those data, STRING lists Predicted Functional Partners under the Legend button.

For CYP3A4 this list is formed of (i) mainly proteins from the UDP glucuronosyltransferase family, which share the same xenobiotic and endobiotic metabolic pathways; (ii) nuclear receptors familyII, which are known to regulate the expression of CYP3A4 from the literature and (iii) flavin-containing monooxygenase 1, which again share the same metabolic pathways as CYP3A4. Notice that upon filtering in the Data settings to only experimental evidence, only proteins involved in the CYP3A4 ubiquitinylation degradation pathway are shown.

9. The Structure section lists all 24 available crystal structures for CYP3A4. The structure of PDB ID 4d6z has the best resolution. It is predicted to be in a monomeric state.
10. PDBeFold found 88 hits, while the first 24 are CYP3A4 structures that are structurally and sequentially almost identical (above 93 % sequence identity because of gaps). Interestingly, structurally the most similar structure is PDB ID 2ve3, which is cyanobacterial CYP120A1, which shares only 25 % sequence identity with human CYP3A4.
11. The active site can be found around the heme residue (Note: the GO molecular function mentions heme binding)
12. No CYP3A4 structures cover the full length of the sequence (see Fig. 9.4) – when the structure is more disordered, it might not be possible to crystalize it. There are 4 disordered regions predicted within the structure (around residue 200, between residues 250 and 270, between residues 280 and 290 and between residues 410 and 425). An additional region which is not resolved is the N-terminal transmembrane helix (which is also part of the first exon of CYP3A4).
13. PDB Flex found 33 structures within the same cluster as PDB ID 4d6z. The structures are almost identical, but there is large local flexibility between residues 171–200 and 235–252. These are the first two regions that RCSB predicts to be disordered.

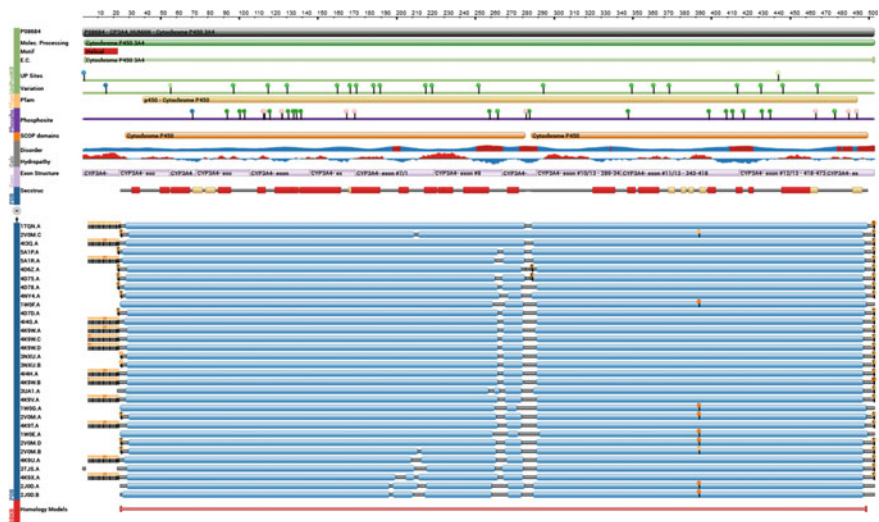


Fig. 9.4 Protein view from RCSB database of CYP3A4 – UNIPROT ID P08684

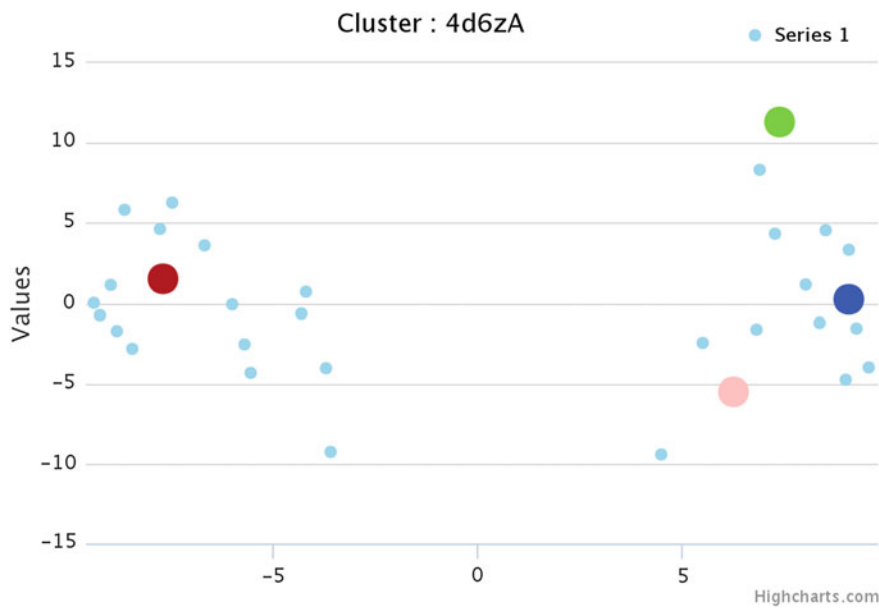


Fig. 9.5 Global flexibility analysis by PDB Flex (color figure online)

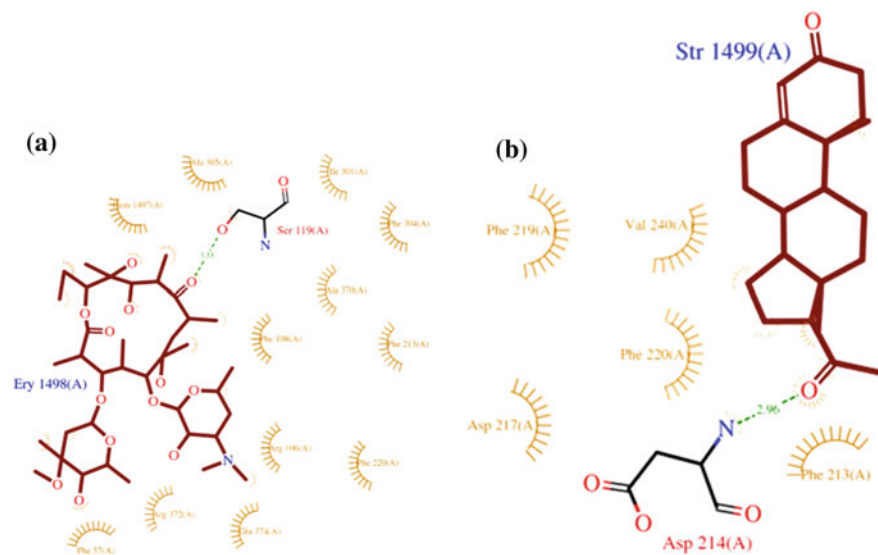


Fig. 9.6 **a** LigPlot view of amino acids interacting with erythromycin (ERY; PDB ID 2j0d). **b** LigPlot view of amino acids interacting with progesterone (STR; PDB ID 1w0f)

14. Based on global flexibility analysis (see Fig. 9.5), the most representative structures are PDB ID 4i4hA (pink), 4k9wB (green), 4d78A (blue), and 2v0mD (red). The final letter denotes the protein chain.

The most distant structures are 1w0fA and 2j0dA.

15. The most distant structures are occupied by erythromycin (2j0d) and progesterone (1w0f).
16. PDBsum shows the surroundings of both ligands on the Ligand page in the Lig-Plot visualization (see Fig. 9.6).

The catalytic residues are listed on the Protein page.

The resulting comparison of residues in all categories can be seen in Table 9.2. As can be seen, both ligands bind in contact with different amino acids, with only F220 in common. Closer inspection of the position in the visualization (e.g. in PDBe or offline) reveals that this is not so surprising, since progesterone is bound to the CYP exterior, while erythromycin is bound within the structure.

17. MOLEonline 2.0 analysis shows (Fig. 9.7) that with the default settings, the channels calculated within those two structures are quite different – the structure containing erythromycin (2j0d) only exhibits a short channel, whereas the progesterone-containing structure (1w0f) contains two relatively long channels. This is quite a common situation, as channels are void spaces which can be influenced by the binding of different molecules.

Table 9.2 Comparison of amino acid residues around ligands with catalytic site residues

Type	Residues
Catalytic site	T309, C442, F435 or E308, T309
Around ERY (PDB ID 2j0d)	F57, R106, F108, S119, F220 , A305, R372, E374
Around STR (PDB ID 1w0f)	F213, D214, D217, F219, F220 , N237, I238, R243

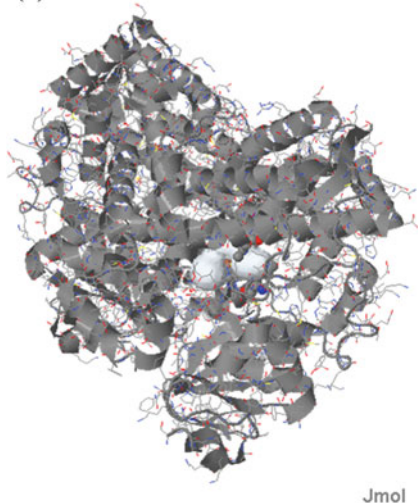
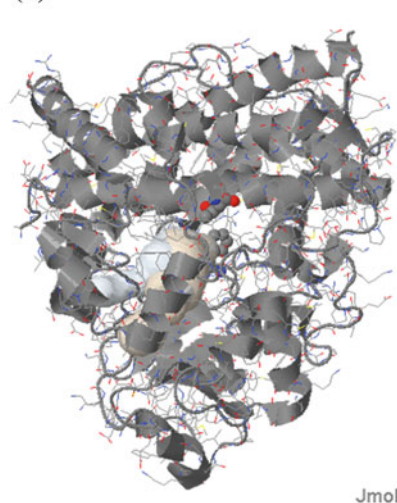
(a) Job review #17067**(b) Job review #17065**

Fig. 9.7 **a** MOLEonline visualization of one channel leading from catalytic site of CYP3A4 structure with erythromycin (ERY; PDB ID 2j0d) with default parameters. **b** MOLEonline visualization of two channels leading from catalytic site of CYP3A4 structure with progesterone (STR; PDB ID 1w0f) with default parameters

18. Enlarging the *ProbeRadius* enabled new channels to be seen within the structure with erythromycin, while there is virtually no change within the structure with progesterone. There is now one overlapping channel – shown in cyan in Fig. 9.8.
19. Indeed, there is one overlapping channel, as can be seen from the table containing the lining (Table 9.3). The channels are however still partly different in that their differences increase as they approach the surface (later in line). They have the same charge, and they are both hydrophilic as can be seen from very low value of hydrophathy and relatively high polarity.
20. From the comparison of the tables of residues with a known mutation effect (Table 9.1) and of lining residues (Table 9.3), it is possible to point to residue L373, whose mutation to phenylalanine leads to an altered testosterone hydroxylase activity. Comparing the table with residues around ligands or catalytic ones finds no common residue.

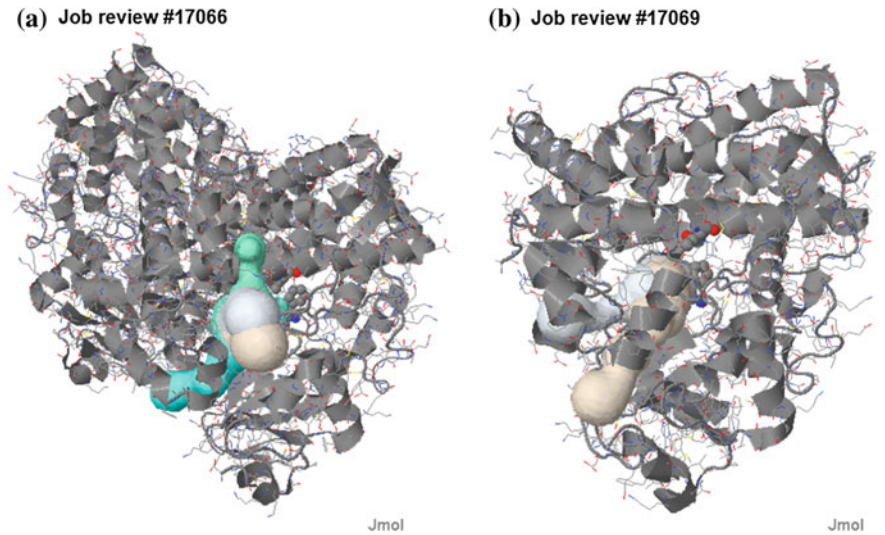


Fig. 9.8 **a** MOLEonline visualization of one channel leading from catalytic site of CYP3A4 structure with erythromycin (ERY; PDB ID 2j0d) with enlarged *ProbeRadius*. **b** MOLEonline visualization of two channels leading from catalytic site of CYP3A4 structure with progesterone (STR; PDB ID 1w0f) with enlarged *ProbeRadius* (color figure online)

Table 9.3 Similar channels from both 3A4 structures. Amino acids shared by both structures are in bold

PDB ID	Lining residues - sidechains	Charge	Hydropathy	Polarity
2j0d #3	435 PHE A, 442 CYS A, 370 ALA A, 373 LEU A, 105 ARG A, 305 ALA A, 482 LEU A, 119 SER A, 108 PHE A, 374 GLU A, 106 ARG A, 57 PHE A, 220 PHE A, 224 THR A, 223 ILE A, 227 PRO A, 230 ILE A, 107 PRO A, 231 PRO A	1	0.1	11.74
1w0f #2	435 PHE A, 442 CYS A, 370 ALA A, 305 ALA A, 105 ARG A, 119 SER A, 304 PHE A, 215 PHE A, 374 GLU A, 108 PHE A, 106 ARG A, 224 THR A, 79 GLN A, 227 PRO A, 78 GLN A, 25 TYR A, 28 HIS A	1	−0.6	14.2

This exercise was intended to show the current capabilities of current databases and online tools for questioning scientific hypotheses connected with macromolecular structures.

Part V

Conclusion

Chapter 10

Concluding Remarks

Jaroslav Koča, Radka Svobodová Vařeková, Lukáš Pravda, Karel Berka, Stanislav Geidl, David Sehnal and Michal Otyepka

At the end of this book, we would like to summarize its purpose and goals.

The purpose of the book is to provide instruction in how to benefit from the richness of the currently available biomacromolecular structural data and understand them via the resources of structural bioinformatics. The main application field of the book is drug design, but the methodologies introduced can be also utilized in other domains such as structural biology, biochemistry, bioinformatics, chemoinformatics, etc.

Here we describe key databases of structural data and the most important steps in their analyses. Specifically, we introduce the validation of biomacromolecular structures, extraction of biomacromolecular structural fragments (bearers of the biomacromolecule's functionality) and their characterization. The book includes practical examples of all the data-processing steps, based on state-of-the-art, online and freely available software tools and databases.

Chapter 11

Exercises Solution

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11.1 Structural Bioinformatics Databases of General Use

1. Na^+/K^+ ATPase

- (a) Upon searching for Na^+/K^+ ATPase, the database returns a list of 43 PDB ID entries sorted by quality. Sorting by Resolution (desc.) gives *PDB ID* 2zxe with a 2.4 Å resolution from *Squalus acanthias* (spiny dogfish).
- (b) The default view or Ligands and Environment view shows that the structure of 2zxe contain several ligands – a MgF_4^{2-} anion mimicking a phosphate group in interaction with a Mg^{2+} cation, 3 K^+ cations for exchange, 1 cholesterol molecule (CLR) exhibiting a probable specific membrane-interaction and 3 sugar moieties (2×NAG and 1×NDG) which are attached to Asn 114 residue in an extracellular part of the sodium-potassium pump on chain B (subunit β , see next answer).
- (c) The Structure Analysis view shows that the structure contains 3 distinct polypeptide chains – chain A is Sodium/potassium-transporting ATPase subunit α , chain B is Sodium/potassium-transporting ATPase subunit β and chain G is a phospholemman-like protein also known as subunit γ or FXD motif protein.
- (d) A closer look at subunit gamma in 1D or 2D visualizations shows that this short 74-amino-acid-long peptide is mostly α -helical.
- (e) The Function and Biology view grabs GO terms associated with 2zxe – its main *biochemical function* is sodium:potassium-exchanging ATPase activity; its main *biological process* is ion transport; and finally its *cellular component* indicates that it is an integral component of the membrane. In addition to this information, the view also lists additional GO terms about other biochemical functions as well as links to more information in other databases such as CATH and UniProt.

2. Larger structures

- (a) The search through the PDBe database gives 8 structures of canine parvovirus viral capsid protein. Three of them are empty (PDB IDs 1c8h, 1p5y, and 2cas). Structure analysis provides a list of possible assemblies for all of them, but in all cases the most preferred assembly is a 60-mer, i.e. 60 canine parvovirus capsid proteins form the whole viral capsid (see Fig. 11.1).
- (b) The entries in PDBe can be filtered by experimental method. Here, Refine query → Experimental methods → Electron microscopy yields 3 structures. The structure with PDB ID 3jcx from the list contains canine parvovirus complexed with Fab E. This structure is the result from EMDB entry EMD-6629. The Volume viewer option shows that the structure of parvovirus is indeed formed of a 60-mer covered with 60 molecules of Fab E light chain and 60 molecules of Fab E heavy chain, as can be seen in the below screenshot of Astex viewer on the page (requires Java). Overall, there is a preferred hetero 180-mer (see Fig. 11.2).

3. Kinase inhibitor example – roscovitine

- (a) Upon searching for the ligand roscovitine, a list of chemical names is shown during typing. If *Find all* is selected, 3 ligands are shown – R6C, RMC and RRC.

From these only RRC is R-roscovitine, while the others are its derivatives. Its structure with annotated atoms can be downloaded and seen here. Note that only some hydrogens are shown (see Fig. 11.3).

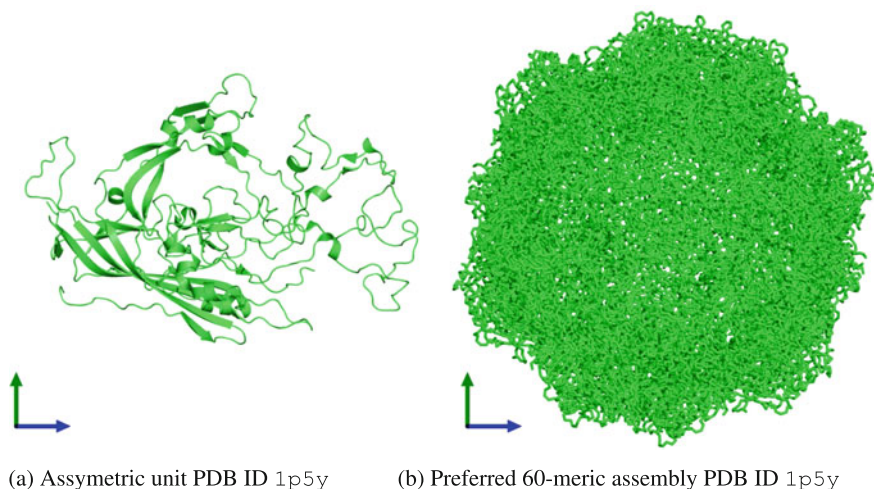


Fig. 11.1 Canine parvovirus capsid protein and the whole assembly

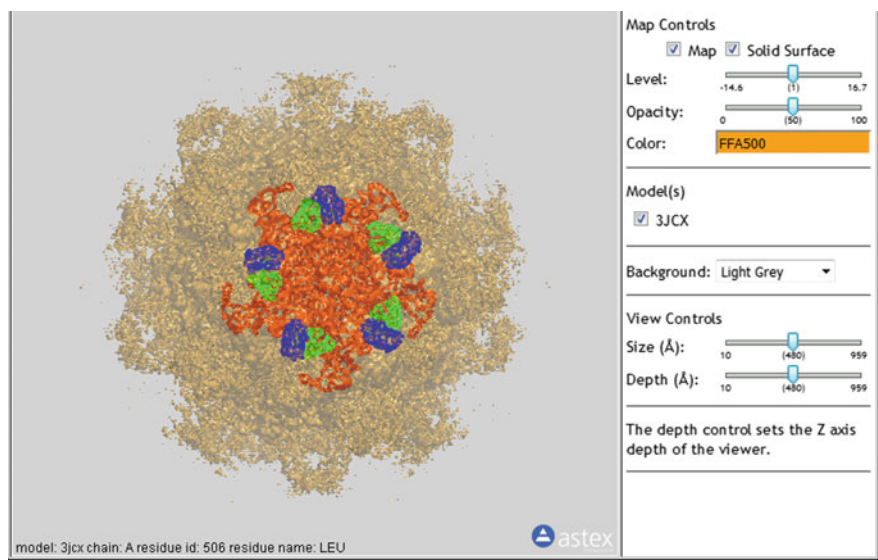
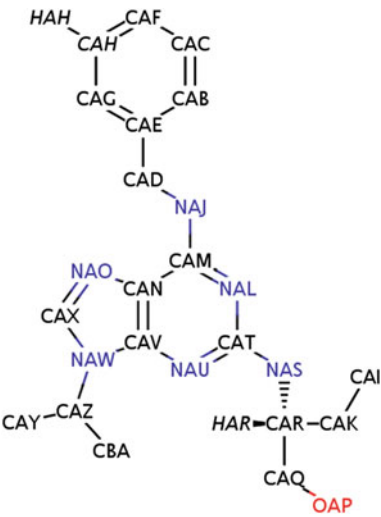


Fig. 11.2 Astex viewer view of 180-mer of canine parvovirus capsid protein complexed with Fab E (PDB ID 3jcx)

Fig. 11.3 2D structure model of roscovitine



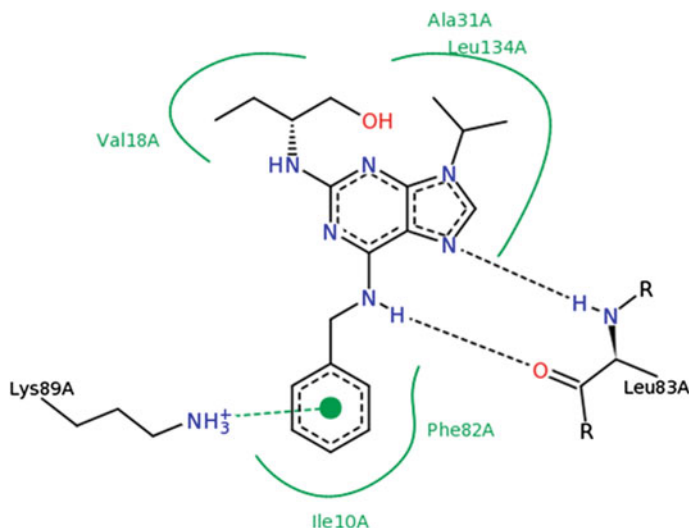


Fig. 11.4 Poseview image of roscovitine in CDK2 binding site (PDB ID 2a41)

(b) *RRC* was found in 4 entries:

PDB ID Proteins	
3ddq	cyclin-dependent kinase 2 + cyclin-A2
2a41	cyclin-dependent kinase 2
1ygk	pyridoxal kinase
1un1	cyclin-dependent kinase 5 + cyclin-dependent kinase 5 activator 1

- (c) PDB ID 2a41 viewed in the Small molecules box shows roscovitine in a Poseview image. From this image it is possible to discern the following interactions: (i) hydrogen bonding of amine and purine nitrogens with backbone of Leu83; (ii) π -cation interaction between the phenyl group and Lys89; and finally (iii) hydrophobic interactions with three distinct cavities – Ile10 with Phe82, Val18 and Ala31 with Leu134 (see Fig. 11.4).
- (d) The small molecules box also shows External Ligand Annotations with binding affinities from external sources – BindingDB, BindingMOAD and PDBbind. For roscovitine it shows values for IC₅₀ ranging from 0.6 nM (BindingDB) to 700 nM (BindingMOAD). Such a large range of IC₅₀ values¹ are relatively common, since IC₅₀ values are sensitive to concentrations of protein and other peculiarities of measurement, and thus in general they are not comparable between individual labs but rather within one paper. K_d

¹IC₅₀ is the concentration leading to a 50% inhibition of reaction - these values are usually the easiest to measure for any inhibitor of a given protein target – an enzyme with a known reaction.

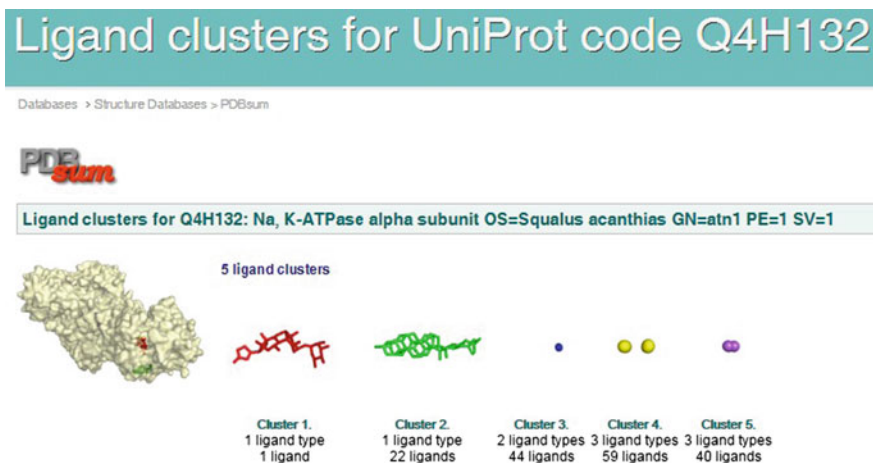


Fig. 11.5 Ligand cluster view on PDBsum for alpha subunit of sodium-potassium pump

measurements² are usually less sensitive, but in this example they seem to have a rather large range of 2900–3400000 nM, however a closer inspection of the BindingDB itself (the values in the RCSB are in the link) shows that there are 3 values in the BindingDB – 2900, 3400 and 3400000 nM, of which the largest value is off by three orders of magnitude, which is suspiciously close to a simple conversion of units from μM to nM. For this reason, it seems that roscovitine binds CDK2 with an affinity of $\sim 3 \mu\text{M}$.

- (e) A simple search in ChEMBL for CDK2 as a target points to several ChEMBL IDs while ChEMBL301 is CDK2 alone. ChEMBL not only stores about 5,757 bioactivities and about 500 assays for this protein target (see, it is a popular one), but it also lists all approved drugs and clinical candidates. None of them have been approved yet (data from December 2015), but two are in the final third clinical phase (alvocidib, dinaciclib), four are in phase two (seliciclib, RG-547, milciclib and (7S)-hydroxylstaurosporine) and six are in phase one (BMS-387032, indirubin sulfate, TG-02, RGB-286638, R-547 and AG-24322).

4. Na^+/K^+ ATPase

- (a) In the Top page view of PDBsum structure PDB ID 2zxe, the ligands icon above the chain leads to the Ligand cluster³ webpage for ligands within structures with the same UNIPROT ID, Q4H132 (α subunit of sodium-potassium pump). There are five individual clusters that show ligands collected from 22 structures with this sequence (see Fig. 11.5).

² K_d is a dissociation constant and it corresponds to the concentration of ligand at which the binding site of a protein is half occupied.

³<https://www.ebi.ac.uk/thornton-srv/databases/cgi-bin/pdbsum/GetPage.pl?pdbcode=2zxe&template=ligclust.html¶m1=Q4H132>.

The first cluster contains the drug ouabain, inhibitor of Na^+/K^+ - ATPase, which is linked to the DrugPort database. The second contains a cholesterol molecule, which is present in all structures at the same position. The third cluster contains 2 ligand types – MgF_4^{2-} and Mg^{2+} - this is the catalytic site of ATP hydrolysis and MgF_4^{2-} is used as mimic of the cleaved phosphate group, whereas Mg^{2+} serves as a reaction catalyzer. The fourth and fifth cluster contain metal cations (K^+ , Tl^+ , Rb^+), which serve as mimics of potassium or sodium cations for crystallization.

- (b) The ligand page lists four individual ligands and four metals. Cholesterol (CLR 3001) is shown not only in 3D or 2D representation, but also in a contact LigPlot diagram similar to the Poseview one in the RCSB database. In addition, PDBsum shows that the cholesterol molecule in PDB ID 2zxe is complete and its chirality is correct.
- (c) The Protein page lists the catalytic residue motifs taken from Catalytic Site Atlas. PDB ID 2zxe contains one catalytic residue – Asp376 on the α subunit. Catalytic residues are also annotated on the secondary structure diagram on the same page.
- (d) On the Protein page, it is possible to select residue conservation on chain B either by selecting the secondary structure listing from the default wiring diagram or there is a link to the sequence's residue conservation in the footnote of the webpage. From the conservation annotation, it is obvious that the least conserved region is between amino acids residues D129 and Q145 (as this is the region with lowest conservation scores marked by blue color – see Fig. 11.6).

It is worth noting that this part of the protein is a turn without much secondary structure.

- (e) The Prot-prot page shows protein-protein interactions and interfaces. The FXYP protein is chain G. As such, it has more protein-protein contacts with the larger alpha subunit (14 res) than with the smaller beta subunit (9 res) (see Fig. 11.7).

The interface between chains G and A is mainly formed from aliphatic amino acids with leucines followed by isoleucines and alanines. The interface between chains G and B is more diverse with a minimal majority of aromatic residues - tyrosines and phenylalanines.

This is reflected in the fact that most of the contacts between the chains are non-bonded (orange links in schematic diagram on the right) with only a small number of hydrogen bonds (blue links). However no inter-protein disulphide bonds are present.

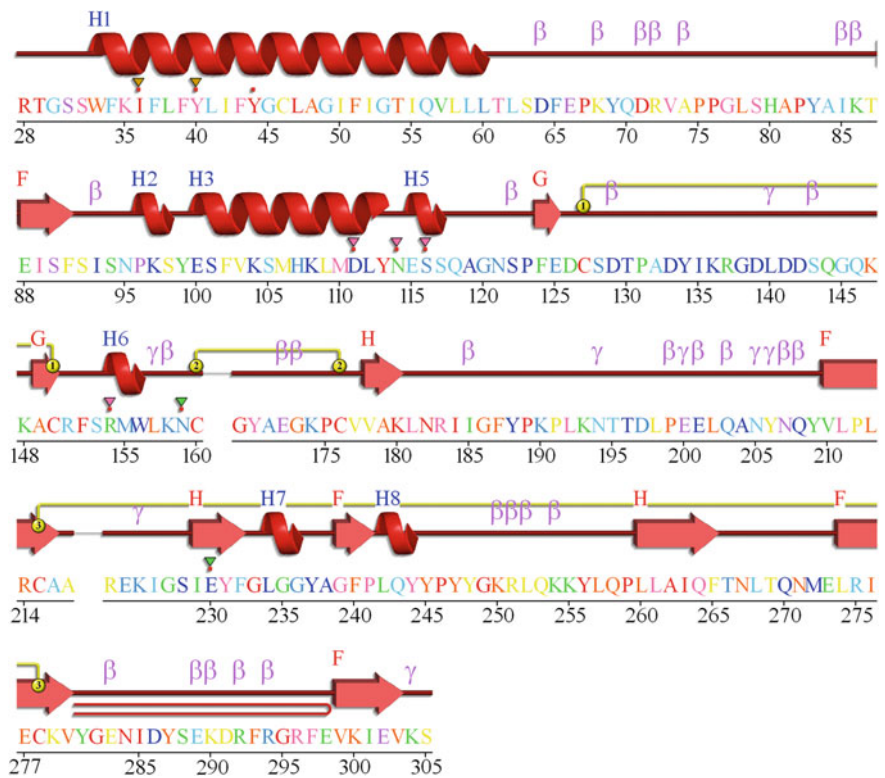
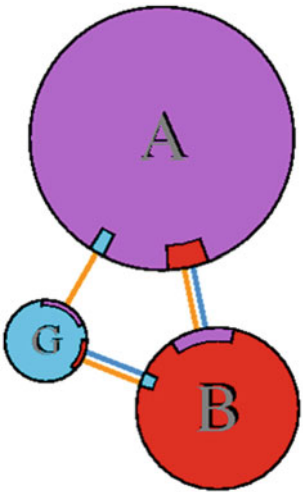


Fig. 11.6 Sequence annotation of beta subunit of sodium-potassium pump. Secondary structure is annotated by numbers above sequence together with secondary structure depiction and sequence conservation is shown by colors of amino acids (color figure online)

Fig. 11.7 Draft of protein-protein interactions within sodium-potassium pump



5. Cytochrome P450 proteins

- (a) Upon submitting cytochrome P450 into the keyword search, CATH lists 8 superfamilies, whereas superfamily 1.10.630.10 is annotated cytochrome P450 itself, other superfamilies either interact with cytochromes P450 (1.20.990.10, NADPH-cytochrome p450 Reductase) or regulate their expression (2.40.30.10 Translation factors).
- (b) The *Classification/Domains* page not only shows clusters of similar sequences, but it also gives the CATH code for cytochrome P450.

Class 1 means mainly the alpha secondary structure is present. Architecture 1.10 code orthogonal bundle tertiary structure Topology 1.10.630 code directly cytochrome P450 fold Homologous Superfamily 1.10.630.10 signifies that this topology has only cytochromes P450, which can be checked on the CATH-Gene3D Hierarchy webpage for this entry.

- (c) GO diversity can be either viewed in graphical form on the *Summary page* or listed in the Functional annotations. Cytochromes P450 are mainly annotated with a heme-binding molecular function (GO:0020037), iron ion binding (GO:0005506) and monooxygenase activity (GO:0004497), xenobiotic metabolic biological process (GO:0006805) and finally to be present mainly within the Endoplasmic reticulum membrane (GO:0005789)
- (d) EC annotation can be also seen on the Summary page or in Functional annotations. The most typical reaction is Unspecific monooxygenase. [EC: 1.14.14.1]:



This reaction acts on a wide range of substrates including many xenobiotics, steroids, fatty acids, vitamins and prostaglandins, whereas catalyzed reactions include hydroxylation, epoxidation, N-oxidation, sulfoxidation, N-, S- and O-dealkylations, desulfation, deamination, and the reduction of azo, nitro, and N-oxide groups.

- (e) The Gene3D button leads to the Gene3D database, which links CATH domains to genomes. Domain frequency show that the cytochrome P450 domain is quite popular among all known genomes, as it is found in more genes than 92.2 % of other domain families. It is especially popular among plants, where this domain's frequency is found in the 99.1 % percentile of all domain families.
- (f) The Summary page shows its structural diversity, where we can identify not only that the smallest domain has just 366 amino acids (PDB ID 1io7) and the largest has 480 amino acids (PDB ID 2hi4), but also 42 representative members of this family, which can be downloaded from the Superfamily *superposition* webpage.

11.2 Validation

1. Demo exercise: Detecting problematic ligands in nipah G attachment glycoprotein

Open ValidatorDB, use: Search → PDB Entry → 3d12 → Quick Search. Browse the bookmarks *Overview*, *Summary* and *Details* to obtain all the required information.

Solution:

- From Overview:
 - 3d12 contains 30 ligands
 - all of them are complete (no missing atoms)
 - 13 of them have at least one chirality issue (in all cases it is a chirality at a carbon atom)
- From Summary:
 - 3d12 contains 11 types of ligands
 - 7 of them contains chirality errors: BGC, GL0, GLC, GXL, NAG, NGA and NGZ
- From Details:
 - BGC: 2 of 5 molecules have wrong chirality (both at C5 atom)
 - GL0: 2 of 2 molecules have wrong chirality (both at C1 atom)
 - GLC: 3 of 3 molecules have wrong chirality (all at C1, C2, C3, C4 and C5 atoms)
 - GXL: 1 of 1 molecule has wrong chirality (at C4 atom)
 - NAG: 3 of 10 molecules have wrong chirality (NAG 1314 A and NAG 1414 D at C1 atom; NAG 1413 D at C1 and C2 atom)
 - NGA: 1 of 1 molecule has wrong chirality (at C1, C2, C3, C4 and C5 atoms)
 - NGZ: 1 of 1 molecule has wrong chirality (at C1 atom)

2. Validation of sucrose (SUC) in Plant Photosystem I – missing atoms

The missing atom is O6 (Fig. 11.8).

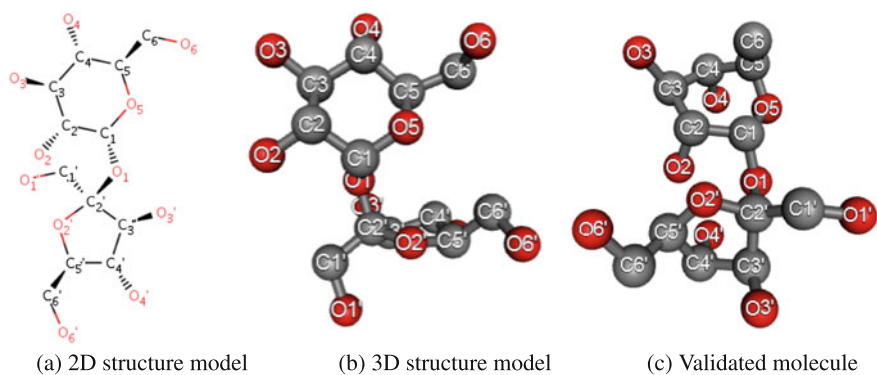
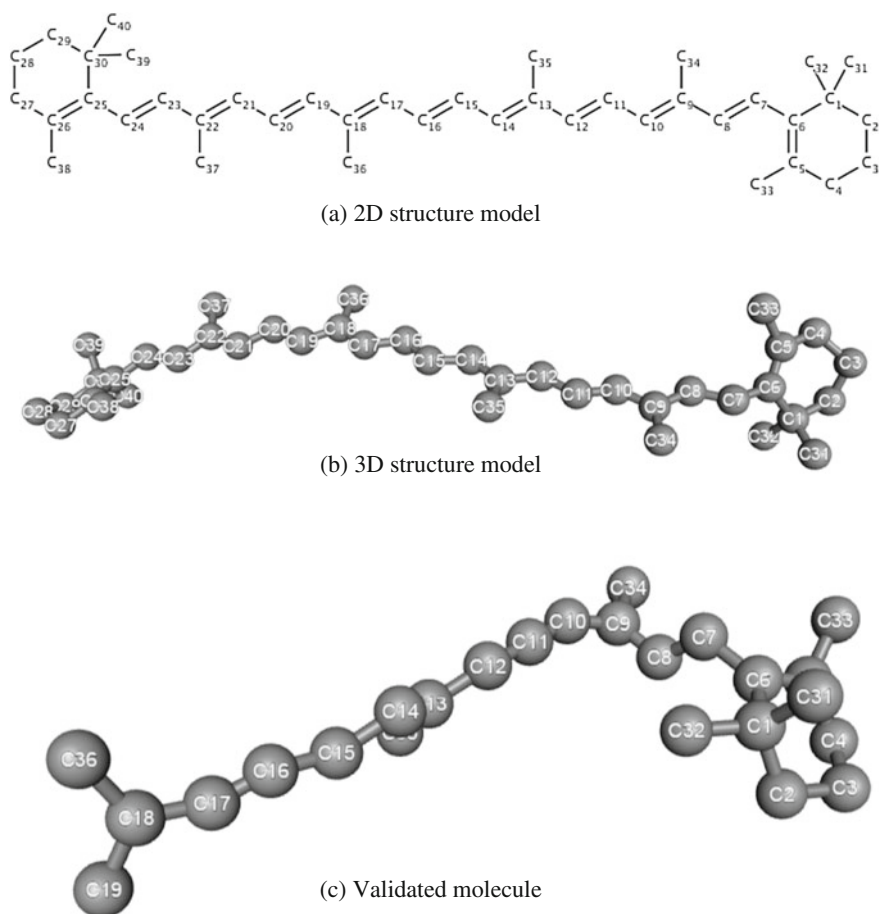
3. Validation of retinal (BCR) in Plant Photosystem I – missing rings

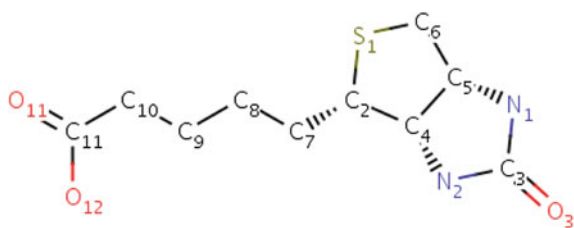
The missing ring atoms are C25–C30 (Fig. 11.9).

4. Demo exercise: Validation of maltose (MAL) ligands in Protein Data Bank – missing atoms and rings

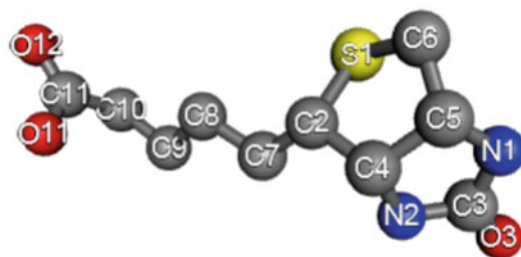
Open ValidatorDB, use: Search → Molecule Annotation → MAL → Quick Search. Browse the bookmarks *Overview*, *Summary* and *Details* to obtain all the required information.

Currently (August 2016), PDB contains 366 maltose ligands, 4 of them have missing atoms and 1 of them has missing rings (Fig. 11.10).

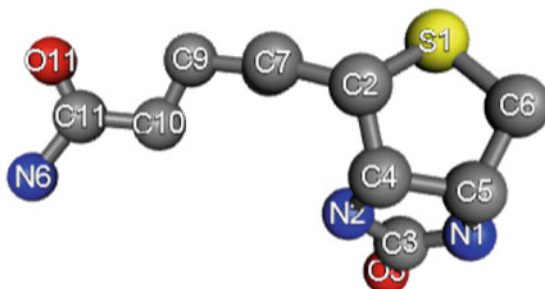
**Fig. 11.8** Sucrose validation**Fig. 11.9** Retinal validation



(a) 2D structure model



(b) 3D structure model



(c) Validated molecule

Fig. 11.10 Maltose validation

5. Detection of atom substitution in biotin (**BTN**) from 50S Complex (PDB ID **1kqs**)

The atom O12 is substituted with the N6 atom, originating from the connected 6-aminohexanoic acid (residuum ACA 78).

6. Detection of chirality problems in all sialic acids (**SIA**) from Protein Data Bank

Sialic acid occurs in 932 ligands from 344 PDB entries (information from August 2016). 13 of the molecules contain chirality errors, 11 chirality errors are at C2 and 2 at C8.

7. Demo exercise: Detection of chirality problems in testosterone derivatives

Open ValidatorDB, use: Search → Molecule Annotation → TH2, TES, FFA, DHT, BDT → Quick Search. Browse the bookmarks *Overview*, *Summary* and *Details* to obtain all the required information.

There are 2 samples of testosterone derivatives which contain chirality errors: residue DHT 201 from 1i37 (rat androgen receptor ligand binding domain) and residue DHT 201 from 1i38 (rat androgen receptor ligand binding domain mutant), both have a chirality error at C5.

8. Demo exercise: Obtaining structure validation data for a rhodostomin

Open Protein Data Bank Europe → 4rqq → click on the entry found → Experiments and Validation → Full validation report

- How many clashes are in the structure? This is given in the section *Too close contacts*. 8 pairs of atoms are too close.
- Which atoms in a clash are the closest? This is given in the same section (there is a table describing all the clashes). The closest atoms are HG2 (from GLU 3) and H (from CYS 4) in chain A.
- How many bond length outliers are in the structure? This is given in the section *Standard geometry*. 8 bonds are outliers.
- Which bond length outlier is the highest? This is given in the same section (there is a table describing all the bond length outliers). It is the bond between CB and CG (difference 0.15 Å from an ideal structure) in ASP 51 in chain B.
- How many bond angle outliers are in the structure? This is given in the section *Standard geometry*. 4 bond angles are outliers.
- Which bond angle outlier is the highest? This is given in the same section (there is a table describing all the bond angle outliers). It is the angle CA-C-N (a difference of more than 11° from the ideal structure) in ARG 46 in chain A.

9. Obtaining structure validation data for a mutant of cytochrome P450cam

- How many clashes are in the structure? 793 pairs of atoms are too close.
- Are there some atoms closer than 1 Å? Yes, there are atoms closer than 1 Å. For example the atoms HH21 (from ARG 277) and HH21 (from ARG 280) in chain C.
- How many bond length outliers are in the structure? The structure contains 567 bond length outliers.
- Are there any bond length outliers >0.3 Å? Yes, for example the atoms CB and CG2 in VAL 399 in chain C.
- How many bond angle outliers are in the structure? The structure contains 353 bond angle outliers.
- Which bond angle outlier >20°? For example the angle C-N-CD in ARG 143 in chain C.

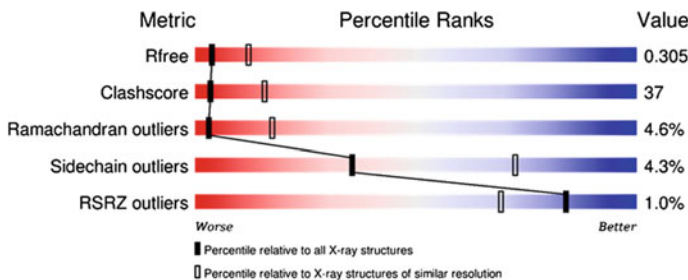


Fig. 11.11 Validation report for PDB entry 1l1fz

10. Demo exercise: Understanding PDB validation report summary criteria for oxy-hemoglobine in methanol

Open Protein Data Bank Europe → 1l1fz → click on the entry found → Experiments and Validation → you will see the following summary criteria (Fig. 11.11).

A description of the summary criteria can be found here: http://wwpdb.org/validation/legacy/XrayValidationReportHelp#overall_quality.

11.3 Detection and Extraction of Fragments

11.3.1 PatternQuery

1. Glycine residues in the tubulin protein (PDB ID 1tub)

- `Residues("Gly")` - all 68 glycine residues
- `Residues("Gly").Inside(Sheets())` - 7 residues inside β -sheets
- `Residues("Gly").Inside(Helices())` - 14 residues inside α -helices

In total 47 glycine residues are not present in any type of secondary structure.

2. All experimental binding sites of the doxycycline antibiotic

Constructing the following query is straightforward and the result is:

```
1 Residues("DXT").AmbientResidues(4)
```

We simply include any residue which is at most 4 Å from any DXT atom. The identification can be sped up by processing only the PDB entries containing a DXT residue, either selected from the Protein Data Bank, or by applying restriction criteria with PQ. The testing set contains 2 different patterns in 2 PDB entries, which is also the overall number of these patterns found in the PDB.

3. Rubredoxin active site

Locate the iron ion together with its surrounding amino acids and verify that it contains exactly 4 cysteine residues. The length of the Fe-S bond is usually around 2.3 Å, however in order to cover certain structural discrepancies we will overestimate the length to 2.5 Å.

```
1 Atoms("Fe")
2 .AmbientResidues(2.5)
3 .Filter(lambda m: m.Count(Residues("Cys")) == 4)
```

The identification can be sped up by only processing the PDB entries containing an Fe ion, either selected from the Protein Data Bank, or by applying restriction criteria with PQ. The testing set contains 5 different patterns in 5 PDB entries. As of August 2016, the entire Protein Data Bank contains 142 patterns in 93 PDB entries as well.

To evaluate the structural conservation of the pattern, load them all in the Site-Binder application and leave the default settings for the alignment. Using Site-Binder, it has been determined that the overall RMSD is 0.49 Å with the majority (116/142) overall RMSD within the range of a single standard deviation. Overall, the rubredoxin binding site is structurally well conserved with the cysteine residues arranged in an almost ideal tetrahedron.

4. Query constructions – identify following patterns

a)

```
1 Atoms("Fe")
2 .Inside(Residues("HEM"))
3 .AmbientResidues(2.5)
```

The testing set contains 22 patterns in 5 PDB entries.

b)

```
1 HetResidues()
2 .Filter(lambda m: m.IsNotConnectedTo(AminoAcids()))
```

The testing set contains 208 patterns in 32 PDB entries.

c)

```
1 ModifiedResidues()
2 .Filter(lambda m: m.Count(Atoms("P")) > 0)
```

The testing set contains 21 patterns in 6 PDB entries.

d)

```

1 Atoms("Pt")
2   .ConnectedResidues(0)
3   .ConnectedResidues(1)

```

The testing set contains 40 patterns in 9 PDB entries.

5. **HET** residues in the vicinity of modified residues

```

1 HetResidues().Inside(ModifiedResidues().AmbientResidues(5))

```

As of August 2016, there are approximately 72,000 ligands, of which the most abundant is N-Acetyl- β -D-glucosamine (NAG), with 21,000 instances originating from 4k individual structures. Approximately 1,000 ligands are incomplete and 2,000 contain a chirality issue.

The count of the most represented ligand can be done after downloading the results in textual form, either with the spreadsheet editor of your choice, or with a simple single-purpose script. In order to evaluate the chirality issue, click on the drop-down list, which is located on the results page in the pattern details tab next to the pattern id list filter. Here you can filter the whole view based on structural discrepancies. The number and type of discrepancy located in the patterns are available there.

6. Manganese ion in a single residue

```

1 Cluster(2, Atoms("Mn"), Atoms("O"))
2   .ConnectedResidues(0)
3   .Filter(lambda m: m.Count(Residues()) == 1)

```

We have to use the `Cluster()` query, rather than the `Near()` query. The `Near()` query checks that there is exactly one occurrence of the defined patterns (manganese and oxygene ion) within the given range of 2 Å. When the `Near()` query is used, the OEC residue is completely missed. First, we create a tuple of Mn and oxygen atoms which are at most 2 Å away, next we add all the residues which are covalently connected to these atoms and verify that the subsequent pattern has exactly one residue.

The PDB archive currently (August 2016) contains the OEC, OEX, and OER residues.

7. RNA k-turn motif and its AAs surrounding

Find in the literature the regular expression identifying the individual parts of the nucleotide chain and encode them into the following query:

```

1 Near(4,
2   RegularMotifs('GGCGAGAC', Type='Nucleotide'),
3   RegularMotifs('GGGAGCC', Type='Nucleotide'))
4   .AmbientResidues(3)

```

The testing set contains 1 pattern in the 4bw0 PDB entry. As of August 2016, the entire Protein Data Bank contains 79 patterns in 75 PDB entries.

8. Classical C₂H₂ zinc finger binding motif

The primary sequence of the motif can be expressed as the regular expression:

```
1 ".{2}C.{2,4}C.{3}[FIY].{5}[AIFPGV].{2}H.{3,5}H"
```

This structure is stabilized by a zinc ion, which forms covalent bonds with two cysteine and two histidine residues. First, it is important to identify the primary sequence motif together with the stabilizing zinc ion and then verify that the zinc is bound by 2×Cys and 2×His residues.

```
1 RegularMotifs(".{2}C.{2,4}C.{3}[FIY].{5}[AIFPGV].{2}H.{3,5}H")
2 .ConnectedAtoms(1)
3 .Filter(lambda m:
4     m.Find(Atoms("Zn"))
5     .AmbientResidues(3)
6     .Filter(lambda n:
7         (n.Count(Residues("Cys")) == 2) &
8         (n.Count(Residues("His")) == 2)))
9     .SeqCount(>0)
```

The logic behind constructing this advanced query is as follows. First, the regular motif using the `RegularMotifs()` query is identified. Second, this pattern is extended by all the atoms connected to the former. Next, the filtering process starts. In the identified pattern, we want to detect a sub pattern - a zinc ion together with all the residues which are at most 3 Å away. Subsequently, it is verified that there are exactly two cysteine and two histidine residues. Finally, we verify that there is at least a single sub pattern.

The identification can be sped up by only processing the PDB entries containing a Zn ion, either selected from the Protein Data Bank, or by applying restriction criteria with PQ. The testing set contains 28 different patterns in 11 PDB entries. As of August 2016, the entire Protein Data Bank contains 381 patterns in 245 PDB entries.

9. C₂H₂ zinc finger binding motif with the zinc substitution

The following query is very similar to the previous example. We can slightly modify the former solution in order to not detect the stabilizing zinc ion, while maintaining everything else. On top of that we will modify the `RegularMotif()` query. The result is the following:

```
1 RegularMotifs(".{2}C.{2,4}C.{3}[FIY].{5}[AIFPGV].{2}H.{3,5}H")
2 .ConnectedAtoms(1)
3 .Filter(lambda m:
4     m.Find(NotAtoms("C", "H", "N", "O", "P", "S", "Zn")).
5     .AmbientResidues(3)
6     .Filter(lambda n:
```

```

7      (n.Count(Residues("Cys")) == 2 ) &
8      (n.Count(Residues("His")) == 2 )))
9      .SeqCount(>0)

```

The only difference is that we are not specifically searching for Zn ions. Instead, we are interested in anything apart from zinc ions and biogenic atoms.

The test set reveals 2 such substitutions 2g1i – where zinc is substituted with cobalt, whilst 2l1o has a cadmium substitution instead.

10. *Pseudomonas aeruginosa* LecB sugar binding site

We can encode the information obtained so far into the first part of the query, i.e. a unique pair of calcium ions at most 4 Å apart. Then all the residues connected to this tuple. The first filtering function filters out residues not possessing a carbohydrate moiety, while the second ensures that nucleotides are discarded as well.

```

1  Near(4, Atoms("Ca"), Atoms("Ca"))
2  .ConnectedResidues(1)
3  .Filter(lambda m:
4      m.Count(Or(Rings(5 * ["C"] + ["O"]),
5                  Rings(4 * ["C"] + ["O"]))) > 0)
6  .Filter(lambda m: m.Count(Atoms("P")) == 0)

```

The identification can be speed up by only processing PDB entries containing a calcium ion and pyran or furan ring, by applying restriction criteria with PQ. The testing set contains 37 different patterns in 10 PDB entries. As of August 2016, there are 158 patterns in 49 entries in the entire Protein Data Bank. These binding sites are located in a variety of pathogens and their usual composition is 3×Asp, 2×Asn, Glu and Gly residues.

11.3.2 *MetaPocket*

1. HIV-protease

The most prominent binding pocket identified by this consensus classifier is the one hosting an inhibitor in the structure and is visualized below (Fig. 11.12). The determined binding pocket correctly contains the catalytic triad as well as other substrate-stabilizing residues. The other two putative binding pockets are in the regions presumably responsible for the protein's flexibility.

2. COX-2 protein binding site

At first, we will have to restrict the search to just a single chain, e.g. A, as the protein complex is very large for this service. However, even after the results are retrieved, neither of the pockets seem to correspond to the naproxen – anti-inflammatory drug (NPS) ligand we want to identify. This is caused by the fact

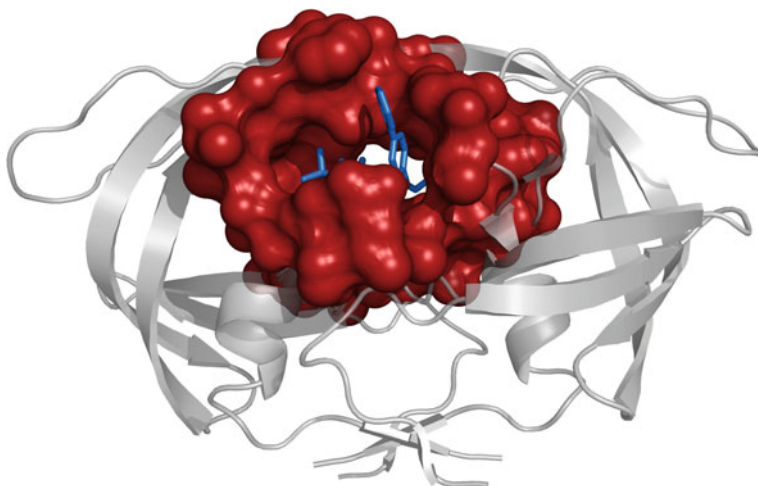


Fig. 11.12 Visualization of the ligand binding pocket of the HIV-1 protease inhibitor

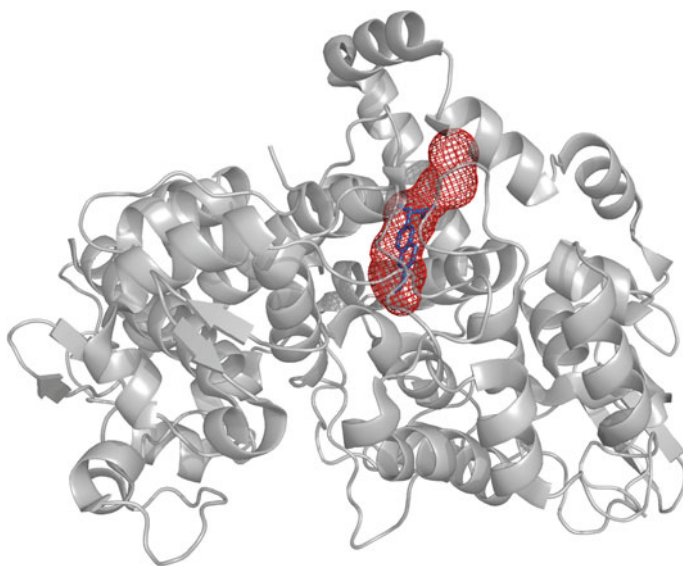


Fig. 11.13 Possible accessible channel visualized for the anti-inflammatory drug naproxen in the COX-2 structure (PDB ID 3n11)

that the ligand is buried deep below the surface of the protein, and therefore the majority of the pocket-predicting services will fail to identify the binding site properly.

In these situations, it is often useful to try to calculate channels leading to the buried ligand in the structure, as shown in Fig. 11.13. The available algorithms

and services are thoroughly described in Chaps. 6 and 8. Using PatternQuery we can easily identify residues Arg 120 and Tyr 355 as playing a major role in naproxen stabilization, as both of them are close enough ($< 3 \text{ \AA}$) to create favorable hydrogen bonds with the substrate.

3. Apicoplast DNA polymerase

Since the 5dkt structure is composed of polymerase as well as exonuclease domains, we need to detect at least 5 binding sites, due to the fact that some of them are identified for the polymerase domain. In total 3 out of the 5 binding sites identified for this protein are within the exonuclease domain – Fig. 11.14.

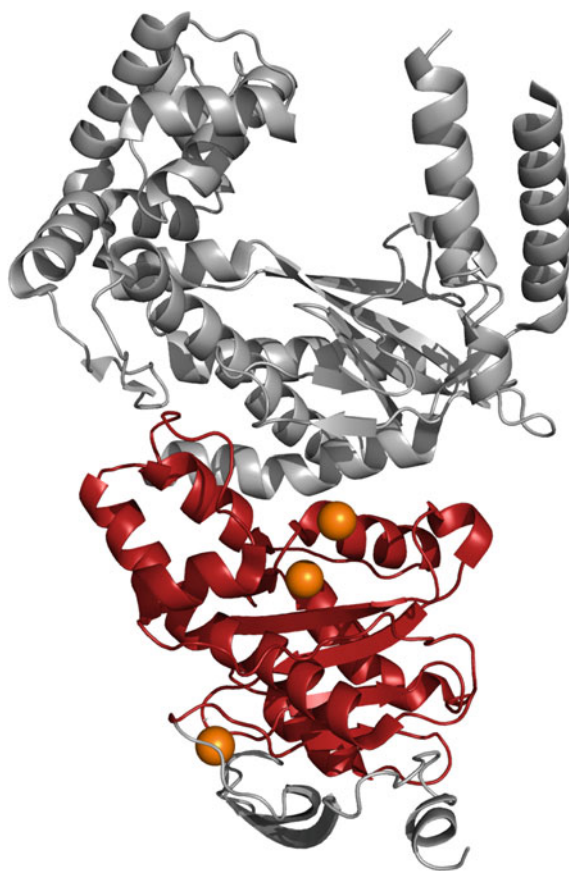


Fig. 11.14 Apicoplast DNA polymerase PDB ID 5dkt with the exonuclease domain highlighted in dark red and the centers of mass for the binding sites identified with MetaPocket as orange spheres (color figure online)

The first two of them seems to be the most promising. Both of them are close to each other and, moreover, to the ion in the structure. The ion coordinates the phosphodiester backbone of DNA in the otherwise electropositive pocket, which compensates for the negative charge of the DNA backbone [3].

4. Detection of caffeine binding site

Since PatternQuery can't detect the binding pocket a priori from the PDB ID. First, we need to *define* the binding pocket and compose the appropriate query. We will consider the binding pocket as all the residues within a radius of 5 Å from the caffeine ligand. Next, we can compose our query - `Residues("CFF").AmbientResidues(5)`. The binding pattern contains just the 11 closest residues responsible for caffeine binding, while MetaPocket returns a larger portion of the structure, since the binding site is composed of 66 residues in total.

Clearly PatternQuery returned just the residues directly involved in the residue binding, including Asn 253 and Ile 274, which are crucial [2], while MetaPocket provides the entire pocket and the majority of the residues will be irrelevant for the residue binding. To conclude, MetaPocket provided the entire hydrophobic cleft, where receptor inhibitors can bind, while PatternQuery restricted the area only to the residues with a direct contact with the caffeine inhibitor (Fig. 11.15).

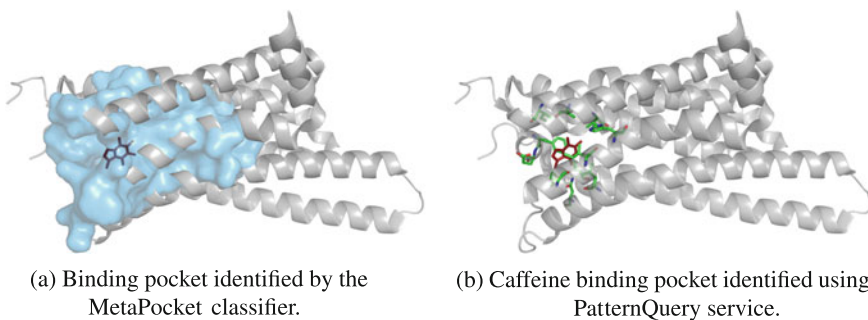


Fig. 11.15 Comparison of caffeine binding site pockets for the adenosine receptor (PDB ID 3rfm). MetaPocket versus PatternQuery. PatternQuery binding pocket is defined as all the residues within the radius of 5 Å from the caffeine residue. The query used for identification is the following: `Residues("CFF").AmbientResidues(5)`

11.4 Detection of Channels

1. Gramicidin D channel

The pore is roughly 30 Å in length with its narrowest sites located close to the channel endings (~1.3 Å wide).

2. *ProbeRadius* and *InteriorThreshold* settings for channel identification

- (a) In order to only identify the transmembrane region of the pore, a *ProbeRadius* value of 5 is sufficient. Otherwise values of over 9 have to be provided. *InteriorThreshold* can be kept to the default value. The resulting length of the pore will highly vary, depending on the length of extracellular region involved in the calculation. The pore spanning through the entire structure has a length of around 150 Å.
- (b) *ProbeRadius* has to be slightly elevated (values over 5 are sufficient), whereas *InteriorThreshold* should be reduced to 1.1.
- (c) The default values are sufficient, however one filtering criterion needs to be diminished (*BottleneckRadius* set to 0.8).

3. Channel starting point

There are a number of best practices for identifying a rational starting point.

- (a) Channels leading to a buried volume of a protein body usually traffic a number of different compounds to and from the active site. On the one hand, you can go to the literature and identify the composition of an active site, on the other hand you can use specialized databases, such as the CSA database for inferring catalytic residues. Finally, ligands are usually either stuck inside an active site, or in a channel during the structure determination experiment. Therefore, they can also be a logical starting point for channel identification.

When none of the above work, MOLE can provide an automatic starting point from the deepest spots inside the cavities, hence providing a potential clue about the location of active sites.

- (b) For identifying transmembrane pores, it is often the best practice to let MOLE automatically identify pores and then select the relevant one. Another approach involves identifying a 3D point inside the transmembrane pore region and starting the calculation there.

4. Polyamine oxidase binding channel

First, it is necessary to only use biological assembly 1 for the calculation, as the asymmetric unit contains multiple biological assemblies. Next, all the HET atoms should be discarded prior to the calculation. In accordance with the paper, the channel starting point was set to the position of the N5H atom of the FAD

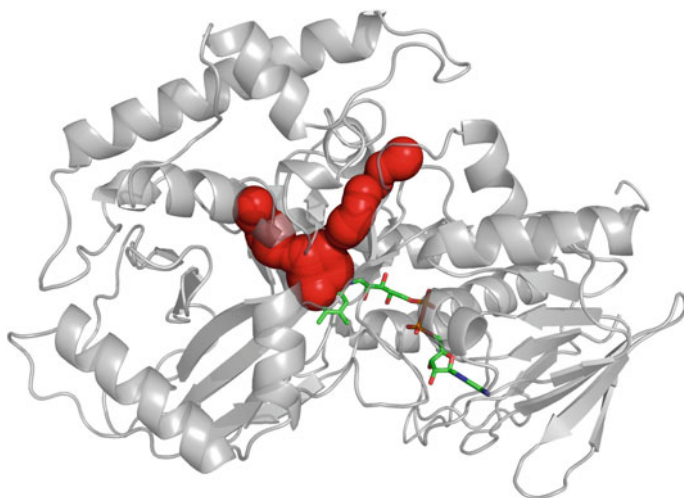


Fig. 11.16 Visualization of U-shaped channel binding tunnel

800 residue [132.437, 56.146, -7.436]. Parameters for *ProbeRadius* and *InteriorThreshold* were set to their defaults. The detected channel is visualized in Fig. 11.16.

5. Polypeptide channel in ribosome

Depending on your setup, a number of channels can be identified. However, the one considered to be a polypeptide exit channel is the widest one. Your result should be similar to Fig. 11.17

11.5 Characterization via Charges

1. Demo exercise: Detection of the first dissociating hydrogen in 3-hydroxybenzoic acid

Download the 3D structure of 3-hydroxybenzoic acid (in SDF format) from PubChem. It has PubChem CID 7420 and its name is Structure3D_CID_7420.sdf. Open ACC → Submit a Computation → Select file → Structure3D_CID_7420.sdf → Upload. In the *EEM Parameter Sets* click on *more* and select *Bult2002_mpa* parameters and then click on *Compute*. Click on *Structure3D_CID_7420* and then on *3D model*. To make the charge differences more visible, check *Min. Value* and *Max. Value* to set the colors from the minimum to maximum charge value in the structure. When you hover the mouse over the hydrogens from the

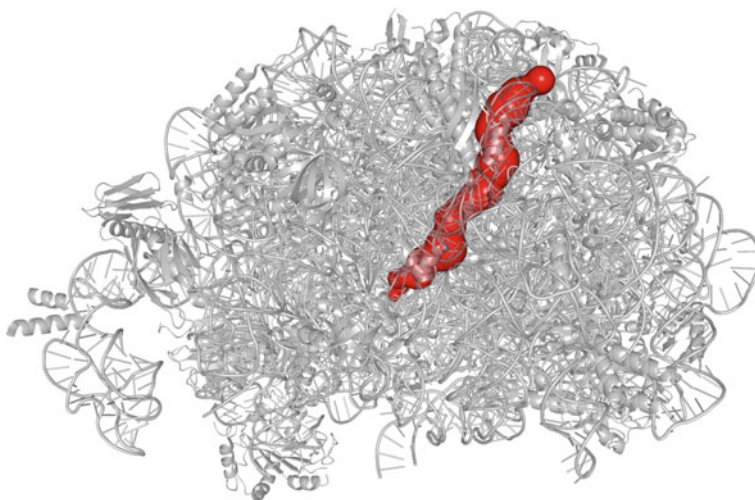


Fig. 11.17 Polypeptide exit channel for the 1jj2 system. In order to identify the full length of the channel, a larger *ProbeRadius* has to be used (e.g. >5)

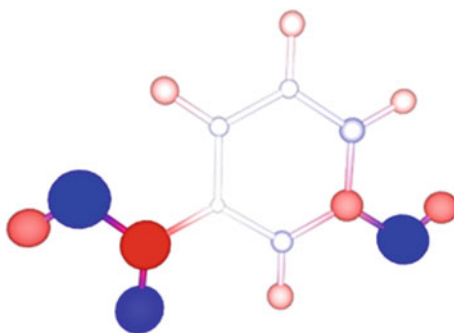
COOH and OH group, you will also see the particular values of their charges. The charge on the H (H 16) from the COOH is 0.33, the charge on the H (H 15) from the OH is 0.3. Therefore the H from COOH has a higher charge (see also Fig. 11.18) and should dissociate first. This agrees with organic chemistry findings.

2. Comparison of charges in phenol molecules and detection of correlation between charges and pK_a

There are the following trends between charges and pK_a (Fig. 11.19).

- The lower the pK_a , the higher the positive charge at H
- The higher the pK_a , the more the negative charge at O

Fig. 11.18 Charge distribution on 3-hydroxybenzoic acid. The H from the COOH group is the most positively charged H atom in the molecule. *Note* The color scale is from -0.6 (blue) to 0.4 (red) (color figure online)



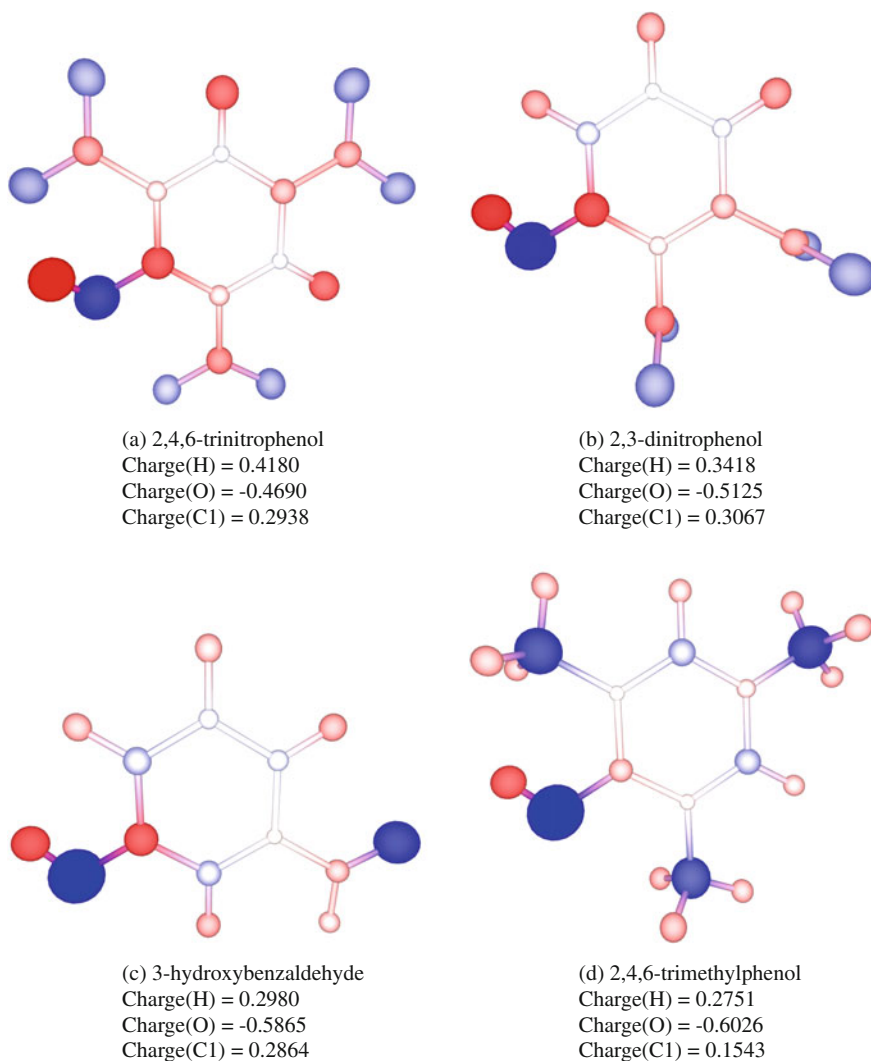


Fig. 11.19 Comparison of charges in phenol molecules. *Note* The color scale is from -0.6 (blue) to 0.4 (red) (color figure online)

- The lower the pK_a , the higher the positive charge at C1 (weak trend)
- Other atoms have no relation with pK_a .

3. Comparison of charge distributions in cocaine binding sites

Both binding sites are similar from a charge point of view – neutral or slightly negatively charged (Fig. 11.20).

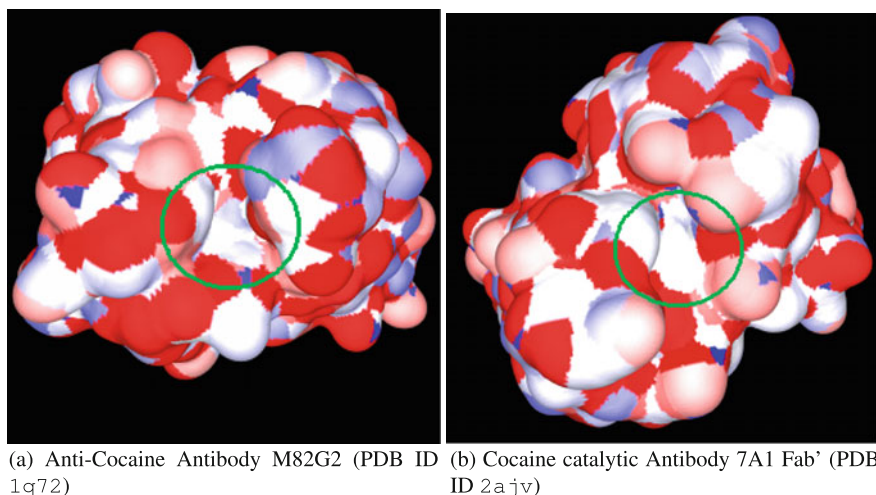


Fig. 11.20 Comparison of charge distributions in cocaine binding sites

4. Comparison of charge distribution in activated and inhibited apoptotic proteins

- Inactive BAX with and without the inhibitor exhibit a very similar charge distribution
- Activation of BAX is performed via binding a strongly charged activator
- The binding of this activator causes a vanishing of charge in Helices 1 and 5 (= white in the figure of activated BAX).
- Afterwards, the ends of the C domain depolarize (they became neutral = they are white in the figure of activated BAX).
- This causes a release of the C domain after the activation of BAX.

The scheme is highlighted in Fig. 11.21.

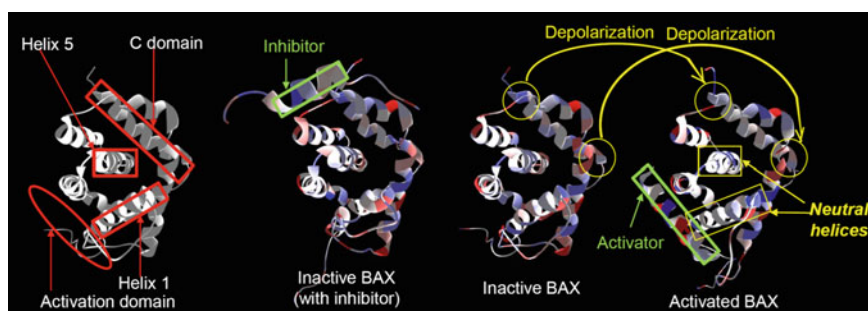


Fig. 11.21 Comparison of charge distribution in activated and inhibited apoptotic proteins

11.6 Channel Characteristics

1. Gramicidin D pore

The hydrophathy index of the channel is -0.4 . This value shifted towards the negative range of spectra suggests that the channel is relatively hydrophilic and in terms of its size allows the permeation of ions.

2. Cytochrome P450 3A4

Using the default settings of the service, there are 3 channels leading to the active site, which is located close to the hemoglobin prosthetic group and is formed by residues Glu 308 and Thr 309. The water channel is the shortest one with a Hydrophathy index of -2 , as shown in Fig. 11.22.

3. Substrate tunnel

In order to identify the access channel, it is crucial to discard the posaconazole residue prior to the calculation, while keeping the heme cofactor in the structure. Afterwards, a channel that is hydrophobic in its behavior can be identified.

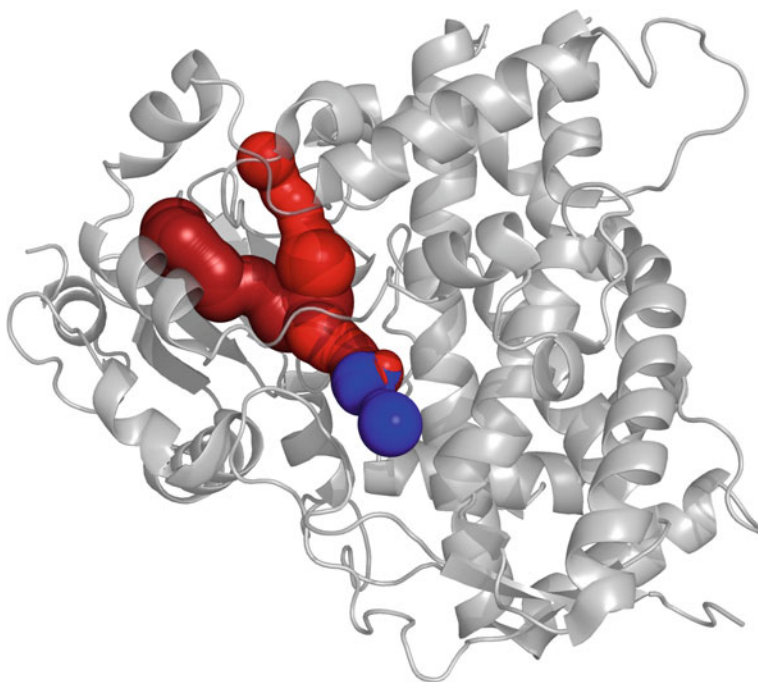


Fig. 11.22 Results of channel analysis of Cytochrome P450 3A4 (CYP3A4). Three channels found from a user-specified starting point (calculation started from Glu 308 and Thr 309 according to the CSA) are shown – the solvent channel is in blue [1] (color figure online)

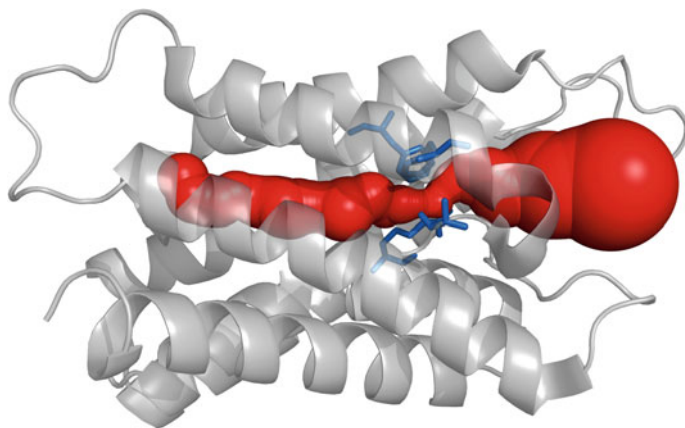


Fig. 11.23 The aquaporin water channel identified in the 3gd8 protein, bottleneck is highlighted with stick residues in blue (color figure online)

4. Bottleneck

First, for our convenience we will use only the asymmetric unit for channel identification, as the biological unit contains the same copy four times, each having a channel in its structure. Next, we have to slightly tamper with the user settings. *ProbeRadius*: 5; *InteriorThreshold*: 1.1 and remove HET atoms. Since the structure contains water molecules passing through the channel, we can place the starting point approximately in the middle of the channel to the position of one of the water molecules (e.g. [7.659, -25.452, 22.465]). As the bottleneck of the channel is quite narrow, in order to pass water molecules in single file, we have to decrease the *BottleneckRadius* parameter to 1.0 as well. Finally, we can calculate channels from this point and merge them into a single pore.

The pore (see Fig. 11.23) is ~ 40 Å in length with a bottleneck as wide as 1.1 Å in radius formed by Histidine, Arginine, Alanine (backbone) and phenylalanine, which forms a selectivity filter preventing the passage of oxonium ions. Its Hydropathy index is -1.33 .

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Glossary

3D	Three-dimensional space
5-HT3	Serotonin receptor
AA	Amino acid
ACC	Atomic Charge Calculator
AIDS	Acquired Immune Deficiency Syndrome
ALIX	apoptosis-linked gene 2 interacting protein X
AQP	Aquaporin
ar/R	Aromatic/arginine constriction region
ATP	Adenosine triphosphate
BMRB	Biological Magnetic Resonance Data Bank
CDK2	Cyclin-dependent kinase 2
COX	Cyclooxygenase
CSA	Catalytic Site Atlas
DNA	Deoxyribonucleic acid
EBI	European Bioinformatics Institute
EEM	Electronegativity Equalization Method
EMDB	Electron Microscopy Data Bank
EMPIAR	Electron Microscopy Pilot Image Archive
GFDB	Glycan Fragment Database
GO	Gene Ontology
HIV	Human Immunodeficiency Virus
HsaF-HsaG	Aldose-Dehydrogenase complex
MD	Molecular dynamics
NDB	Nucleic Acids Database
NMR	Nuclear magnetic resonance spectroscopy
OPM	Orientation of Proteins in Membranes
PDB	Protein Data Bank
PDB format	Protein Data Bank structure file format
PDBe	PDB in Europe
PDBj	PDB Japan

PED	Protein Ensemble Database
PQ	PatternQuery
QM	Quantum mechanics
RCSB PDB	Research Collaboratory for Structural Bioinformatics PDB
RMSD	Root Mean Square Deviation
RNA	Ribonucleic acid
SAXS	Small Angle X-ray Scattering
UBD	Ubiquitin binding domain
vdW	van der Waals
wwPDB	Worldwide PDB
ZnF	Zinc finger

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